

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12384809
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Shih; Wei-Chun et al.

Organic electroluminescent materials and devices

Abstract

Provided are compounds of Formula Ir(L.sub.A).sub.x(L.sub.C).sub.y wherein: ligand L.sub.A has Formula I' ##STR00001## and ligand L.sub.C has Formula II' ##STR00002##

Inventors: Shih; Wei-Chun (Lawrenceville, NJ), Ji; Zhiqiang (Chalfont, PA), Boudreault; Pierre-Luc T. (Pennington, NJ), Chen; Hsiao-Fan (Lawrence Township, NJ), Lu; Tongxiang (Lawrenceville, NJ)

Applicant: Universal Display Corporation (Ewing, NJ)

Family ID: 74833671

Assignee: UNIVERSAL DISPLAY CORPORATION (Ewing, NJ)

Appl. No.: 18/483721

Filed: October 10, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240076307 A1	Mar. 07, 2024

Related U.S. Application Data

continuation parent-doc US 16986953 20200806 US 11820783 child-doc US 18483721
us-provisional-application US 62896857 20190906

Publication Classification

Int. Cl.: C07F15/00 (20060101); C09K11/06 (20060101); H10K85/30 (20230101)

U.S. Cl.:

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4769292	12/1987	Tang et al.	N/A	N/A
5061569	12/1990	VanSlyke et al.	N/A	N/A
5247190	12/1992	Friend et al.	N/A	N/A
5703436	12/1996	Forrest et al.	N/A	N/A
5707745	12/1997	Forrest et al.	N/A	N/A
5834893	12/1997	Bulovic et al.	N/A	N/A
5844363	12/1997	Gu et al.	N/A	N/A
6013982	12/1999	Thompson et al.	N/A	N/A
6087196	12/1999	Sturm et al.	N/A	N/A
6091195	12/1999	Forrest et al.	N/A	N/A
6097147	12/1999	Baldo et al.	N/A	N/A
6294398	12/2000	Kim et al.	N/A	N/A
6303238	12/2000	Thompson et al.	N/A	N/A
6337102	12/2001	Forrest et al.	N/A	N/A
6468819	12/2001	Kim et al.	N/A	N/A
6528187	12/2002	Okada	N/A	N/A
6687266	12/2003	Ma et al.	N/A	N/A
6835469	12/2003	Kwong et al.	N/A	N/A
6921915	12/2004	Takiguchi et al.	N/A	N/A
7087321	12/2005	Kwong et al.	N/A	N/A
7090928	12/2005	Thompson et al.	N/A	N/A
7154114	12/2005	Brooks et al.	N/A	N/A
7250226	12/2006	Tokito et al.	N/A	N/A
7279704	12/2006	Walters et al.	N/A	N/A
7332232	12/2007	Ma et al.	N/A	N/A
7338722	12/2007	Thompson et al.	N/A	N/A
7393599	12/2007	Thompson et al.	N/A	N/A
7396598	12/2007	Takeuchi et al.	N/A	N/A
7431968	12/2007	Shtein et al.	N/A	N/A
7445855	12/2007	Mackenzie et al.	N/A	N/A
7534505	12/2008	Lin et al.	N/A	N/A
8883322	12/2013	Wu	428/917	C09K 11/06
10003033	12/2017	Boudreault et al.	N/A	N/A
2002/0034656	12/2001	Thompson et al.	N/A	N/A
2002/0134984	12/2001	Igarashi	N/A	N/A
2002/0158242	12/2001	Son et al.	N/A	N/A
2003/0068526	12/2002	Kamatani	N/A	N/A
2003/0138657	12/2002	Li et al.	N/A	N/A

2003/0152802	12/2002	Tsuboyama et al.	N/A	N/A
2003/0162053	12/2002	Marks et al.	N/A	N/A
2003/0175553	12/2002	Thompson et al.	N/A	N/A
2003/0230980	12/2002	Forrest et al.	N/A	N/A
2004/0036077	12/2003	Ise	N/A	N/A
2004/0137267	12/2003	Igarashi et al.	N/A	N/A
2004/0137268	12/2003	Igarashi et al.	N/A	N/A
2004/0174116	12/2003	Lu et al.	N/A	N/A
2005/0025993	12/2004	Thompson et al.	N/A	N/A
2005/0112407	12/2004	Ogasawara et al.	N/A	N/A
2005/0191519	12/2004	Mishima et al.	N/A	N/A
2005/0238919	12/2004	Ogasawara	N/A	N/A
2005/0244673	12/2004	Satoh et al.	N/A	N/A
2005/0260441	12/2004	Thompson et al.	N/A	N/A
2005/0260449	12/2004	Walters et al.	N/A	N/A
2006/0008670	12/2005	Lin et al.	N/A	N/A
2006/0202194	12/2005	Jeong et al.	N/A	N/A
2006/0240279	12/2005	Adamovich et al.	N/A	N/A
2006/0251923	12/2005	Lin et al.	N/A	N/A
2006/0263635	12/2005	Ise	N/A	N/A
2006/0280965	12/2005	Kwong et al.	N/A	N/A
2007/0190359	12/2006	Knowles et al.	N/A	N/A
2007/0278938	12/2006	Yabunouchi et al.	N/A	N/A
2008/0015355	12/2007	Schafer et al.	N/A	N/A
2008/0018221	12/2007	Egen et al.	N/A	N/A
2008/0106190	12/2007	Yabunouchi et al.	N/A	N/A
2008/0124572	12/2007	Mizuki et al.	N/A	N/A
2008/0220265	12/2007	Xia et al.	N/A	N/A
2008/0297033	12/2007	Knowles et al.	N/A	N/A
2009/0008605	12/2008	Kawamura et al.	N/A	N/A
2009/0009065	12/2008	Nishimura et al.	N/A	N/A
2009/0017330	12/2008	Iwakuma et al.	N/A	N/A
2009/0030202	12/2008	Iwakuma et al.	N/A	N/A
2009/0039776	12/2008	Yamada et al.	N/A	N/A
2009/0045730	12/2008	Nishimura et al.	N/A	N/A
2009/0045731	12/2008	Nishimura et al.	N/A	N/A
2009/0101870	12/2008	Prakash et al.	N/A	N/A
2009/0108737	12/2008	Kwong et al.	N/A	N/A
2009/0115316	12/2008	Zheng et al.	N/A	N/A
2009/0165846	12/2008	Johannes et al.	N/A	N/A
2009/0167162	12/2008	Lin et al.	N/A	N/A
2009/0179554	12/2008	Kuma et al.	N/A	N/A
2013/0299795	12/2012	Xia	546/4	H10K 85/342
2015/0236277	12/2014	Boudreault	257/40	H10K 85/342
2019/0067601	12/2018	Ji	N/A	N/A
2019/0218240	12/2018	Yoo	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
106397340	12/2016	CN	N/A

106397489	12/2016	CN	N/A
106397490	12/2016	CN	N/A
106397494	12/2016	CN	N/A
106588993	12/2016	CN	N/A
108358971	12/2017	CN	N/A
108383879	12/2017	CN	N/A
108409792	12/2017	CN	N/A
108484678	12/2017	CN	N/A
108484679	12/2017	CN	N/A
108484680	12/2017	CN	N/A
0650955	12/1994	EP	N/A
1725079	12/2005	EP	N/A
2034538	12/2008	EP	N/A
200511610	12/2004	JP	N/A
2007123392	12/2006	JP	N/A
2007254297	12/2006	JP	N/A
2008074939	12/2007	JP	N/A
01/39234	12/2000	WO	N/A
02/02714	12/2001	WO	N/A
02015654	12/2001	WO	N/A
03040257	12/2002	WO	N/A
03060956	12/2002	WO	N/A
2004093207	12/2003	WO	N/A
2004107822	12/2003	WO	N/A
2005014551	12/2004	WO	N/A
2005019373	12/2004	WO	N/A
2005030900	12/2004	WO	N/A
2005089025	12/2004	WO	N/A
2005123873	12/2004	WO	N/A
2006009024	12/2005	WO	N/A
2006056418	12/2005	WO	N/A
2006072002	12/2005	WO	N/A
2006082742	12/2005	WO	N/A
2006098120	12/2005	WO	N/A
2006100298	12/2005	WO	N/A
2006103874	12/2005	WO	N/A
2006114966	12/2005	WO	N/A
2006132173	12/2005	WO	N/A
2007002683	12/2006	WO	N/A
2007004380	12/2006	WO	N/A
2007063754	12/2006	WO	N/A
2007063796	12/2006	WO	N/A
2008056746	12/2007	WO	N/A
2008101842	12/2007	WO	N/A
2008132085	12/2007	WO	N/A
2009000673	12/2007	WO	N/A
2009003898	12/2008	WO	N/A
2009008311	12/2008	WO	N/A
2009018009	12/2008	WO	N/A
2009021126	12/2008	WO	N/A

2009050290	12/2008	WO	N/A
2009062578	12/2008	WO	N/A
2009063833	12/2008	WO	N/A
2009066778	12/2008	WO	N/A
2009066779	12/2008	WO	N/A
2009086028	12/2008	WO	N/A
2009100991	12/2008	WO	N/A

OTHER PUBLICATIONS

Adachi, Chihaya et al., "Organic Electroluminescent Device Having a Hole Conductor as an Emitting Layer," Appl. Phys. Lett., 55(15): 1489-1491 (1989). cited by applicant

Adachi, Chihaya et al., "Nearly 100% Internal Phosphorescence Efficiency in an Organic Light Emitting Device," J. Appl. Phys., 90(10): 5048-5051 (2001). cited by applicant

Adachi, Chihaya et al., "High-Efficiency Red Electrophosphorescence Devices," Appl. Phys. Lett., 78(11)1622-1624 (2001). cited by applicant

Aonuma, Masaki et al., "Material Design of Hole Transport Materials Capable of Thick-Film Formation in Organic Light Emitting Diodes," Appl. Phys. Lett., 90, Apr. 30, 2007, 183503-1-183503-3. cited by applicant

Baldo et al., Highly Efficient Phosphorescent Emission from Organic Electroluminescent Devices, Nature, vol. 395, 151-154, (1998). cited by applicant

Baldo et al., Very high-efficiency green organic light-emitting devices based on electrophosphorescence, Appl. Phys. Lett., vol. 75, No. 1, 4-6 (1999). cited by applicant

Gao, Zhiqiang et al., "Bright-Blue Electroluminescence From a Silyl-Substituted ter-(phenylene-vinylene) derivative," Appl. Phys. Lett., 74(6): 865-867 (1999). cited by applicant

Guo, Tzung-Fang et al., "Highly Efficient Electrophosphorescent Polymer Light-Emitting Devices," Organic Electronics, 1: 15-20 (2000). cited by applicant

Hamada, Yuji et al., "High Luminance in Organic Electroluminescent Devices with Bis(10-hydroxybenzo[h]quinolinato) beryllium as an Emitter," Chem. Lett., 905-906 (1993). cited by applicant

Holmes, R.J. et al., "Blue Organic Electrophosphorescence Using Exothermic Host-Guest Energy Transfer," Appl. Phys. Lett., 82(15):2422-2424 (2003). cited by applicant

Hu, Nan-Xing et al., "Novel High Tg Hole-Transport Molecules Based on Indolo[3,2-b]carbazoles for Organic Light-Emitting Devices," Synthetic Metals, 111-112:421-424 (2000). cited by applicant

Huang, Jinsong et al., "Highly Efficient Red-Emission Polymer Phosphorescent Light-Emitting Diodes Based on Two Novel Tris(1-phenylisoquinolinato-C2,N)iridium(III) Derivatives," Adv. Mater., 19:739-743 (2007). cited by applicant

Huang, Wei-Sheng et al., "Highly Phosphorescent Bis-Cyclometalated Iridium Complexes Containing Benzoimidazole-Based Ligands," Chem. Mater., 16(12):2480-2488 (2004). cited by applicant

Hung, L.S. et al., "Anode Modification in Organic Light-Emitting Diodes by Low-Frequency Plasma Polymerization of CHF₃," Appl. Phys. Lett., 78(5):673-675 (2001). cited by applicant

Ikai, Masamichi et al., "Highly Efficient Phosphorescence From Organic Light-Emitting Devices with an Exciton-Block Layer," Appl. Phys. Lett., 79(2):156-158 (2001). cited by applicant

Ikeda, Hisao et al., "P-185 Low-Drive-Voltage OLEDs with a Buffer Layer Having Molybdenum Oxide," SID Symposium Digest, 37:923-926 (2006). cited by applicant

Inada, Hiroshi and Shirota, Yasuhiko, "1,3,5-Tris[4-(diphenylamino)phenyl]benzene and its Methylsubstituted Derivatives as a Novel Class of Amorphous Molecular Materials," J. Mater. Chem., 3(3):319-320 (1993). cited by applicant

Kanno, Hiroshi et al., "Highly Efficient and Stable Red Phosphorescent Organic Light-Emitting Device Using bis[2-(2-benzothiazoyl)phenolato]zinc(II) as host material," Appl. Phys. Lett.,

90:123509-1-123509-3 (2007). cited by applicant

Kido, Junji et al., 1,2,4-Triazole Derivative as an Electron Transport Layer in Organic Electroluminescent Devices, *Jpn. J. Appl. Phys.*, 32:L917-L920 (1993). cited by applicant

Kuwabara, Yoshiyuki et al., "Thermally Stable Multilayered Organic Electroluminescent Devices Using Novel Starburst Molecules, 4,4',4''-Tri(N-carbazolyl)triphenylamine (TCTA) and 4,4',4''-Tris(3-methylphenylphenyl-amino) triphenylamine (m-MTDATA), as Hole-Transport Materials," *Adv. Mater.*, 6(9):677-679 (1994). cited by applicant

Kwong, Raymond C. et al., "High Operational Stability of Electrophosphorescent Devices," *Appl. Phys. Lett.*, 81(1) 162-164 (2002). cited by applicant

Lamansky, Sergey et al., "Synthesis and Characterization of Phosphorescent Cyclometalated Iridium Complexes," *Inorg. Chem.*, 40(7):1704-1711 (2001). cited by applicant

Lee, Chang-Lyoul et al., "Polymer Phosphorescent Light-Emitting Devices Doped with Tris(2-phenylpyridine) Iridium as a Triplet Emitter," *Appl. Phys. Lett.*, 77(15):2280-2282 (2000). cited by applicant

Lo, Shih-Chun et al., "Blue Phosphorescence from Iridium(III) Complexes at Room Temperature," *Chem. Mater.*, 18 (21)5119-5129 (2006). cited by applicant

Ma, Yuguang et al., "Triplet Luminescent Dinuclear-Gold(I) Complex-Based Light-Emitting Diodes with Low Turn-On voltage," *Appl. Phys. Lett.*, 74(10):1361-1363 (1999). cited by applicant

Mi, Bao-Xiu et al., "Thermally Stable Hole-Transporting Material for Organic Light-Emitting Diode an Isoindole Derivative," *Chem. Mater.*, 15(16):3148-3151 (2003). cited by applicant

Nishida, Jun-ichi et al., "Preparation, Characterization, and Electroluminescence Characteristics of α -Diimine-type Platinum(II) Complexes with Perfluorinated Phenyl Groups as Ligands," *Chem. Lett.*, 34(4): 592-593 (2005). cited by applicant

Niu, Yu-Hua et al., "Highly Efficient Electrophosphorescent Devices with Saturated Red Emission from a Neutral Osmium Complex," *Chem. Mater.*, 17(13):3532-3536 (2005). cited by applicant

Noda, Tetsuya and Shirota, Yasuhiko, "5,5'-Bis(dimesitylboryl)-2,2'-bithiophene and 5,5'-Bis(dimesitylboryl)-2,2'5',2''-terthiophene as a Novel Family of Electron-Transporting Amorphous Molecular Materials," *J. Am. Chem. Soc.*, 120 (37):9714-9715 (1998). cited by applicant

Okumoto, Kenji et al., "Green Fluorescent Organic Light-Emitting Device with External Quantum Efficiency of Nearly 10%," *Appl. Phys. Lett.*, 89:063504-1-063504-3 (2006). cited by applicant

Palilis, Leonidas C., "High Efficiency Molecular Organic Light-Emitting Diodes Based On Silole Derivatives And Their Exciplexes," *Organic Electronics*, 4:113-121 (2003). cited by applicant

Paulose, Betty Marie Jennifer S. et al., "First Examples of Alkenyl Pyridines as Organic Ligands for Phosphorescent Iridium Complexes," *Adv. Mater.*, 16(22):2003-2007 (2004). cited by applicant

Ranjan, Sudhir et al., "Realizing Green Phosphorescent Light-Emitting Materials from Rhenium(I) Pyrazolato Diimine Complexes," *Inorg. Chem.*, 42(4):1248-1255 (2003). cited by applicant

Sakamoto, Youichi et al., "Synthesis, Characterization, and Electron-Transport Property of Perfluorinated Phenylene Dendrimers," *J. Am. Chem. Soc.*, 122(8):1832-1833 (2000). cited by applicant

Salbeck, J. et al., "Low Molecular Organic Glasses for Blue Electroluminescence," *Synthetic Metals*, 91: 209-215 (1997). cited by applicant

Shirota, Yasuhiko et al., "Starburst Molecules Based on pi-Electron Systems as Materials for Organic Electroluminescent Devices," *Journal of Luminescence*, 72-74:985-991 (1997). cited by applicant

Sotoyama, Wataru et al., "Efficient Organic Light-Emitting Diodes with Phosphorescent Platinum Complexes Containing N^C^N-Coordinating Tridentate Ligand," *Appl. Phys. Lett.*, 86:153505-1-153505-3 (2005). cited by applicant

Sun, Yiru and Forrest, Stephen R., "High-Efficiency White Organic Light Emitting Devices with Three Separate Phosphorescent Emission Layers," *Appl. Phys. Lett.*, 91:263503-1-263503-3 (2007). cited by applicant

T. Östergård et al., “Langmuir-Blodgett Light-Emitting Diodes Of Poly(3-Hexylthiophene) Electro-Optical Characteristics Related to Structure,” *Synthetic Metals*, 88:171-177 (1997). cited by applicant

Takizawa, Shin-ya et al., “Phosphorescent Iridium Complexes Based on 2-Phenylimidazo[1,2- α]pyridine Ligands Tuning of Emission Color toward the Blue Region and Application to Polymer Light-Emitting Devices,” *Inorg. Chem.*, 46(10):4308-4319 (2007). cited by applicant

Tang, C.W. and VanSlyke, S.A., “Organic Electroluminescent Diodes,” *Appl. Phys. Lett.*, 51(12):913-915 (1987). cited by applicant

Tung, Yung-Liang et al., “Organic Light-Emitting Diodes Based on Charge-Neutral Ru II PHosphorescent Emitters,” *Adv. Mater.*, 17(8):1059-1064 (2005). cited by applicant

Van Slyke, S. A. et al., “Organic Electroluminescent Devices with Improved Stability,” *Appl. Phys. Lett.*, 69 (15):2160-2162 (1996). cited by applicant

Wang, Y. et al., “Highly Efficient Electroluminescent Materials Based on Fluorinated Organometallic Iridium Compounds,” *Appl. Phys. Lett.*, 79(4):449-451 (2001). cited by applicant

Wong, Keith Man-Chung et al., A Novel Class of Phosphorescent Gold(III) Alkynyl-Based Organic Light-Emitting Devices with Tunable Colour, *Chem. Commun.*, 2906-2908 (2005). cited by applicant

Wong, Wai-Yeung, “Multifunctional Iridium Complexes Based on Carbazole Modules as Highly Efficient Electrophosphors,” *Angew. Chem. Int. Ed.*, 45:7800-7803 (2006). cited by applicant

Primary Examiner: Rodd; Christopher M

Attorney, Agent or Firm: DUANE MORRIS LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of co-pending U.S. patent application Ser. No. 16/986,953, filed Aug. 6, 2020, which claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Application No. 62/896,857, filed on Sep. 6, 2019, the entire contents of which are incorporated herein by reference. FIELD (2) The present disclosure generally relates to organometallic compounds and formulations and their various uses including as emitters in devices such as organic light emitting diodes and related electronic devices.

BACKGROUND

(1) Opto-electronic devices that make use of organic materials are becoming increasingly desirable for various reasons. Many of the materials used to make such devices are relatively inexpensive, so organic opto-electronic devices have the potential for cost advantages over inorganic devices. In addition, the inherent properties of organic materials, such as their flexibility, may make them well suited for particular applications such as fabrication on a flexible substrate. Examples of organic opto-electronic devices include organic light emitting diodes/devices (OLEDs), organic phototransistors, organic photovoltaic cells, and organic photodetectors. For OLEDs, the organic materials may have performance advantages over conventional materials.

(2) OLEDs make use of thin organic films that emit light when voltage is applied across the device. OLEDs are becoming an increasingly interesting technology for use in applications such as flat panel displays, illumination, and backlighting.

(3) One application for phosphorescent emissive molecules is a full color display. Industry standards for such a display call for pixels adapted to emit particular colors, referred to as “saturated” colors. In particular, these standards call for saturated red, green, and blue pixels.

Alternatively, the OLED can be designed to emit white light. In conventional liquid crystal displays emission from a white backlight is filtered using absorption filters to produce red, green and blue emission. The same technique can also be used with OLEDs. The white OLED can be either a single emissive layer (EML) device or a stack structure. Color may be measured using CIE coordinates, which are well known to the art.

SUMMARY

(4) The present invention discloses transition metal compounds comprising new polyaza-substituted ligands as emissive dopants for improving device performance of OLED devices.

(5) In one aspect, the present disclosure provides a compound of Formula

$\text{Ir}(\text{L.sub.A})_{\text{sub.x}}(\text{L.sub.C})_{\text{sub.y}}$, wherein: ligand L.sub.A has Formula I'

(6) ##STR00003##

and ligand L.sub.C has Formula II

(7) ##STR00004##

wherein: x is 1, 2, or 3; y is 0, 1, or 2; $x+y=3$; X¹-X⁸ are each independently C, N, or CR, and at least four of X¹-X⁸ are N; the X¹-X⁴ that connects to ring A is C, and the X¹-X⁴ that coordinates to Ir is N; ring A is an unsaturated 5-membered or 6-membered carbocyclic or heterocyclic ring; R^A represents zero, mono, or up to the maximum number of allowed substitutions to its associated ring; each R, R¹-R³, and R^A is independently a hydrogen or a substituent selected from the group consisting of deuterium, halogen, alkyl, cycloalkyl, heteroalkyl, heterocycloalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, boryl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carboxylic acid, ether, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof; and two substituents can be joined or fused together to form a ring, wherein the ligand L.sub.A coordinates to Ir to form a 5-membered chelate ring as indicated by the two dashed lines of the Formula I', with the proviso that the compound of Formula $\text{Ir}(\text{L.sub.A})_{\text{sub.x}}(\text{L.sub.C})_{\text{sub.y}}$ is not

(8) ##STR00005##

(9) In another aspect, the present disclosure provides a formulation of the compound of the present disclosure.

(10) In yet another aspect, the present disclosure provides an OLED having an organic layer comprising the compound of the present disclosure.

(11) In yet another aspect, the present disclosure provides a consumer product comprising an OLED with an organic layer comprising the compound of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 shows an organic light emitting device.

(2) FIG. 2 shows an inverted organic light emitting device that does not have a separate electron transport layer.

DETAILED DESCRIPTION

A. Terminology

(3) Unless otherwise specified, the below terms used herein are defined as follows:

(4) As used herein, the term “organic” includes polymeric materials as well as small molecule organic materials that may be used to fabricate organic opto-electronic devices. “Small molecule” refers to any organic material that is not a polymer, and “small molecules” may actually be quite large. Small molecules may include repeat units in some circumstances. For example, using a long chain alkyl group as a substituent does not remove a molecule from the “small molecule” class. Small molecules may also be incorporated into polymers, for example as a pendent group on a polymer backbone or as a part of the backbone. Small molecules may also serve as the core moiety

of a dendrimer, which consists of a series of chemical shells built on the core moiety. The core moiety of a dendrimer may be a fluorescent or phosphorescent small molecule emitter. A dendrimer may be a “small molecule,” and it is believed that all dendrimers currently used in the field of OLEDs are small molecules.

(5) As used herein, “top” means furthest away from the substrate, while “bottom” means closest to the substrate. Where a first layer is described as “disposed over” a second layer, the first layer is disposed further away from substrate. There may be other layers between the first and second layer, unless it is specified that the first layer is “in contact with” the second layer. For example, a cathode may be described as “disposed over” an anode, even though there are various organic layers in between.

(6) As used herein, “solution processable” means capable of being dissolved, dispersed, or transported in and/or deposited from a liquid medium, either in solution or suspension form.

(7) A ligand may be referred to as “photoactive” when it is believed that the ligand directly contributes to the photoactive properties of an emissive material. A ligand may be referred to as “ancillary” when it is believed that the ligand does not contribute to the photoactive properties of an emissive material, although an ancillary ligand may alter the properties of a photoactive ligand

(8) As used herein, and as would be generally understood by one skilled in the art, a first “Highest Occupied Molecular Orbital” (HOMO) or “Lowest Unoccupied Molecular Orbital” (LUMO) energy level is “greater than” or “higher than” a second HOMO or LUMO energy level if the first energy level is closer to the vacuum energy level. Since ionization potentials (IP) are measured as a negative energy relative to a vacuum level, a higher HOMO energy level corresponds to an IP having a smaller absolute value (an IP that is less negative). Similarly, a higher LUMO energy level corresponds to an electron affinity (EA) having a smaller absolute value (an EA that is less negative). On a conventional energy level diagram, with the vacuum level at the top, the LUMO energy level of a material is higher than the HOMO energy level of the same material. A “higher” HOMO or LUMO energy level appears closer to the top of such a diagram than a “lower” HOMO or LUMO energy level.

(9) As used herein, and as would be generally understood by one skilled in the art, a first work function is “greater than” or “higher than” a second work function if the first work function has a higher absolute value. Because work functions are generally measured as negative numbers relative to vacuum level, this means that a “higher” work function is more negative. On a conventional energy level diagram, with the vacuum level at the top, a “higher” work function is illustrated as further away from the vacuum level in the downward direction. Thus, the definitions of HOMO and LUMO energy levels follow a different convention than work functions.

(10) The terms “halo,” “halogen,” and “halide” are used interchangeably and refer to fluorine, chlorine, bromine, and iodine.

(11) The term “acyl” refers to a substituted carbonyl radical ($\text{C}(\text{O})\text{—R.sub.s}$).

(12) The term “ester” refers to a substituted oxycarbonyl ($\text{—O—C}(\text{O})\text{—R.sub.s}$ or $\text{—C}(\text{O})\text{—O—R.sub.s}$) radical.

(13) The term “ether” refers to an —OR , radical.

(14) The terms “sulfanyl” or “thio-ether” are used interchangeably and refer to a —SR.sub.s radical.

(15) The term “sulfinyl” refers to a $\text{—S}(\text{O})\text{—R.sub.s}$ radical.

(16) The term “sulfonyl” refers to a —SO.sub.2—R.sub.s radical.

(17) The term “phosphino” refers to a —P(R.sub.s).sub.3 radical, wherein each R.sub.s can be same or different.

(18) The term “silyl” refers to a $\text{—Si(R.sub.s).sub.3}$ radical, wherein each R.sub.s can be same or different.

(19) The term “boryl” refers to a —B(R.sub.s).sub.2 radical or its Lewis adduct —B(R.sub.s).sub.3 radical, wherein R.sub.s can be same or different.

- (20) In each of the above, R.sub.s can be hydrogen or a substituent selected from the group consisting of deuterium, halogen, alkyl, cycloalkyl, heteroalkyl, heterocycloalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, and combination thereof. Preferred R.sub.s is selected from the group consisting of alkyl, cycloalkyl, aryl, heteroaryl, and combination thereof.
- (21) The term “alkyl” refers to and includes both straight and branched chain alkyl radicals. Preferred alkyl groups are those containing from one to fifteen carbon atoms and includes methyl, ethyl, propyl, 1-methylethyl, butyl, 1-methylpropyl, 2-methylpropyl, pentyl, 1-methylbutyl, 2-methylbutyl, 3-methylbutyl, 1,1-dimethylpropyl, 1,2-dimethylpropyl, 2,2-dimethylpropyl, and the like. Additionally, the alkyl group may be optionally substituted.
- (22) The term “cycloalkyl” refers to and includes monocyclic, polycyclic, and spiro alkyl radicals. Preferred cycloalkyl groups are those containing 3 to 12 ring carbon atoms and includes cyclopropyl, cyclopentyl, cyclohexyl, bicyclo[3.1.1]heptyl, spiro[4.5]decyl, spiro[5.5]undecyl, adamantyl, and the like. Additionally, the cycloalkyl group may be optionally substituted.
- (23) The terms “heteroalkyl” or “heterocycloalkyl” refer to an alkyl or a cycloalkyl radical, respectively, having at least one carbon atom replaced by a heteroatom. Optionally the at least one heteroatom is selected from O, S, N, P, B, Si and Se, preferably, O, S or N. Additionally, the heteroalkyl or heterocycloalkyl group may be optionally substituted.
- (24) The term “alkenyl” refers to and includes both straight and branched chain alkene radicals. Alkenyl groups are essentially alkyl groups that include at least one carbon-carbon double bond in the alkyl chain. Cycloalkenyl groups are essentially cycloalkyl groups that include at least one carbon-carbon double bond in the cycloalkyl ring. The term “heteroalkenyl” as used herein refers to an alkenyl radical having at least one carbon atom replaced by a heteroatom. Optionally the at least one heteroatom is selected from O, S, N, P, B, Si, and Se, preferably, O, S, or N. Preferred alkenyl, cycloalkenyl, or heteroalkenyl groups are those containing two to fifteen carbon atoms. Additionally, the alkenyl, cycloalkenyl, or heteroalkenyl group may be optionally substituted.
- (25) The term “alkynyl” refers to and includes both straight and branched chain alkyne radicals. Alkynyl groups are essentially alkyl groups that include at least one carbon-carbon triple bond in the alkyl chain. Preferred alkynyl groups are those containing two to fifteen carbon atoms. Additionally, the alkynyl group may be optionally substituted.
- (26) The terms “aralkyl” or “arylalkyl” are used interchangeably and refer to an alkyl group that is substituted with an aryl group. Additionally, the aralkyl group may be optionally substituted.
- (27) The term “heterocyclic group” refers to and includes aromatic and non-aromatic cyclic radicals containing at least one heteroatom. Optionally the at least one heteroatom is selected from O, S, N, P, B, Si, and Se, preferably, O, S, or N. Hetero-aromatic cyclic radicals may be used interchangeably with heteroaryl. Preferred hetero-non-aromatic cyclic groups are those containing 3 to 7 ring atoms which includes at least one hetero atom, and includes cyclic amines such as morpholino, piperidino, pyrrolidino, and the like, and cyclic ethers/thio-ethers, such as tetrahydrofuran, tetrahydropyran, tetrahydrothiophene, and the like. Additionally, the heterocyclic group may be optionally substituted.
- (28) The term “aryl” refers to and includes both single-ring aromatic hydrocarbyl groups and polycyclic aromatic ring systems. The polycyclic rings may have two or more rings in which two carbons are common to two adjoining rings (the rings are “fused”) wherein at least one of the rings is an aromatic hydrocarbyl group, e.g., the other rings can be cycloalkyls, cycloalkenyls, aryl, heterocycles, and/or heteroaryls. Preferred aryl groups are those containing six to thirty carbon atoms, preferably six to twenty carbon atoms, more preferably six to twelve carbon atoms. Especially preferred is an aryl group having six carbons, ten carbons or twelve carbons. Suitable aryl groups include phenyl, biphenyl, triphenyl, triphenylene, tetraphenylene, naphthalene, anthracene, phenalene, phenanthrene, fluorene, pyrene, chrysene, perylene, and azulene, preferably phenyl, biphenyl, triphenyl, triphenylene, fluorene, and naphthalene. Additionally, the aryl group

may be optionally substituted.

(29) The term “heteroaryl” refers to and includes both single-ring aromatic groups and polycyclic aromatic ring systems that include at least one heteroatom. The heteroatoms include, but are not limited to O, S, N, P, B, Si, and Se. In many instances, O, S, or N are the preferred heteroatoms. Hetero-single ring aromatic systems are preferably single rings with 5 or 6 ring atoms, and the ring can have from one to six heteroatoms. The hetero-polycyclic ring systems can have two or more rings in which two atoms are common to two adjoining rings (the rings are “fused”) wherein at least one of the rings is a heteroaryl, e.g., the other rings can be cycloalkyls, cycloalkenyls, aryl, heterocycles, and/or heteroaryls. The hetero-polycyclic aromatic ring systems can have from one to six heteroatoms per ring of the polycyclic aromatic ring system. Preferred heteroaryl groups are those containing three to thirty carbon atoms, preferably three to twenty carbon atoms, more preferably three to twelve carbon atoms. Suitable heteroaryl groups include dibenzothiophene, dibenzofuran, dibenzoselenophene, furan, thiophene, benzofuran, benzothiophene, benzoselenophene, carbazole, indolocarbazole, pyridylindole, pyrrolodipyridine, pyrazole, imidazole, triazole, oxazole, thiazole, oxadiazole, oxatriazole, dioxazole, thiadiazole, pyridine, pyridazine, pyrimidine, pyrazine, triazine, oxazine, oxathiazine, oxadiazine, indole, benzimidazole, indazole, indoxazine, benzoxazole, benzisoxazole, benzothiazole, quinoline, isoquinoline, cinnoline, quinazoline, quinoxaline, naphthyridine, phthalazine, pteridine, xanthene, acridine, phenazine, phenothiazine, phenoxazine, benzofuroypyridine, furodipyridine, benzothienopyridine, thienodipyridine, benzoselenophenopyridine, and selenophenodipyridine, preferably dibenzothiophene, dibenzofuran, dibenzoselenophene, carbazole, indolocarbazole, imidazole, pyridine, triazine, benzimidazole, 1,2-azaborine, 1,3-azaborine, 1,4-azaborine, borazine, and aza-analogs thereof. Additionally, the heteroaryl group may be optionally substituted.

(30) Of the aryl and heteroaryl groups listed above, the groups of triphenylene, naphthalene, anthracene, dibenzothiophene, dibenzofuran, dibenzoselenophene, carbazole, indolocarbazole, imidazole, pyridine, pyrazine, pyrimidine, triazine, and benzimidazole, and the respective aza-analogs of each thereof are of particular interest.

(31) The terms alkyl, cycloalkyl, heteroalkyl, heterocycloalkyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aralkyl, heterocyclic group, aryl, and heteroaryl, as used herein, are independently unsubstituted, or independently substituted, with one or more general substituents.

(32) In many instances, the general substituents are selected from the group consisting of deuterium, halogen, alkyl, cycloalkyl, heteroalkyl, heterocycloalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carboxylic acid, ether, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, boryl, and combinations thereof.

(33) In some instances, the preferred general substituents are selected from the group consisting of deuterium, fluorine, alkyl, cycloalkyl, heteroalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, aryl, heteroaryl, nitrile, isonitrile, sulfanyl, boryl, and combinations thereof.

(34) In some instances, the more preferred general substituents are selected from the group consisting of deuterium, fluorine, alkyl, cycloalkyl, alkoxy, aryloxy, amino, silyl, boryl, aryl, heteroaryl, sulfanyl, and combinations thereof.

(35) In yet other instances, the most preferred general substituents are selected from the group consisting of deuterium, fluorine, alkyl, cycloalkyl, aryl, heteroaryl, and combinations thereof.

(36) The terms “substituted” and “substitution” refer to a substituent other than H that is bonded to the relevant position, e.g., a carbon or nitrogen. For example, when R^{sup.1} represents mono-substitution, then one R^{sup.1} must be other than H (i.e., a substitution). Similarly, when R^{sup.1} represents di-substitution, then two of R^{sup.1} must be other than H. Similarly, when R^{sup.1} represents zero or no substitution, R^{sup.1}, for example, can be a hydrogen for available valencies of ring atoms, as in carbon atoms for benzene and the nitrogen atom in pyrrole, or simply

represents nothing for ring atoms with fully filled valencies, e.g., the nitrogen atom in pyridine. The maximum number of substitutions possible in a ring structure will depend on the total number of available valencies in the ring atoms.

(37) As used herein, “combinations thereof” indicates that one or more members of the applicable list are combined to form a known or chemically stable arrangement that one of ordinary skill in the art can envision from the applicable list. For example, an alkyl and deuterium can be combined to form a partial or fully deuterated alkyl group; a halogen and alkyl can be combined to form a halogenated alkyl substituent; and a halogen, alkyl, and aryl can be combined to form a halogenated arylalkyl. In one instance, the term substitution includes a combination of two to four of the listed groups. In another instance, the term substitution includes a combination of two to three groups. In yet another instance, the term substitution includes a combination of two groups. Preferred combinations of substituent groups are those that contain up to fifty atoms that are not hydrogen or deuterium, or those which include up to forty atoms that are not hydrogen or deuterium, or those that include up to thirty atoms that are not hydrogen or deuterium. In many instances, a preferred combination of substituent groups will include up to twenty atoms that are not hydrogen or deuterium.

(38) The “aza” designation in the fragments described herein, i.e. aza-dibenzofuran, aza-dibenzothiophene, etc. means that one or more of the C—H groups in the respective aromatic ring can be replaced by a nitrogen atom, for example, and without any limitation, azatriphenylene encompasses both dibenzo[f,h]quinoxaline and dibenzo[f,h]quinoline. One of ordinary skill in the art can readily envision other nitrogen analogs of the aza-derivatives described above, and all such analogs are intended to be encompassed by the terms as set forth herein.

(39) As used herein, “deuterium” refers to an isotope of hydrogen. Deuterated compounds can be readily prepared using methods known in the art. For example, U.S. Pat. No. 8,557,400, Patent Pub. No. WO 2006/095951, and U.S. Pat. Application Pub. No. US 2011/0037057, which are hereby incorporated by reference in their entireties, describe the making of deuterium-substituted organometallic complexes. Further reference is made to Ming Yan, et al., *Tetrahedron* 2015, 71, 1425-30 and Atzrodt et al., *Angew. Chem. Int. Ed. (Reviews)* 2007, 46, 7744-65, which are incorporated by reference in their entireties, describe the deuteration of the methylene hydrogens in benzyl amines and efficient pathways to replace aromatic ring hydrogens with deuterium, respectively.

(40) It is to be understood that when a molecular fragment is described as being a substituent or otherwise attached to another moiety, its name may be written as if it were a fragment (e.g. phenyl, phenylene, naphthyl, dibenzofuryl) or as if it were the whole molecule (e.g. benzene, naphthalene, dibenzofuran). As used herein, these different ways of designating a substituent or attached fragment are considered to be equivalent.

(41) In some instance, a pair of adjacent substituents can be optionally joined or fused into a ring. The preferred ring is a five, six, or seven-membered carbocyclic or heterocyclic ring, includes both instances where the portion of the ring formed by the pair of substituents is saturated and where the portion of the ring formed by the pair of substituents is unsaturated. As used herein, “adjacent” means that the two substituents involved can be on the same ring next to each other, or on two neighboring rings having the two closest available substitutable positions, such as 2,2' positions in a biphenyl, or 1,8 position in a naphthalene, as long as they can form a stable fused ring system.

B. The Compounds of the Present Disclosure

(42) In one aspect, the present disclosure provides a compound of Formula
 $\text{Ir}(\text{L.sub.A})_{\text{sub.x}}(\text{L.sub.C})_{\text{sub.y}}$, wherein: ligand L.sub.A has Formula I'

(43) ##STR00006##

and ligand L.sub.C has Formula II' R^{sup.2}

(44) ##STR00007##

wherein: x is 1, 2, or 3; y is 0, 1, or 2; x+y=3; X^{sup.1}-X^{sup.8} are each independently C, N, or CR,

and at least four of X¹-X⁸ are N; the X¹-X⁴ that connects to ring A is C, and the X¹-X⁴ that coordinates to Ir is N; ring A is an unsaturated 5-membered or 6-membered carbocyclic or heterocyclic ring; R^A represents zero, mono, or up to the maximum number of allowed substitutions to its associated ring; each R, R¹-R³, and R^A is independently a hydrogen or a substituent selected from the group consisting of the general substituents defined herein; and two substituents can be joined or fused together to form a ring, wherein the ligand L_A coordinates to Ir to form a 5-membered chelate ring as indicated by the two dashed lines of the Formula I', with the proviso that the compound of Formula Ir(L_A).sub.x(L_C).sub.y is not

(45) ##STR00008##

(46) In some embodiments, each R, R¹-R³, and R^A each can independently be a hydrogen or a substituent selected from the group consisting of the preferred general substituents defined herein.

(47) In some embodiments, four of X¹-X⁸ are N.

(48) In some embodiments, five or more of X¹-X⁸ are N. In some embodiments, the maximum number of N atoms that are bonded to one another within a ring is two.

(49) In some embodiments, X² and X⁴ are N, and X' and X³ are C.

(50) In some embodiments, X² and X⁴ are C, and X' and X³ are N.

(51) In some embodiments, X² and X³ are N, and X' and X⁴ are C.

(52) In some embodiments, X² and X³ are C, and X' and X⁴ are N.

(53) In some embodiments, X², X⁴, X⁵, and X⁸ are N, and X' and X³ are C.

(54) In some embodiments, X⁶ and X⁷ are CR wherein the two R substituents are joined to form a fused ring.

(55) In some embodiments, two of the X⁵-X⁸ are CR and adjacent to each other with the two R substituents joined to form a fused 6-membered aromatic ring.

(56) In some embodiments, X⁶ and X⁷ are CR with at least one R substituent being F.

(57) In some embodiments, X⁶ and X⁷ are CR with at least one R substituent being deuterium.

(58) In some embodiments, ring A is a 6-membered aromatic ring.

(59) In some embodiments, ring A is a 5-membered heteroaryl ring.

(60) In some embodiments, two adjacent R^A substituents are joined to form a 6-membered aromatic ring fused to ring A.

(61) In some embodiments, R¹ and R² are independently an alkyl, cycloalkyl, partially or fully deuterated variants thereof, partially or fully fluorinated variants thereof, and combinations thereof. In some embodiments, R¹ comprises at least two carbon atoms. In some embodiments, R¹ comprises at least three carbon atoms. In some embodiments, R¹ and R² both comprise at least two carbon atoms. In some embodiments, R¹ and R² both comprise at least three carbon atoms.

(62) In some embodiments, R³ is H.

(63) In some embodiments, the compound has Formula Ir(L_A).sub.2(L_C).

(64) In some embodiments, the ligand L_A is selected from the group consisting of:

(65) ##STR00009## ##STR00010## ##STR00011##

wherein: Z¹—Z⁶ are each independently N or CR^F; Y is O or S; each of R^E and R^F is independently a hydrogen or a substituent selected from the group consisting of deuterium, fluorine, alkyl, cycloalkyl, heteroalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, aryl, heteroaryl, nitrile, isonitrile, sulfanyl, and combinations thereof.

(66) In some embodiments, the ligand L_A is L_{Ai-m}, wherein i is an integer from 1 to 200 and m is an integer from 1 to 20, wherein the ligand L_A is selected from the group consisting of L_{A1-1} through L_{A200-20}, wherein the structure of each of L_{Ai-1} through

L.sub.Ai-20 is defined below:

(67) ##STR00012## ##STR00013## ##STR00014## ##STR00015##

wherein for each L.sub.Ai in L.sub.Ai-1 to L.sub.Ai-20, R.sup.E and G in formulas 1 to formula 20 are defined as follows:

(68) TABLE-US-00001 L.sub.Ai R.sup.E G L.sub.A1 R.sup.E1 G.sup.1 L.sub.A2 R.sup.E2 G.sup.1 L.sub.A3 R.sup.E3 G.sup.1 L.sub.A4 R.sup.E4 G.sup.1 L.sub.A5 R.sup.E5 G.sup.1 L.sub.A6 R.sup.E6 G.sup.1 L.sub.A7 R.sup.E7 G.sup.1 L.sub.A8 R.sup.E8 G.sup.1 L.sub.A9 R.sup.E9 G.sup.1 L.sub.A10 R.sup.E10 G.sup.1 L.sub.A11 R.sup.E11 G.sup.1 L.sub.A12 R.sup.E12 G.sup.1 L.sub.A13 R.sup.E13 G.sup.1 L.sub.A14 R.sup.E14 G.sup.1 L.sub.A15 R.sup.E15 G.sup.1 L.sub.A16 R.sup.E16 G.sup.1 L.sub.A17 R.sup.E17 G.sup.1 L.sub.A18 R.sup.E18 G.sup.1 L.sub.A19 R.sup.E19 G.sup.1 L.sub.A20 R.sup.E20 G.sup.1 L.sub.A21 R.sup.E1 G.sup.2 L.sub.A22 R.sup.E2 G.sup.2 L.sub.A23 R.sup.E3 G.sup.2 L.sub.A24 R.sup.E4 G.sup.2 L.sub.A25 R.sup.E5 G.sup.2 L.sub.A26 R.sup.E6 G.sup.2 L.sub.A27 R.sup.E7 G.sup.2 L.sub.A28 R.sup.E8 G.sup.2 L.sub.A29 R.sup.E9 G.sup.2 L.sub.A30 R.sup.E10 G.sup.2 L.sub.A31 R.sup.E11 G.sup.2 L.sub.A32 R.sup.E12 G.sup.2 L.sub.A33 R.sup.E13 G.sup.2 L.sub.A34 R.sup.E14 G.sup.2 L.sub.A35 R.sup.E15 G.sup.2 L.sub.A36 R.sup.E16 G.sup.2 L.sub.A37 R.sup.E17 G.sup.2 L.sub.A38 R.sup.E18 G.sup.2 L.sub.A39 R.sup.E19 G.sup.2 L.sub.A40 R.sup.E20 G.sup.2 L.sub.A41 R.sup.E1 G.sup.3 L.sub.A42 R.sup.E2 G.sup.3 L.sub.A43 R.sup.E3 G.sup.3 L.sub.A44 R.sup.E4 G.sup.3 L.sub.A45 R.sup.E5 G.sup.3 L.sub.A46 R.sup.E6 G.sup.3 L.sub.A47 R.sup.E7 G.sup.3 L.sub.A48 R.sup.E8 G.sup.3 L.sub.A49 R.sup.E9 G.sup.3 L.sub.A50 R.sup.E10 G.sup.3 L.sub.A51 R.sup.E11 G.sup.3 L.sub.A52 R.sup.E12 G.sup.3 L.sub.A53 R.sup.E13 G.sup.3 L.sub.A54 R.sup.E14 G.sup.3 L.sub.A55 R.sup.E15 G.sup.3 L.sub.A56 R.sup.E16 G.sup.3 L.sub.A57 R.sup.E17 G.sup.3 L.sub.A58 R.sup.E15 G.sup.3 L.sub.A59 R.sup.E19 G.sup.3 L.sub.A60 R.sup.E20 G.sup.3 L.sub.A61 R.sup.E1 G.sup.4 L.sub.A62 R.sup.E2 G.sup.4 L.sub.A63 R.sup.E3 G.sup.4 L.sub.A64 R.sup.E4 G.sup.4 L.sub.A65 R.sup.E5 G.sup.4 L.sub.A66 R.sup.E6 G.sup.4 L.sub.A67 R.sup.E7 G.sup.4 L.sub.A68 R.sup.E8 G.sup.4 L.sub.A69 R.sup.E9 G.sup.4 L.sub.A70 R.sup.E10 G.sup.4 L.sub.A71 R.sup.E11 G.sup.4 L.sub.A72 R.sup.E12 G.sup.4 L.sub.A73 R.sup.113 G.sup.4 L.sub.A74 R.sup.E14 G.sup.4 L.sub.A75 R.sup.E15 G.sup.4 L.sub.A76 R.sup.E16 G.sup.4 L.sub.A77 R.sup.E17 G.sup.4 L.sub.A78 R.sup.E18 G.sup.4 L.sub.A79 R.sup.E19 G.sup.4 L.sub.A80 R.sup.E20 G.sup.4 L.sub.A81 R.sup.E1 G.sup.5 L.sub.A82 R.sup.E2 G.sup.5 L.sub.A83 R.sup.E3 G.sup.5 L.sub.A84 R.sup.E4 G.sup.5 L.sub.A85 R.sup.E5 G.sup.5 L.sub.A86 R.sup.E6 G.sup.5 L.sub.A87 R.sup.E7 G.sup.5 L.sub.A88 R.sup.E8 G.sup.5 L.sub.A89 R.sup.E9 G.sup.5 L.sub.A90 R.sup.E10 G.sup.5 L.sub.A91 R.sup.E11 G.sup.5 L.sub.A92 R.sup.E12 G.sup.5 L.sub.A93 R.sup.E13 G.sup.5 L.sub.A94 R.sup.E14 G.sup.5 L.sub.A95 R.sup.E15 G.sup.5 L.sub.A96 R.sup.E16 G.sup.5 L.sub.A97 R.sup.E17 G.sup.5 L.sub.A98 R.sup.E18 G.sup.5 L.sub.A99 R.sup.E19 G.sup.5 L.sub.A100 R.sup.E20 G.sup.5 L.sub.A101 R.sup.E1 G.sup.6 L.sub.A102 R.sup.E2 G.sup.6 L.sub.A103 R.sup.E3 G.sup.6 L.sub.A104 R.sup.E4 G.sup.6 L.sub.A105 R.sup.E5 G.sup.6 L.sub.A106 R.sup.E6 G.sup.6 L.sub.A107 R.sup.E7 G.sup.6 L.sub.A108 R.sup.E8 G.sup.6 L.sub.A109 R.sup.E9 G.sup.6 L.sub.A110 R.sup.E10 G.sup.6 L.sub.A111 R.sup.E11 G.sup.6 L.sub.A112 R.sup.E12 G.sup.6 L.sub.A113 R.sup.E13 G.sup.6 L.sub.A114 R.sup.E14 G.sup.6 L.sub.A115 R.sup.E15 G.sup.6 L.sub.A116 R.sup.E16 G.sup.6 L.sub.A117 R.sup.E17 G.sup.6 L.sub.A118 R.sup.E18 G.sup.6 L.sub.A119 R.sup.E19 G.sup.6 L.sub.A120 R.sup.E20 G.sup.6 L.sub.A121 R.sup.E1 G.sup.7 L.sub.A122 R.sup.E2 G.sup.7 L.sub.A123 R.sup.E3 G.sup.7 L.sub.A124 R.sup.E4 G.sup.7 L.sub.A125 R.sup.E5 G.sup.7 L.sub.A126 R.sup.E6 G.sup.7 L.sub.A127 R.sup.E7 G.sup.7 L.sub.A128 R.sup.E8 G.sup.7 L.sub.A129 R.sup.E9 G.sup.7 L.sub.A130 R.sup.E10 G.sup.7 L.sub.A131 R.sup.E11 G.sup.7 L.sub.A132 R.sup.E12 G.sup.7 L.sub.A133 R.sup.E13 G.sup.7 L.sub.A134 R.sup.E14 G.sup.7 L.sub.A135 R.sup.E15 G.sup.7 L.sub.A136 R.sup.E16 G.sup.7 L.sub.A137 R.sup.E17 G.sup.7 L.sub.A138 R.sup.E18 G.sup.7 L.sub.A139 R.sup.E19 G.sup.7 L.sub.A140 R.sup.E20 G.sup.7 L.sub.A141 R.sup.E1 G.sup.8

L.sub.A142 R.sup.E2 G.sup.8 L.sub.A143 R.sup.E3 G.sup.8 L.sub.A144 R.sup.E4 G.sup.8
L.sub.A145 R.sup.E5 G.sup.8 L.sub.A146 R.sup.E6 G.sup.8 L.sub.A147 R.sup.E7 G.sup.8
L.sub.A148 R.sup.E8 G.sup.8 L.sub.A149 R.sup.E9 G.sup.8 L.sub.A150 R.sup.E10 G.sup.8
L.sub.A151 R.sup.E11 G.sup.8 L.sub.A152 R.sup.E12 G.sup.8 L.sub.A153 R.sup.E13 G.sup.8
L.sub.A154 R.sup.E14 G.sup.8 L.sub.A155 R.sup.E15 G.sup.8 L.sub.A156 R.sup.E16 G.sup.8
L.sub.A157 R.sup.E17 G.sup.8 L.sub.A158 R.sup.E18 G.sup.8 L.sub.A159 R.sup.E19 G.sup.8
L.sub.A160 R.sup.E20 G.sup.8 L.sub.A161 R.sup.E1 G.sup.9 L.sub.A162 R.sup.E2 G.sup.9
L.sub.A163 R.sup.E3 G.sup.9 L.sub.A164 R.sup.E4 G.sup.9 L.sub.A165 R.sup.E5 G.sup.9
L.sub.A166 R.sup.E6 G.sup.9 L.sub.A167 R.sup.E7 G.sup.9 L.sub.A168 R.sup.E8 G.sup.9
L.sub.A169 R.sup.E9 G.sup.9 L.sub.A170 R.sup.E10 G.sup.9 L.sub.A171 R.sup.E11 G.sup.9
L.sub.A172 R.sup.E12 G.sup.9 L.sub.A173 R.sup.E13 G.sup.9 L.sub.A174 R.sup.E14 G.sup.9
L.sub.A175 R.sup.E15 G.sup.9 L.sub.A176 R.sup.E16 G.sup.9 L.sub.A177 R.sup.E17 G.sup.9
L.sub.A178 R.sup.E18 G.sup.9 L.sub.A179 R.sup.E19 G.sup.9 L.sub.A180 R.sup.E20 G.sup.9
L.sub.A181 R.sup.E1 G.sup.10 L.sub.A182 R.sup.E2 G.sup.10 L.sub.A183 R.sup.E3 G.sup.10
L.sub.A184 R.sup.E4 G.sup.10 L.sub.A185 R.sup.E5 G.sup.10 L.sub.A186 R.sup.E6 G.sup.10
L.sub.A187 R.sup.E7 G.sup.10 L.sub.A188 R.sup.E8 G.sup.10 L.sub.A189 R.sup.E9 G.sup.10
L.sub.A190 R.sup.E10 G.sup.10 L.sub.A191 R.sup.E11 G.sup.10 L.sub.A192 R.sup.E12 G.sup.10
L.sub.A193 R.sup.E13 G.sup.10 L.sub.A194 R.sup.E14 G.sup.10 L.sub.A195 R.sup.E15 G.sup.10
L.sub.A196 R.sup.E16 G.sup.10 L.sub.A197 R.sup.E17 G.sup.10 L.sub.A198 R.sup.E18 G.sup.10
L.sub.A199 R.sup.E19 G.sup.10 L.sub.A200 R.sup.E20 G.sup.10

wherein R.sup.1 to R.sup.20 have the following structures:

(69) ##STR00016## ##STR00017##

and

wherein G.sup.1 to G.sup.10 have the following structures:

(70) ##STR00018## ##STR00019##

(71) In some embodiments, the ligand L.sub.A is selected from the group consisting of:

(72) ##STR00020## ##STR00021## ##STR00022## ##STR00023## ##STR00024##

##STR00025## ##STR00026## ##STR00027## ##STR00028## ##STR00029## ##STR00030##

##STR00031## ##STR00032## ##STR00033##

(73) In some embodiments of the compound, L.sub.C is selected from the group consisting of:

L.sub.Cj-1 based on a structure of

(74) ##STR00034## and L.sub.Cj-11 based on a structure of

(75) ##STR00035## wherein j is an integer from 1 to 768; wherein for each L.sub.C; in

L.sub.Cj-I and L.sub.Cj-II, R.sup.1 and R.sup.2 are as defined below:

(76) TABLE-US-00002 L.sub.Cj R.sup.1 R.sup.2 L.sub.Cjd R.sup.1 R.sup.2 L.sub.Cj R.sup.1
R.sup.2 L.sub.Cj R.sup.1 R.sup.2 L.sub.C1 R.sup.D1 R.sup.D1 L.sub.C193 R.sup.D1 R.sup.D3
L.sub.C385 R.sup.D17 R.sup.D40 L.sub.C577 R.sup.D143 R.sup.D120 L.sub.C2 R.sup.D2
R.sup.D2 L.sub.C194 R.sup.D1 R.sup.D4 L.sub.C386 R.sup.D17 R.sup.D41 L.sub.C578
R.sup.D143 R.sup.D133 L.sub.C3 R.sup.D3 R.sup.D3 L.sub.C195 R.sup.D1 R.sup.D5 L.sub.C387
R.sup.D17 R.sup.D42 L.sub.C579 R.sup.D143 R.sup.D134 L.sub.C4 R.sup.D4 R.sup.D4
L.sub.C196 R.sup.D1 R.sup.D9 L.sub.C388 R.sup.D17 R.sup.D43 L.sub.C580 R.sup.D143
R.sup.D135 L.sub.C5 R.sup.D5 R.sup.D5 L.sub.C197 R.sup.D1 R.sup.D10 L.sub.C389 R.sup.D17
R.sup.D48 L.sub.C581 R.sup.D143 R.sup.D136 L.sub.C6 R.sup.D6 R.sup.D6 L.sub.C198
R.sup.D1 R.sup.D17 L.sub.C390 R.sup.D17 R.sup.D49 L.sub.C582 R.sup.D143 R.sup.D144
L.sub.C7 R.sup.D7 R.sup.D7 L.sub.C199 R.sup.D1 R.sup.D18 L.sub.C391 R.sup.D17 R.sup.D50
L.sub.C583 R.sup.D143 R.sup.D145 L.sub.C8 R.sup.D8 R.sup.D8 L.sub.C200 R.sup.D1
R.sup.D20 L.sub.C392 R.sup.D17 R.sup.D54 L.sub.C584 R.sup.D143 R.sup.D146 L.sub.C9
R.sup.D9 R.sup.D9 L.sub.C201 R.sup.D1 R.sup.D22 L.sub.C393 R.sup.D17 R.sup.D55
L.sub.C585 R.sup.D143 R.sup.D147 L.sub.C10 R.sup.D10 R.sup.D10 L.sub.C202 R.sup.D1
R.sup.D37 L.sub.C394 R.sup.D17 R.sup.D58 L.sub.C586 R.sup.D143 R.sup.D149 L.sub.C11

R.sup.D11 R.sup.D11 L.sub.C203 R.sup.D1 R.sup.D40 L.sub.C395 R.sup.D17 R.sup.D59
L.sub.C587 R.sup.D143 R.sup.D151 L.sub.C12 R.sup.D12 R.sup.D12 L.sub.C204 R.sup.D1
R.sup.D41 L.sub.C396 R.sup.D17 R.sup.D78 L.sub.C588 R.sup.D143 R.sup.D154 L.sub.C13
R.sup.D13 R.sup.D13 L.sub.C205 R.sup.D1 R.sup.D42 L.sub.C397 R.sup.D17 R.sup.D79
L.sub.C589 R.sup.D143 R.sup.D155 L.sub.C14 R.sup.D14 R.sup.D14 L.sub.C206 R.sup.D1
R.sup.D43 L.sub.C398 R.sup.D17 R.sup.D81 L.sub.C590 R.sup.D143 R.sup.D161 L.sub.C15
R.sup.D15 R.sup.D15 L.sub.C207 R.sup.D1 R.sup.D48 L.sub.C399 R.sup.D17 R.sup.D87
L.sub.C591 R.sup.D143 R.sup.D175 L.sub.C16 R.sup.D16 R.sup.D16 L.sub.C208 R.sup.D1
R.sup.D49 L.sub.C400 R.sup.D17 R.sup.D88 L.sub.C592 R.sup.D144 R.sup.D3 L.sub.C17
R.sup.D17 R.sup.D17 L.sub.C209 R.sup.D1 R.sup.D50 L.sub.C401 R.sup.D17 R.sup.D89
L.sub.C593 R.sup.D144 R.sup.D5 L.sub.C18 R.sup.D18 R.sup.D18 L.sub.C210 R.sup.D1
R.sup.D54 L.sub.C402 R.sup.D17 R.sup.D93 L.sub.C594 R.sup.D144 R.sup.D17 L.sub.C19
R.sup.D19 R.sup.D19 L.sub.C211 R.sup.D1 R.sup.D55 L.sub.C403 R.sup.D17 R.sup.D116
L.sub.C595 R.sup.D144 R.sup.D18 L.sub.C20 R.sup.D20 R.sup.D20 L.sub.C212 R.sup.D1
R.sup.D58 L.sub.C404 R.sup.D17 R.sup.D117 L.sub.C596 R.sup.D144 R.sup.D20 L.sub.C21
R.sup.D21 R.sup.D21 L.sub.C213 R.sup.D1 R.sup.D59 L.sub.C405 R.sup.D17 R.sup.D118
L.sub.C597 R.sup.D144 R.sup.D22 L.sub.C22 R.sup.D22 R.sup.D22 L.sub.C214 R.sup.D1
R.sup.D78 L.sub.C406 R.sup.D17 R.sup.D119 L.sub.C598 R.sup.D144 R.sup.D37 L.sub.C23
R.sup.D23 R.sup.D23 L.sub.C215 R.sup.D1 R.sup.D79 L.sub.C407 R.sup.D17 R.sup.D120
L.sub.C599 R.sup.D144 R.sup.D40 L.sub.C24 R.sup.D24 R.sup.D24 L.sub.C216 R.sup.D1
R.sup.D81 L.sub.C408 R.sup.D17 R.sup.D133 L.sub.C600 R.sup.D144 R.sup.D41 L.sub.C25
R.sup.D25 R.sup.D25 L.sub.C217 R.sup.D1 R.sup.D87 L.sub.C409 R.sup.D17 R.sup.D134
L.sub.C601 R.sup.D144 R.sup.D42 L.sub.C26 R.sup.D26 R.sup.D26 L.sub.C218 R.sup.D1
R.sup.D88 L.sub.C410 R.sup.D17 R.sup.D135 L.sub.C602 R.sup.D144 R.sup.D43 L.sub.C27
R.sup.D27 R.sup.D27 L.sub.C219 R.sup.D1 R.sup.D89 L.sub.C411 R.sup.D17 R.sup.D136
L.sub.C603 R.sup.D144 R.sup.D48 L.sub.C28 R.sup.D28 R.sup.D28 L.sub.C220 R.sup.D1
R.sup.D93 L.sub.C412 R.sup.D17 R.sup.D143 L.sub.C604 R.sup.D144 R.sup.D49 L.sub.C29
R.sup.D29 R.sup.D29 L.sub.C221 R.sup.D1 R.sup.D116 L.sub.C413 R.sup.D17 R.sup.D144
L.sub.C605 R.sup.D144 R.sup.D54 L.sub.C30 R.sup.D30 R.sup.D30 L.sub.C222 R.sup.D1
R.sup.D117 L.sub.C414 R.sup.D17 R.sup.D145 L.sub.C606 R.sup.D144 R.sup.D58 L.sub.C31
R.sup.D31 R.sup.D31 L.sub.C223 R.sup.D1 R.sup.D118 L.sub.C415 R.sup.D17 R.sup.D146
L.sub.C607 R.sup.D144 R.sup.D59 L.sub.C32 R.sup.D32 R.sup.D32 L.sub.C224 R.sup.D1
R.sup.D119 L.sub.C416 R.sup.D17 R.sup.D147 L.sub.C608 R.sup.D144 R.sup.D78 L.sub.C33
R.sup.D33 R.sup.D33 L.sub.C225 R.sup.D1 R.sup.D120 L.sub.C417 R.sup.D17 R.sup.D149
L.sub.C609 R.sup.D144 R.sup.D79 L.sub.C34 R.sup.D34 R.sup.D34 L.sub.C226 R.sup.D1
R.sup.D133 L.sub.C418 R.sup.D17 R.sup.D151 L.sub.C610 R.sup.D144 R.sup.D81 L.sub.C35
R.sup.D35 R.sup.D35 L.sub.C227 R.sup.D1 R.sup.D134 L.sub.C419 R.sup.D17 R.sup.D154
L.sub.C611 R.sup.D144 R.sup.D87 L.sub.C36 R.sup.D36 R.sup.D36 L.sub.C228 R.sup.D1
R.sup.D135 L.sub.C420 R.sup.D17 R.sup.D155 L.sub.C612 R.sup.D144 R.sup.D88 L.sub.C37
R.sup.D37 R.sup.D37 L.sub.C229 R.sup.D1 R.sup.D136 L.sub.C421 R.sup.D17 R.sup.D161
L.sub.C613 R.sup.D144 R.sup.D89 L.sub.C38 R.sup.D38 R.sup.D38 L.sub.C230 R.sup.D1
R.sup.D143 L.sub.C422 R.sup.D17 R.sup.D175 L.sub.C614 R.sup.D144 R.sup.D93 L.sub.C39
R.sup.D39 R.sup.D39 L.sub.C231 R.sup.D1 R.sup.D144 L.sub.C423 R.sup.D50 R.sup.D3
L.sub.C615 R.sup.D144 R.sup.D116 L.sub.C40 R.sup.D40 R.sup.D40 L.sub.C232 R.sup.D1
R.sup.D145 L.sub.C424 R.sup.D50 R.sup.D5 L.sub.C616 R.sup.D144 R.sup.D117 L.sub.C41
R.sup.D41 R.sup.D41 L.sub.C233 R.sup.D1 R.sup.D146 L.sub.C425 R.sup.D50 R.sup.D18
L.sub.C617 R.sup.D144 R.sup.D118 L.sub.C42 R.sup.D42 R.sup.D42 L.sub.C234 R.sup.D1
R.sup.D147 L.sub.C426 R.sup.D50 R.sup.D20 L.sub.C618 R.sup.D144 R.sup.D119 L.sub.C43
R.sup.D43 R.sup.D43 L.sub.C235 R.sup.D1 R.sup.D149 L.sub.C427 R.sup.D50 R.sup.D22
L.sub.C619 R.sup.D144 R.sup.D120 L.sub.C44 R.sup.D44 R.sup.D44 L.sub.C236 R.sup.D1

R.sup.D151 L.sub.C428 R.sup.D50 R.sup.D37 L.sub.C620 R.sup.D144 R.sup.D133 L.sub.C45
R.sup.D45 R.sup.D45 L.sub.C237 R.sup.D1 R.sup.D154 L.sub.C429 R.sup.D50 R.sup.D40
L.sub.C621 R.sup.D144 R.sup.D134 L.sub.C46 R.sup.D46 R.sup.D46 L.sub.C238 R.sup.D1
R.sup.D155 L.sub.C430 R.sup.D50 R.sup.D41 L.sub.C622 R.sup.D144 R.sup.D135 L.sub.C47
R.sup.D47 R.sup.D47 L.sub.C239 R.sup.D1 R.sup.D161 L.sub.C431 R.sup.D50 R.sup.D42
L.sub.C623 R.sup.D144 R.sup.D136 L.sub.C48 R.sup.D48 R.sup.D48 L.sub.C240 R.sup.D1
R.sup.D175 L.sub.C432 R.sup.D50 R.sup.D43 L.sub.C624 R.sup.D144 R.sup.D145 L.sub.C49
R.sup.D49 R.sup.D49 L.sub.C241 R.sup.D4 R.sup.D3 L.sub.C433 R.sup.D50 R.sup.D48
L.sub.C625 R.sup.D144 R.sup.D146 L.sub.C50 R.sup.D50 R.sup.D50 L.sub.C242 R.sup.D4
R.sup.D5 L.sub.C434 R.sup.D50 R.sup.D49 L.sub.C626 R.sup.D144 R.sup.D147 L.sub.C51
R.sup.D51 R.sup.D51 L.sub.C243 R.sup.D4 R.sup.D9 L.sub.C435 R.sup.D50 R.sup.D54
L.sub.C627 R.sup.D144 R.sup.D149 L.sub.C52 R.sup.D52 R.sup.D52 L.sub.C244 R.sup.D4
R.sup.D10 L.sub.C436 R.sup.D50 R.sup.D55 L.sub.C628 R.sup.D144 R.sup.D151 L.sub.C53
R.sup.D53 R.sup.D53 L.sub.C245 R.sup.D4 R.sup.D17 L.sub.C437 R.sup.D50 R.sup.D58
L.sub.C629 R.sup.D144 R.sup.D154 L.sub.C54 R.sup.D54 R.sup.D54 L.sub.C246 R.sup.D4
R.sup.D18 L.sub.C438 R.sup.D50 R.sup.D59 L.sub.C630 R.sup.D144 R.sup.D155 L.sub.C55
R.sup.D55 R.sup.D55 L.sub.C247 R.sup.D4 R.sup.D20 L.sub.C439 R.sup.D50 R.sup.D78
L.sub.C631 R.sup.D144 R.sup.D161 L.sub.C56 R.sup.D56 R.sup.D56 L.sub.C248 R.sup.D4
R.sup.D22 L.sub.C440 R.sup.D50 R.sup.D79 L.sub.C632 R.sup.D144 R.sup.D175 L.sub.C57
R.sup.D57 R.sup.D57 L.sub.C249 R.sup.D4 R.sup.D37 L.sub.C441 R.sup.D50 R.sup.D81
L.sub.C633 R.sup.D145 R.sup.D3 L.sub.C58 R.sup.D58 R.sup.D58 L.sub.C250 R.sup.D4
R.sup.D40 L.sub.C442 R.sup.D50 R.sup.D87 L.sub.C634 R.sup.D145 R.sup.D5 L.sub.C59
R.sup.D59 R.sup.D59 L.sub.C251 R.sup.D4 R.sup.D41 L.sub.C443 R.sup.D50 R.sup.D88
L.sub.C635 R.sup.D145 R.sup.D17 L.sub.C60 R.sup.D60 R.sup.D60 L.sub.C252 R.sup.D4
R.sup.D42 L.sub.C444 R.sup.D50 R.sup.D89 L.sub.C636 R.sup.D145 R.sup.D18 L.sub.C61
R.sup.D61 R.sup.D61 L.sub.C253 R.sup.D4 R.sup.D43 L.sub.C445 R.sup.D50 R.sup.D93
L.sub.C637 R.sup.D145 R.sup.D20 L.sub.C62 R.sup.D62 R.sup.D62 L.sub.C254 R.sup.D4
R.sup.D48 L.sub.C446 R.sup.D50 R.sup.D116 L.sub.C638 R.sup.D145 R.sup.D22 L.sub.C63
R.sup.D63 R.sup.D63 L.sub.C255 R.sup.D4 R.sup.D49 L.sub.C447 R.sup.D50 R.sup.D117
L.sub.C639 R.sup.D145 R.sup.D37 L.sub.C64 R.sup.D64 R.sup.D64 L.sub.C256 R.sup.D4
R.sup.D50 L.sub.C448 R.sup.D50 R.sup.D118 L.sub.C640 R.sup.D145 R.sup.D40 L.sub.C65
R.sup.D65 R.sup.D65 L.sub.C257 R.sup.D4 R.sup.D54 L.sub.C449 R.sup.D50 R.sup.D119
L.sub.C641 R.sup.D145 R.sup.D41 L.sub.C66 R.sup.D66 R.sup.D66 L.sub.C258 R.sup.D4
R.sup.D55 L.sub.C450 R.sup.D50 R.sup.D120 L.sub.C642 R.sup.D145 R.sup.D42 L.sub.C67
R.sup.D67 R.sup.D67 L.sub.C259 R.sup.D4 R.sup.D58 L.sub.C451 R.sup.D50 R.sup.D133
L.sub.C643 R.sup.D145 R.sup.D43 L.sub.C68 R.sup.D68 R.sup.D68 L.sub.C260 R.sup.D4
R.sup.D59 L.sub.C452 R.sup.D50 R.sup.D134 L.sub.C644 R.sup.D145 R.sup.D48 L.sub.C69
R.sup.D69 R.sup.D69 L.sub.C261 R.sup.D4 R.sup.D78 L.sub.C453 R.sup.D50 R.sup.D135
L.sub.C645 R.sup.D145 R.sup.D49 L.sub.C70 R.sup.D70 R.sup.D70 L.sub.C262 R.sup.D4
R.sup.D79 L.sub.C454 R.sup.D50 R.sup.D136 L.sub.C646 R.sup.D145 R.sup.D54 L.sub.C71
R.sup.D71 R.sup.D71 L.sub.C263 R.sup.D4 R.sup.D81 L.sub.C455 R.sup.D50 R.sup.D143
L.sub.C647 R.sup.D145 R.sup.D58 L.sub.C72 R.sup.D72 R.sup.D72 L.sub.C264 R.sup.D4
R.sup.D87 L.sub.C456 R.sup.D50 R.sup.D144 L.sub.C648 R.sup.D145 R.sup.D59 L.sub.C73
R.sup.D73 R.sup.D73 L.sub.C265 R.sup.D4 R.sup.D88 L.sub.C457 R.sup.D50 R.sup.D145
L.sub.C649 R.sup.D145 R.sup.D78 L.sub.C74 R.sup.D74 R.sup.D74 L.sub.C266 R.sup.D4
R.sup.D89 L.sub.C458 R.sup.D50 R.sup.D146 L.sub.C650 R.sup.D145 R.sup.D79 L.sub.C75
R.sup.D75 R.sup.D75 L.sub.C267 R.sup.D4 R.sup.D93 L.sub.C459 R.sup.D50 R.sup.D147
L.sub.C651 R.sup.D145 R.sup.D81 L.sub.C76 R.sup.D76 R.sup.D76 L.sub.C268 R.sup.D4
R.sup.D116 L.sub.C460 R.sup.D50 R.sup.D149 L.sub.C652 R.sup.D145 R.sup.D87 L.sub.C77
R.sup.D77 R.sup.D77 L.sub.C269 R.sup.D4 R.sup.D117 L.sub.C461 R.sup.D50 R.sup.D151

L.sub.C653 R.sup.D145 R.sup.D78 L.sub.C270 R.sup.D4
R.sup.D118 L.sub.C462 R.sup.D50 R.sup.D154 L.sub.C654 R.sup.D145 R.sup.D89 L.sub.C79
R.sup.D79 R.sup.D79 L.sub.C271 R.sup.D4 R.sup.D119 L.sub.C463 R.sup.D50 R.sup.D155
L.sub.C655 R.sup.D145 R.sup.D93 L.sub.C80 R.sup.D80 R.sup.D80 L.sub.C272 R.sup.D4
R.sup.D120 L.sub.C464 R.sup.D50 R.sup.D161 L.sub.C656 R.sup.D145 R.sup.D116 L.sub.C81
R.sup.D81 R.sup.D81 L.sub.C273 R.sup.D4 R.sup.D133 L.sub.C465 R.sup.D50 R.sup.D175
L.sub.C657 R.sup.D145 R.sup.D117 L.sub.C82 R.sup.D82 R.sup.D82 L.sub.C274 R.sup.D4
R.sup.D134 L.sub.C466 R.sup.D55 R.sup.D3 L.sub.C658 R.sup.D145 R.sup.D118 L.sub.C83
R.sup.D83 R.sup.D83 L.sub.C275 R.sup.D4 R.sup.D135 L.sub.C467 R.sup.D55 R.sup.D5
L.sub.C659 R.sup.D145 R.sup.D119 L.sub.C84 R.sup.D84 R.sup.D84 L.sub.C276 R.sup.D4
R.sup.D136 L.sub.C468 R.sup.D55 R.sup.D18 L.sub.C660 R.sup.D145 R.sup.D120 L.sub.C85
R.sup.D85 R.sup.D85 L.sub.C277 R.sup.D4 R.sup.D143 L.sub.C469 R.sup.D55 R.sup.D20
L.sub.C661 R.sup.D145 R.sup.D133 L.sub.C86 R.sup.D86 R.sup.D86 L.sub.C278 R.sup.D4
R.sup.D144 L.sub.C470 R.sup.D55 R.sup.D22 L.sub.C662 R.sup.D145 R.sup.D134 L.sub.C87
R.sup.D87 R.sup.D87 L.sub.C279 R.sup.D4 R.sup.D145 L.sub.C471 R.sup.D55 R.sup.D37
L.sub.C663 R.sup.D145 R.sup.D135 L.sub.C88 R.sup.D88 R.sup.D88 L.sub.C280 R.sup.D4
R.sup.D146 L.sub.C472 R.sup.D55 R.sup.D40 L.sub.C664 R.sup.D145 R.sup.D136 L.sub.C89
R.sup.D89 R.sup.D89 L.sub.C281 R.sup.D4 R.sup.D147 L.sub.C473 R.sup.D55 R.sup.D41
L.sub.C665 R.sup.D145 R.sup.D146 L.sub.C90 R.sup.D90 R.sup.D90 L.sub.C282 R.sup.D4
R.sup.D149 L.sub.C474 R.sup.D55 R.sup.D42 L.sub.C666 R.sup.D145 R.sup.D147 L.sub.C91
R.sup.D91 R.sup.D91 L.sub.C283 R.sup.D4 R.sup.D151 L.sub.C475 R.sup.D55 R.sup.D43
L.sub.C667 R.sup.D145 R.sup.D149 L.sub.C92 R.sup.D92 R.sup.D92 L.sub.C284 R.sup.D4
R.sup.D154 L.sub.C476 R.sup.D55 R.sup.D48 L.sub.C668 R.sup.D145 R.sup.D151 L.sub.C93
R.sup.D93 R.sup.D93 L.sub.C285 R.sup.D4 R.sup.D155 L.sub.C477 R.sup.D55 R.sup.D49
L.sub.C669 R.sup.D145 R.sup.D154 L.sub.C94 R.sup.D94 R.sup.D94 L.sub.C286 R.sup.D4
R.sup.D161 L.sub.C478 R.sup.D55 R.sup.D54 L.sub.C670 R.sup.D145 R.sup.D155 L.sub.C95
R.sup.D95 R.sup.D95 L.sub.C287 R.sup.D4 R.sup.D175 L.sub.C479 R.sup.D55 R.sup.D58
L.sub.C671 R.sup.D145 R.sup.D161 L.sub.C96 R.sup.D96 R.sup.D96 L.sub.C288 R.sup.D9
R.sup.D3 L.sub.C480 R.sup.D55 R.sup.D59 L.sub.C672 R.sup.D145 R.sup.D175 L.sub.C97
R.sup.D97 R.sup.D97 L.sub.C289 R.sup.D9 R.sup.D5 L.sub.C481 R.sup.D55 R.sup.D78
L.sub.C673 R.sup.D146 R.sup.D3 L.sub.C98 R.sup.D98 R.sup.D98 L.sub.C290 R.sup.D9
R.sup.D10 L.sub.C482 R.sup.D55 R.sup.D79 L.sub.C674 R.sup.D146 R.sup.D5 L.sub.C99
R.sup.D99 R.sup.D99 L.sub.C291 R.sup.D9 R.sup.D17 L.sub.C483 R.sup.D55 R.sup.D81
L.sub.C675 R.sup.D146 R.sup.D17 L.sub.C100 R.sup.D100 R.sup.D100 L.sub.C292 R.sup.D9
R.sup.D18 L.sub.C484 R.sup.D55 R.sup.D87 L.sub.C676 R.sup.D146 R.sup.D18 L.sub.C101
R.sup.D101 R.sup.D101 L.sub.C293 R.sup.D9 R.sup.D20 L.sub.C485 R.sup.D55 R.sup.D88
L.sub.C677 R.sup.D146 R.sup.D20 L.sub.C102 R.sup.D102 R.sup.D102 L.sub.C294 R.sup.D9
R.sup.D22 L.sub.C486 R.sup.D55 R.sup.D89 L.sub.C678 R.sup.D146 R.sup.D22 L.sub.C103
R.sup.D103 R.sup.D103 L.sub.C295 R.sup.D9 R.sup.D37 L.sub.C487 R.sup.D55 R.sup.D93
L.sub.C679 R.sup.D146 R.sup.D37 L.sub.C104 R.sup.D104 R.sup.D104 L.sub.C296 R.sup.D9
R.sup.D40 L.sub.C488 R.sup.D55 R.sup.D116 L.sub.C680 R.sup.D146 R.sup.D40 L.sub.C105
R.sup.D105 R.sup.D105 L.sub.C297 R.sup.D9 R.sup.D41 L.sub.C489 R.sup.D55 R.sup.D117
L.sub.C681 R.sup.D146 R.sup.D41 L.sub.C106 R.sup.D106 R.sup.D106 L.sub.C298 R.sup.D9
R.sup.D42 L.sub.C490 R.sup.D55 R.sup.D118 L.sub.C682 R.sup.D146 R.sup.D42 L.sub.C107
R.sup.D107 R.sup.D107 L.sub.C299 R.sup.D9 R.sup.D43 L.sub.C491 R.sup.D55 R.sup.D119
L.sub.C683 R.sup.D146 R.sup.D43 L.sub.C108 R.sup.D108 R.sup.D108 L.sub.C300 R.sup.D9
R.sup.D48 L.sub.C492 R.sup.D55 R.sup.D120 L.sub.C684 R.sup.D146 R.sup.D48 L.sub.C109
R.sup.D109 R.sup.D109 L.sub.C301 R.sup.D9 R.sup.D49 L.sub.C493 R.sup.D55 R.sup.D133
L.sub.C685 R.sup.D146 R.sup.D49 L.sub.C110 R.sup.D110 R.sup.D110 L.sub.C302 R.sup.D9
R.sup.D50 L.sub.C494 R.sup.D55 R.sup.D134 L.sub.C686 R.sup.D146 R.sup.D54 L.sub.C111

R.sup.D111 R.sup.D111 L.sub.C303 R.sup.D9 R.sup.D54 L.sub.C495 R.sup.D55 R.sup.D135
L.sub.C687 R.sup.D146 R.sup.D58 L.sub.C112 R.sup.D112 R.sup.D112 L.sub.C304 R.sup.D9
R.sup.D55 L.sub.C496 R.sup.D55 R.sup.D136 L.sub.C688 R.sup.D146 R.sup.D59 L.sub.C113
R.sup.D113 R.sup.D113 L.sub.C305 R.sup.D9 R.sup.D58 L.sub.C497 R.sup.D55 R.sup.D143
L.sub.C689 R.sup.D146 R.sup.D78 L.sub.C114 R.sup.D114 R.sup.D114 L.sub.C306 R.sup.D9
R.sup.D59 L.sub.C498 R.sup.D55 R.sup.D144 L.sub.C690 R.sup.D146 R.sup.D79 L.sub.C115
R.sup.D115 R.sup.D115 L.sub.C307 R.sup.D9 R.sup.D78 L.sub.C499 R.sup.D55 R.sup.D145
L.sub.C691 R.sup.D146 R.sup.D81 L.sub.C116 R.sup.D116 R.sup.D116 L.sub.C308 R.sup.D9
R.sup.D79 L.sub.C500 R.sup.D55 R.sup.D146 L.sub.C692 R.sup.D146 R.sup.D87 L.sub.C117
R.sup.D117 R.sup.D177 L.sub.C309 R.sup.D9 R.sup.D81 L.sub.C501 R.sup.D55 R.sup.D147
L.sub.C693 R.sup.D146 R.sup.D88 L.sub.C118 R.sup.D118 R.sup.D118 L.sub.C310 R.sup.D9
R.sup.D87 L.sub.C502 R.sup.D55 R.sup.D149 L.sub.C694 R.sup.D146 R.sup.D89 L.sub.C119
R.sup.D119 R.sup.D119 L.sub.C311 R.sup.D9 R.sup.D88 L.sub.C503 R.sup.D55 R.sup.D151
L.sub.C695 R.sup.D146 R.sup.D93 L.sub.C120 R.sup.D120 R.sup.D120 L.sub.C312 R.sup.D9
R.sup.D89 L.sub.C504 R.sup.D55 R.sup.D154 L.sub.C696 R.sup.D146 R.sup.D117 L.sub.C121
R.sup.D121 R.sup.D121 L.sub.C313 R.sup.D9 R.sup.D93 L.sub.C505 R.sup.D55 R.sup.D155
L.sub.C697 R.sup.D146 R.sup.D118 L.sub.C122 R.sup.D122 R.sup.D122 L.sub.C314 R.sup.D9
R.sup.D116 L.sub.C506 R.sup.D55 R.sup.D161 L.sub.C698 R.sup.D146 R.sup.D119 L.sub.C123
R.sup.D123 R.sup.D123 L.sub.C315 R.sup.D9 R.sup.D117 L.sub.C507 R.sup.D55 R.sup.D175
L.sub.C699 R.sup.D146 R.sup.D120 L.sub.C124 R.sup.D124 R.sup.D124 L.sub.C316 R.sup.D9
R.sup.D118 L.sub.C508 R.sup.D116 R.sup.D3 L.sub.C700 R.sup.D146 R.sup.D133 L.sub.C125
R.sup.D125 R.sup.D125 L.sub.C317 R.sup.D9 R.sup.D119 L.sub.C509 R.sup.D116 R.sup.D5
L.sub.C701 R.sup.D146 R.sup.D134 L.sub.C126 R.sup.D126 R.sup.D126 L.sub.C318 R.sup.D9
R.sup.D120 L.sub.C510 R.sup.D116 R.sup.D17 L.sub.C702 R.sup.D146 R.sup.D135 L.sub.C127
R.sup.D127 R.sup.D127 L.sub.C319 R.sup.D9 R.sup.D133 L.sub.C511 R.sup.D116 R.sup.D18
L.sub.C703 R.sup.D146 R.sup.D136 L.sub.C128 R.sup.D128 R.sup.D128 L.sub.C320 R.sup.D9
R.sup.D134 L.sub.C512 R.sup.D116 R.sup.D20 L.sub.C704 R.sup.D146 R.sup.D146 L.sub.C129
R.sup.D129 R.sup.D129 L.sub.C321 R.sup.D9 R.sup.D135 L.sub.C513 R.sup.D116 R.sup.D22
L.sub.C705 R.sup.D146 R.sup.D147 L.sub.C130 R.sup.D130 R.sup.D130 L.sub.C322 R.sup.D9
R.sup.D136 L.sub.C514 R.sup.D116 R.sup.D37 L.sub.C706 R.sup.D146 R.sup.D149 L.sub.C131
R.sup.D131 R.sup.D131 L.sub.C323 R.sup.D9 R.sup.D143 L.sub.C515 R.sup.D116 R.sup.D40
L.sub.C707 R.sup.D146 R.sup.D151 L.sub.C132 R.sup.D132 R.sup.D132 L.sub.C324 R.sup.D9
R.sup.D144 L.sub.C516 R.sup.D116 R.sup.D41 L.sub.C708 R.sup.D146 R.sup.D154 L.sub.C133
R.sup.D133 R.sup.D133 L.sub.C325 R.sup.D9 R.sup.D145 L.sub.C517 R.sup.D116 R.sup.D42
L.sub.C709 R.sup.D146 R.sup.D155 L.sub.C134 R.sup.D134 R.sup.D134 L.sub.C326 R.sup.D9
R.sup.D146 L.sub.C518 R.sup.D116 R.sup.D43 L.sub.C710 R.sup.D146 R.sup.D161 L.sub.C135
R.sup.D135 R.sup.D135 L.sub.C327 R.sup.D9 R.sup.D147 L.sub.C519 R.sup.D116 R.sup.D48
L.sub.C711 R.sup.D146 R.sup.D175 L.sub.C136 R.sup.D136 R.sup.D136 L.sub.C328 R.sup.D9
R.sup.D149 L.sub.C520 R.sup.D116 R.sup.D49 L.sub.C712 R.sup.D133 R.sup.D3 L.sub.C137
R.sup.D137 R.sup.D137 L.sub.C329 R.sup.D9 R.sup.D151 L.sub.C521 R.sup.D116 R.sup.D54
L.sub.C713 R.sup.D133 R.sup.D5 L.sub.C138 R.sup.D138 R.sup.D138 L.sub.C330 R.sup.D9
R.sup.D154 L.sub.C522 R.sup.D116 R.sup.D58 L.sub.C714 R.sup.D133 R.sup.D3 L.sub.C139
R.sup.D139 R.sup.D139 L.sub.C331 R.sup.D9 R.sup.D155 L.sub.C523 R.sup.D116 R .sup.D59
L.sub.C715 R.sup.D133 R.sup.D18 L.sub.C140 R.sup.D140 R.sup.D140 L.sub.C332 R.sup.D9
R.sup.D161 L.sub.C524 R.sup.D116 R.sup.D78 L.sub.C716 R.sup.D133 R.sup.D20 L.sub.C141
R.sup.D141 R.sup.D141 L.sub.C333 R.sup.D9 R.sup.D175 L.sub.C525 R.sup.D116 R.sup.D79
L.sub.C717 R.sup.D133 R.sup.D22 L.sub.C142 R.sup.D142 R.sup.D142 L.sub.C334 R.sup.D10
R.sup.D3 L.sub.C526 R.sup.D116 R.sup.D81 L.sub.C718 R.sup.D133 R.sup.D37 L.sub.C143
R.sup.D143 R.sup.D143 L.sub.C335 R.sup.D10 R.sup.D5 L.sub.C527 R.sup.D116 R.sup.D87
L.sub.C719 R.sup.D133 R.sup.D40 L.sub.C144 R.sup.D144 R.sup.D144 L.sub.C336 R.sup.D10

R.sup.D17 L.sub.C528 R.sup.D116 R.sup.D88 L.sub.C720 R.sup.D133 R.sup.D133 R.sub.C145
R.sup.D145 R.sup.D145 L.sub.C337 R.sup.D10 R.sup.D18 L.sub.C529 R.sup.D116 R.sup.D89
L.sub.C721 R.sup.D133 R.sup.D42 L.sub.C146 R.sup.D146 R.sup.D146 L.sub.C338 R.sup.D10
R.sup.D20 L.sub.C530 R.sup.D116 R.sup.D93 L.sub.C722 R.sup.D133 R.sup.D43 L.sub.C147
R.sup.D147 R.sup.D147 L.sub.C339 R.sup.D10 R.sup.D22 L.sub.C531 R.sup.D116 R.sup.D117
L.sub.C723 R.sup.D133 R.sup.D48 L.sub.C148 R.sup.D148 R.sup.D148 L.sub.C340 R.sup.D10
R.sup.D37 L.sub.C532 R.sup.D116 R.sup.D118 L.sub.C724 R.sup.D133 R.sup.D49 L.sub.C149
R.sup.D149 R.sup.D149 L.sub.C341 R.sup.D10 R.sup.D40 L.sub.C533 R.sup.D116 R.sup.D119
L.sub.C725 R.sup.D133 R.sup.D54 L.sub.C150 R.sup.D150 R.sup.D150 L.sub.C342 R.sup.D10
R.sup.D41 L.sub.C534 R.sup.D116 R.sup.D120 L.sub.C726 R.sup.D133 R.sup.D58 L.sub.C151
R.sup.D151 R.sup.D151 L.sub.C343 R.sup.D10 R.sup.D42 L.sub.C535 R.sup.D116 R.sup.D133
L.sub.C727 R.sup.D133 R.sup.D59 L.sub.C152 R.sup.D152 R.sup.D152 L.sub.C344 R.sup.D10
R.sup.D43 L.sub.C536 R.sup.D116 R.sup.D134 L.sub.C728 R.sup.D133 R.sup.D78 L.sub.C153
R.sup.D153 R.sup.D153 L.sub.C345 R.sup.D10 R.sup.D48 L.sub.C537 R.sup.D116 R.sup.D135
L.sub.C729 R.sup.D133 R.sup.D79 L.sub.C154 R.sup.D154 R.sup.D154 L.sub.C346 R.sup.D10
R.sup.D49 L.sub.C538 R.sup.D116 R.sup.D136 L.sub.C730 R.sup.D133 R.sup.D81 L.sub.C155
R.sup.D155 R.sup.D155 L.sub.C347 R.sup.D10 R.sup.D50 L.sub.C539 R.sup.D116 R.sup.D143
L.sub.C731 R.sup.D133 R.sup.D87 L.sub.C156 R.sup.D156 R.sup.D156 L.sub.C348 R.sup.D10
R.sup.D54 L.sub.C540 R.sup.D116 R.sup.D144 L.sub.C732 R.sup.D133 R.sup.D88 L.sub.C157
R.sup.D157 R.sup.D157 L.sub.C349 R.sup.D10 R.sup.D55 L.sub.C541 R.sup.D116 R.sup.D145
L.sub.C733 R.sup.D133 R.sup.D89 L.sub.C158 R.sup.D158 R.sup.D158 L.sub.C350 R.sup.D10
R.sup.D58 L.sub.C542 R.sup.D116 R.sup.D146 L.sub.C734 R.sup.D133 R.sup.D93 L.sub.C159
R.sup.D159 R.sup.D159 L.sub.C351 R.sup.D10 R.sup.D59 L.sub.C543 R.sup.D116 R.sup.D147
L.sub.C735 R.sup.D133 R.sup.D117 L.sub.C160 R.sup.D160 R.sup.D160 L.sub.C352 R.sup.D10
R.sup.D78 L.sub.C544 R.sup.D116 R.sup.D149 L.sub.C736 R.sup.D133 R.sup.D118 L.sub.C161
R.sup.D161 R.sup.D161 L.sub.C353 R.sup.D10 R.sup.D79 L.sub.C545 R.sup.D116 R.sup.D151
L.sub.C737 R.sup.D133 R.sup.D119 L.sub.C162 R.sup.D162 R.sup.D162 L.sub.C354 R.sup.D10
R.sup.D81 L.sub.C546 R.sup.D116 R.sup.D154 L.sub.C738 R.sup.D133 R.sup.D120 L.sub.C163
R.sup.D163 R.sup.D163 L.sub.C355 R.sup.D10 R.sup.D87 L.sub.C547 R.sup.D116 R.sup.D155
L.sub.C739 R.sup.D133 R.sup.D133 L.sub.C164 R.sup.D164 R.sup.D164 L.sub.C356 R.sup.D10
R.sup.D88 L.sub.C548 R.sup.D116 R.sup.D161 L.sub.C740 R.sup.D133 R.sup.D134 L.sub.C165
R.sup.D165 R.sup.D165 L.sub.C357 R.sup.D10 R.sup.D89 L.sub.C549 R.sup.D116 R.sup.D175
L.sub.C741 R.sup.D133 R.sup.D135 L.sub.C166 R.sup.D166 R.sup.D166 L.sub.C358 R.sup.D10
R.sup.D93 L.sub.C550 R.sup.D143 R.sup.D3 L.sub.C742 R.sup.D133 R.sup.D136 L.sub.C167
R.sup.D167 R.sup.D167 L.sub.C359 R.sup.D10 R.sup.D116 L.sub.C551 R.sup.D143 R.sup.D5
L.sub.C743 R.sup.D133 R.sup.D146 L.sub.C168 R.sup.D168 R.sup.D168 L.sub.C360 R.sup.D10
R.sup.D117 L.sub.C552 R.sup.D143 R.sup.D17 L.sub.C744 R.sup.D133 R.sup.D147 L.sub.C169
R.sup.D169 R.sup.D169 L.sub.C361 R.sup.D10 R.sup.D118 L.sub.C553 R.sup.D143 R.sup.D18
L.sub.C745 R.sup.D133 R.sup.D149 L.sub.C170 R.sup.D170 R.sup.D170 L.sub.C362 R.sup.D10
R.sup.D119 L.sub.C554 R.sup.D143 R.sup.D20 L.sub.C746 R.sup.D133 R.sup.D151 L.sub.C171
R.sup.D171 R.sup.D171 L.sub.C363 R.sup.D10 R.sup.D120 L.sub.C555 R.sup.D143 R.sup.D22
L.sub.C747 R.sup.D133 R.sup.D154 L.sub.C172 R.sup.D172 R.sup.D172 L.sub.C364 R.sup.D10
R.sup.D133 L.sub.C556 R.sup.D143 R.sup.D37 L.sub.C748 R.sup.D133 R.sup.D155 L.sub.C173
R.sup.D173 R.sup.D173 L.sub.C365 R.sup.D10 R.sup.D134 L.sub.C557 R.sup.D143 R.sup.D40
L.sub.C749 R.sup.D133 R.sup.D161 L.sub.C174 R.sup.D174 R.sup.D174 L.sub.C366 R.sup.D10
R.sup.D135 L.sub.C558 R.sup.D143 R.sup.D41 L.sub.C750 R.sup.D133 R.sup.D175 L.sub.C175
R.sup.D175 R.sup.D175 L.sub.C367 R.sup.D10 R.sup.D136 L.sub.C559 R.sup.D143 R.sup.D42
L.sub.C751 R.sup.D175 R.sup.D3 L.sub.C176 R.sup.D176 R.sup.D176 L.sub.C368 R.sup.D10
R.sup.D143 L.sub.C560 R.sup.D143 R.sup.D43 L.sub.C752 R.sup.D175 R.sup.D5 L.sub.C177
R.sup.D177 R.sup.D177 L.sub.C369 R.sup.D10 R.sup.D144 L.sub.C561 R.sup.D143 R.sup.D48

(83) In some embodiments the compound is selected from the group consisting of:

(84) ##STR00059## ##STR00060## ##STR00061## ##STR00062## ##STR00063##
##STR00064## ##STR00065## ##STR00066## ##STR00067## ##STR00068##

C. The OLEDs and the Devices of the Present Disclosure

(85) In another aspect, the present disclosure also provides an OLED device comprising a first organic layer that contains a compound as disclosed in the above compounds section of the present disclosure.

(86) In some embodiments, the OLED comprises an anode, a cathode, and a first organic layer disposed between the anode and the cathode. The first organic layer can comprise a organic light emitting device (OLED) comprising: an anode; a cathode; and an organic layer disposed between the anode and the cathode, wherein the organic layer comprises a compound of Formula

$\text{Ir}(\text{L.sub.A})_{\text{sub.x}}(\text{L.sub.C})_{\text{sub.y}}$, wherein: ligand L.sub.A has Formula I'

(87) ##STR00069## and ligand L.sub.C has Formula II'

(88) ##STR00070## wherein: x is 1, 2, or 3; y is 0, 1, or 2; $x+y=3$; $\text{X}_{\text{sup.1}}-\text{X}_{\text{sup.8}}$ are each independently C, N, or CR, and at least four of $\text{X}_{\text{sup.1}}-\text{X}_{\text{sup.8}}$ are N; the $\text{X}_{\text{sup.1}}-\text{X}_{\text{sup.4}}$ that connects to ring A is C, and the $\text{X}_{\text{sup.1}}-\text{X}_{\text{sup.4}}$ that coordinates to Ir is N; the maximum number of N atoms that are bonded to one another within a ring is two; ring A is an unsaturated 5-membered or 6-membered carbocyclic or heterocyclic ring; R.sub.A represents zero, mono, or up to the maximum number of allowed substitutions to its associated ring; each R, R.sub.1-R.sub.3, and R.sub.A is independently a hydrogen or a substituent selected from the group consisting of the general substituents defined herein; and two substituents can be joined or fused together to form a ring, wherein the ligand L.sub.A coordinates to Ir to form a 5-membered chelate ring as indicated by the two dashed lines of the Formula I', with the proviso that the compound of Formula $\text{Ir}(\text{L.sub.A})_{\text{sub.x}}(\text{L.sub.C})_{\text{sub.y}}$ is not

(89) ##STR00071##

(90) In some embodiments, the organic layer may be an emissive layer and the compound as described herein may be an emissive dopant or a non-emissive dopant.

(91) In some embodiments, the organic layer may further comprise a host, wherein the host comprises a triphenylene containing benzo-fused thiophene or benzo-fused furan, wherein any substituent in the host is an unfused substituent independently selected from the group consisting of $\text{C}_{\text{sub.nH.sub.2n+1}}$, $\text{OC}_{\text{sub.nH.sub.2+1}}$, $\text{OAr}_{\text{sub.1}}$, $\text{N}(\text{C}_{\text{sub.nH.sub.2+1}})_{\text{sub.2}}$, $\text{N}(\text{Ar}_{\text{sub.1}})_{\text{sub.2}}$, $\text{CH}=\text{CH}-\text{C}_{\text{sub.nH.sub.2+1}}$, $\text{C}\equiv\text{CC}_{\text{sub.nH.sub.2+1}}$, $\text{Ar}_{\text{sub.1}}$, $\text{Ar}_{\text{sub.1}}-\text{Ar}_{\text{sub.2}}$, $\text{C}_{\text{sub.nH.sub.2}}-\text{Ar}_{\text{sub.1}}$, or no substitution, wherein n is from 1 to 10; and wherein Ar.sub.1 and Ar.sub.2 are independently selected from the group consisting of benzene, biphenyl, naphthalene, triphenylene, carbazole, and heteroaromatic analogs thereof.

(92) In some embodiments, the organic layer may further comprise a host, wherein host comprises at least one chemical group selected from the group consisting of triphenylene, carbazole, indolocarbazole, dibenzothiophene, dibenzofuran, dibenzoselenophene, 5,9-dioxa-13b-boranaphtho[3,2,1-de]anthracene, aza-triphenylene, aza-carbazole, aza-indolocarbazole, aza-dibenzothiophene, aza-dibenzofuran, aza-dibenzoselenophene, and aza-(5,9-dioxa-13b-boranaphtho[3,2,1-de]anthracene).

(93) In some embodiments, the host may be selected from the HOST Group consisting of:

(94) ##STR00072## ##STR00073## ##STR00074## ##STR00075## ##STR00076##
##STR00077##

and combinations thereof.

(95) In some embodiments, the organic layer may further comprise a host, wherein the host comprises a metal complex.

(96) In some embodiments, the compound as described herein may be a sensitizer; wherein the device may further comprise an acceptor; and wherein the acceptor may be selected from the group consisting of fluorescent emitter, delayed fluorescence emitter, and combination thereof.

(97) In yet another aspect, the OLED of the present disclosure may also comprise an emissive region containing a compound as disclosed in the above compounds section of the present disclosure.

(98) In some embodiments, the emissive region can comprise a compound of Formula $\text{Ir}(\text{L.sub.A})_{\text{sub.x}}(\text{L.sub.C})_{\text{sub.y}}$, wherein: ligand L.sub.A has Formula I'

(99) ##STR00078## and ligand L.sub.C has Formula II'

(100) ##STR00079## wherein: x is 1, 2, or 3; y is 0, 1, or 2; $x+y=3$; X¹-X⁸ are each independently C, N, or CR, and at least four of X¹-X⁸ are N; the X¹-X⁴ that connects to ring A is C, and the X¹-X⁴ that coordinates to Ir is N; the maximum number of N atoms that are bonded to one another within a ring is two; ring A is an unsaturated 5-membered or 6-membered carbocyclic or heterocyclic ring; R^A represents zero, mono, or up to the maximum number of allowed substitutions to its associated ring; each R, R¹-R³, and R^A is independently a hydrogen or a substituent selected from the group consisting of the general substituents defined herein; and two substituents can be joined or fused together to form a ring, wherein the ligand L.sub.A coordinates to Ir to form a 5-membered chelate ring as indicated by the two dashed lines of the Formula I', with the proviso that the compound of Formula $\text{Ir}(\text{L.sub.A})_{\text{sub.x}}(\text{L.sub.C})_{\text{sub.y}}$ is not

(101) ##STR00080##

(102) In some embodiments of the emissive region, the compound can be an emissive dopant or a non-emissive dopant.

(103) In some embodiments, the emissive region further comprises a host, wherein the host contains at least one group selected from the group consisting of metal complex, triphenylene, carbazole, dibenzothiophene, dibenzofuran, dibenzoselenophene, aza-triphenylene, aza-carbazole, aza-dibenzothiophene, aza-dibenzofuran, and aza-dibenzoselenophene. In some embodiments, the emissive region further comprises a host, wherein the host is selected from the Host Group defined above

(104) In yet another aspect, the present disclosure also provides a consumer product comprising an organic light-emitting device (OLED) having an anode; a cathode; and an organic layer disposed between the anode and the cathode, wherein the organic layer may comprise a compound as disclosed in the above compounds section of the present disclosure.

(105) In some embodiments, the consumer product comprises an OLED having an anode; a cathode; and an organic layer disposed between the anode and the cathode, wherein the organic layer can comprise a compound of Formula $\text{Ir}(\text{L.sub.A})_{\text{sub.x}}(\text{L.sub.C})_{\text{sub.y}}$, wherein: ligand L.sub.A has Formula I'

(106) ##STR00081## and ligand L.sub.C has Formula II'

(107) ##STR00082## wherein: x is 1, 2, or 3; y is 0, 1, or 2; $x+y=3$; X¹-X⁸ are each independently C, N, or CR, and at least four of X¹-X⁸ are N; the X¹-X⁴ that connects to ring A is C, and the X¹-X⁴ that coordinates to Ir is N; the maximum number of N atoms that are bonded to one another within a ring is two; ring A is an unsaturated 5-membered or 6-membered carbocyclic or heterocyclic ring; R^A represents zero, mono, or up to the maximum number of allowed substitutions to its associated ring; each R, R¹-R³, and R^A is independently a hydrogen or a substituent selected from the group consisting of the general substituents defined herein; and two substituents can be joined or fused together to form a ring, wherein the ligand L.sub.A coordinates to Ir to form a 5-membered chelate ring as indicated by the two dashed lines of the Formula I', with the proviso that the compound of Formula $\text{Ir}(\text{L.sub.A})_{\text{sub.x}}(\text{L.sub.C})_{\text{sub.y}}$ is not

(108) ##STR00083##

(109) In some embodiments, the consumer product can be one of a flat panel display, a computer monitor, a medical monitor, a television, a billboard, a light for interior or exterior illumination and/or signaling, a heads-up display, a fully or partially transparent display, a flexible display, a

laser printer, a telephone, a cell phone, tablet, a phablet, a personal digital assistant (PDA), a wearable device, a laptop computer, a digital camera, a camcorder, a viewfinder, a micro-display that is less than 2 inches diagonal, a 3-D display, a virtual reality or augmented reality display, a vehicle, a video wall comprising multiple displays tiled together, a theater or stadium screen, a light therapy device, and a sign.

(110) Generally, an OLED comprises at least one organic layer disposed between and electrically connected to an anode and a cathode. When a current is applied, the anode injects holes and the cathode injects electrons into the organic layer(s). The injected holes and electrons each migrate toward the oppositely charged electrode. When an electron and hole localize on the same molecule, an “exciton,” which is a localized electron-hole pair having an excited energy state, is formed. Light is emitted when the exciton relaxes via a photoemissive mechanism. In some cases, the exciton may be localized on an excimer or an exciplex. Non-radiative mechanisms, such as thermal relaxation, may also occur, but are generally considered undesirable.

(111) Several OLED materials and configurations are described in U.S. Pat. Nos. 5,844,363, 6,303,238, and 5,707,745, which are incorporated herein by reference in their entirety.

(112) The initial OLEDs used emissive molecules that emitted light from their singlet states (“fluorescence”) as disclosed, for example, in U.S. Pat. No. 4,769,292, which is incorporated by reference in its entirety. Fluorescent emission generally occurs in a time frame of less than 10 nanoseconds.

(113) More recently, OLEDs having emissive materials that emit light from triplet states (“phosphorescence”) have been demonstrated. Baldo et al., “Highly Efficient Phosphorescent Emission from Organic Electroluminescent Devices,” *Nature*, vol. 395, 151-154, 1998; (“Baldo-I”) and Baldo et al., “Very high-efficiency green organic light-emitting devices based on electrophosphorescence,” *Appl. Phys. Lett.*, vol. 75, No. 3, 4-6 (1999) (“Baldo-II”), are incorporated by reference in their entireties. Phosphorescence is described in more detail in U.S. Pat. No. 7,279,704 at cols. 5-6, which are incorporated by reference.

(114) FIG. 1 shows an organic light emitting device 100. The figures are not necessarily drawn to scale. Device 100 may include a substrate 110, an anode 115, a hole injection layer 120, a hole transport layer 125, an electron blocking layer 130, an emissive layer 135, a hole blocking layer 140, an electron transport layer 145, an electron injection layer 150, a protective layer 155, a cathode 160, and a barrier layer 170. Cathode 160 is a compound cathode having a first conductive layer 162 and a second conductive layer 164. Device 100 may be fabricated by depositing the layers described, in order. The properties and functions of these various layers, as well as example materials, are described in more detail in U.S. Pat. No. 7,279,704 at cols. 6-10, which are incorporated by reference.

(115) More examples for each of these layers are available. For example, a flexible and transparent substrate-anode combination is disclosed in U.S. Pat. No. 5,844,363, which is incorporated by reference in its entirety. An example of a p-doped hole transport layer is m-MTDATA doped with F.sub.4-TCNQ at a molar ratio of 50:1, as disclosed in U.S. Patent Application Publication No. 2003/0230980, which is incorporated by reference in its entirety. Examples of emissive and host materials are disclosed in U.S. Pat. No. 6,303,238 to Thompson et al., which is incorporated by reference in its entirety. An example of an n-doped electron transport layer is BPhen doped with Li at a molar ratio of 1:1, as disclosed in U.S. Patent Application Publication No. 2003/0230980, which is incorporated by reference in its entirety. U.S. Pat. Nos. 5,703,436 and 5,707,745, which are incorporated by reference in their entireties, disclose examples of cathodes including compound cathodes having a thin layer of metal such as Mg:Ag with an overlying transparent, electrically-conductive, sputter-deposited ITO layer. The theory and use of blocking layers is described in more detail in U.S. Pat. No. 6,097,147 and U.S. Patent Application Publication No. 2003/0230980, which are incorporated by reference in their entireties. Examples of injection layers are provided in U.S. Patent Application Publication No. 2004/0174116, which is incorporated by reference in its

entirety. A description of protective layers may be found in U.S. Patent Application Publication No. 2004/0174116, which is incorporated by reference in its entirety.

(116) FIG. 2 shows an inverted OLED 200. The device includes a substrate 210, a cathode 215, an emissive layer 220, a hole transport layer 225, and an anode 230. Device 200 may be fabricated by depositing the layers described, in order. Because the most common OLED configuration has a cathode disposed over the anode, and device 200 has cathode 215 disposed under anode 230, device 200 may be referred to as an “inverted” OLED. Materials similar to those described with respect to device 100 may be used in the corresponding layers of device 200. FIG. 2 provides one example of how some layers may be omitted from the structure of device 100.

(117) The simple layered structure illustrated in FIGS. 1 and 2 is provided by way of non-limiting example, and it is understood that embodiments of the present disclosure may be used in connection with a wide variety of other structures. The specific materials and structures described are exemplary in nature, and other materials and structures may be used. Functional OLEDs may be achieved by combining the various layers described in different ways, or layers may be omitted entirely, based on design, performance, and cost factors. Other layers not specifically described may also be included. Materials other than those specifically described may be used. Although many of the examples provided herein describe various layers as comprising a single material, it is understood that combinations of materials, such as a mixture of host and dopant, or more generally a mixture, may be used. Also, the layers may have various sublayers. The names given to the various layers herein are not intended to be strictly limiting. For example, in device 200, hole transport layer 225 transports holes and injects holes into emissive layer 220, and may be described as a hole transport layer or a hole injection layer. In one embodiment, an OLED may be described as having an “organic layer” disposed between a cathode and an anode. This organic layer may comprise a single layer, or may further comprise multiple layers of different organic materials as described, for example, with respect to FIGS. 1 and 2.

(118) Structures and materials not specifically described may also be used, such as OLEDs comprised of polymeric materials (PLEDs) such as disclosed in U.S. Pat. No. 5,247,190 to Friend et al., which is incorporated by reference in its entirety. By way of further example, OLEDs having a single organic layer may be used. OLEDs may be stacked, for example as described in U.S. Pat. No. 5,707,745 to Forrest et al, which is incorporated by reference in its entirety. The OLED structure may deviate from the simple layered structure illustrated in FIGS. 1 and 2. For example, the substrate may include an angled reflective surface to improve out-coupling, such as a mesa structure as described in U.S. Pat. No. 6,091,195 to Forrest et al., and/or a pit structure as described in U.S. Pat. No. 5,834,893 to Bulovic et al., which are incorporated by reference in their entireties.

(119) Unless otherwise specified, any of the layers of the various embodiments may be deposited by any suitable method. For the organic layers, preferred methods include thermal evaporation, ink jet, such as described in U.S. Pat. Nos. 6,013,982 and 6,087,196, which are incorporated by reference in their entireties, organic vapor phase deposition (OVPD), such as described in U.S. Pat. No. 6,337,102 to Forrest et al., which is incorporated by reference in its entirety, and deposition by organic vapor jet printing (OVJP), such as described in U.S. Pat. No. 7,431,968, which is incorporated by reference in its entirety. Other suitable deposition methods include spin coating and other solution based processes. Solution based processes are preferably carried out in nitrogen or an inert atmosphere. For the other layers, preferred methods include thermal evaporation. Preferred patterning methods include deposition through a mask, cold welding such as described in U.S. Pat. Nos. 6,294,398 and 6,468,819, which are incorporated by reference in their entireties, and patterning associated with some of the deposition methods such as ink jet and organic vapor jet printing (OVJP). Other methods may also be used. The materials to be deposited may be modified to make them compatible with a particular deposition method. For example, substituents such as alkyl and aryl groups, branched or unbranched, and preferably containing at least 3 carbons, may be used in small molecules to enhance their ability to undergo solution processing. Substituents

having 20 carbons or more may be used, and 3-20 carbons are a preferred range. Materials with asymmetric structures may have better solution processability than those having symmetric structures, because asymmetric materials may have a lower tendency to recrystallize. Dendrimer substituents may be used to enhance the ability of small molecules to undergo solution processing.

(120) Devices fabricated in accordance with embodiments of the present disclosure may further optionally comprise a barrier layer. One purpose of the barrier layer is to protect the electrodes and organic layers from damaging exposure to harmful species in the environment including moisture, vapor and/or gases, etc. The barrier layer may be deposited over, under or next to a substrate, an electrode, or over any other parts of a device including an edge. The barrier layer may comprise a single layer, or multiple layers. The barrier layer may be formed by various known chemical vapor deposition techniques and may include compositions having a single phase as well as compositions having multiple phases. Any suitable material or combination of materials may be used for the barrier layer. The barrier layer may incorporate an inorganic or an organic compound or both. The preferred barrier layer comprises a mixture of a polymeric material and a non-polymeric material as described in U.S. Pat. No. 7,968,146, PCT Pat. Application Nos. PCT/US2007/023098 and PCT/US2009/042829, which are herein incorporated by reference in their entireties. To be considered a “mixture”, the aforesaid polymeric and non-polymeric materials comprising the barrier layer should be deposited under the same reaction conditions and/or at the same time. The weight ratio of polymeric to non-polymeric material may be in the range of 95:5 to 5:95. The polymeric material and the non-polymeric material may be created from the same precursor material. In one example, the mixture of a polymeric material and a non-polymeric material consists essentially of polymeric silicon and inorganic silicon.

(121) Devices fabricated in accordance with embodiments of the present disclosure can be incorporated into a wide variety of electronic component modules (or units) that can be incorporated into a variety of electronic products or intermediate components. Examples of such electronic products or intermediate components include display screens, lighting devices such as discrete light source devices or lighting panels, etc. that can be utilized by the end-user product manufacturers. Such electronic component modules can optionally include the driving electronics and/or power source(s). Devices fabricated in accordance with embodiments of the present disclosure can be incorporated into a wide variety of consumer products that have one or more of the electronic component modules (or units) incorporated therein. A consumer product comprising an OLED that includes the compound of the present disclosure in the organic layer in the OLED is disclosed. Such consumer products would include any kind of products that include one or more light source(s) and/or one or more of some type of visual displays. Some examples of such consumer products include flat panel displays, curved displays, computer monitors, medical monitors, televisions, billboards, lights for interior or exterior illumination and/or signaling, heads-up displays, fully or partially transparent displays, flexible displays, rollable displays, foldable displays, stretchable displays, laser printers, telephones, mobile phones, tablets, phablets, personal digital assistants (PDAs), wearable devices, laptop computers, digital cameras, camcorders, viewfinders, micro-displays (displays that are less than 2 inches diagonal), 3-D displays, virtual reality or augmented reality displays, vehicles, video walls comprising multiple displays tiled together, theater or stadium screen, a light therapy device, and a sign. Various control mechanisms may be used to control devices fabricated in accordance with the present disclosure, including passive matrix and active matrix. Many of the devices are intended for use in a temperature range comfortable to humans, such as 18 degrees C. to 30 degrees C., and more preferably at room temperature (20-25° C.), but could be used outside this temperature range, for example, from -40 degree C. to +80° C.

(122) More details on OLEDs, and the definitions described above, can be found in U.S. Pat. No. 7,279,704, which is incorporated herein by reference in its entirety.

(123) The materials and structures described herein may have applications in devices other than

OLEDs. For example, other optoelectronic devices such as organic solar cells and organic photodetectors may employ the materials and structures. More generally, organic devices, such as organic transistors, may employ the materials and structures.

(124) In some embodiments, the OLED has one or more characteristics selected from the group consisting of being flexible, being rollable, being foldable, being stretchable, and being curved. In some embodiments, the OLED is transparent or semi-transparent. In some embodiments, the OLED further comprises a layer comprising carbon nanotubes.

(125) In some embodiments, the OLED further comprises a layer comprising a delayed fluorescent emitter. In some embodiments, the OLED comprises a RGB pixel arrangement or white plus color filter pixel arrangement. In some embodiments, the OLED is a mobile device, a hand held device, or a wearable device. In some embodiments, the OLED is a display panel having less than 10 inch diagonal or 50 square inch area. In some embodiments, the OLED is a display panel having at least 10 inch diagonal or 50 square inch area. In some embodiments, the OLED is a lighting panel.

(126) In some embodiments, the compound can be an emissive dopant. In some embodiments, the compound can produce emissions via phosphorescence, fluorescence, thermally activated delayed fluorescence, i.e., TADF (also referred to as E-type delayed fluorescence; see, e.g., U.S. application Ser. No. 15/700,352, which is hereby incorporated by reference in its entirety), triplet-triplet annihilation, or combinations of these processes. In some embodiments, the emissive dopant can be a racemic mixture, or can be enriched in one enantiomer. In some embodiments, the compound can be homoleptic (each ligand is the same). In some embodiments, the compound can be heteroleptic (at least one ligand is different from others). When there are more than one ligand coordinated to a metal, the ligands can all be the same in some embodiments. In some other embodiments, at least one ligand is different from the other ligands. In some embodiments, every ligand can be different from each other. This is also true in embodiments where a ligand being coordinated to a metal can be linked with other ligands being coordinated to that metal to form a tridentate, tetradentate, pentadentate, or hexadentate ligands. Thus, where the coordinating ligands are being linked together, all of the ligands can be the same in some embodiments, and at least one of the ligands being linked can be different from the other ligand(s) in some other embodiments.

(127) In some embodiments, the compound can be used as a phosphorescent sensitizer in an OLED where one or multiple layers in the OLED contains an acceptor in the form of one or more fluorescent and/or delayed fluorescence emitters. In some embodiments, the compound can be used as one component of an exciplex to be used as a sensitizer. As a phosphorescent sensitizer, the compound must be capable of energy transfer to the acceptor and the acceptor will emit the energy or further transfer energy to a final emitter. The acceptor concentrations can range from 0.001% to 100%. The acceptor could be in either the same layer as the phosphorescent sensitizer or in one or more different layers. In some embodiments, the acceptor is a TADF emitter. In some embodiments, the acceptor is a fluorescent emitter. In some embodiments, the emission can arise from any or all of the sensitizer, acceptor, and final emitter

(128) According to another aspect, a formulation comprising the compound described herein is also disclosed.

(129) The OLED disclosed herein can be incorporated into one or more of a consumer product, an electronic component module, and a lighting panel. The organic layer can be an emissive layer and the compound can be an emissive dopant in some embodiments, while the compound can be a non-emissive dopant in other embodiments.

(130) In yet another aspect of the present disclosure, a formulation that comprises the novel compound disclosed herein is described. The formulation can include one or more components selected from the group consisting of a solvent, a host, a hole injection material, hole transport material, electron blocking material, hole blocking material, and an electron transport material, disclosed herein.

(131) The present disclosure encompasses any chemical structure comprising the novel compound

of the present disclosure, or a monovalent or polyvalent variant thereof. In other words, the inventive compound, or a monovalent or polyvalent variant thereof, can be a part of a larger chemical structure. Such chemical structure can be selected from the group consisting of a monomer, a polymer, a macromolecule, and a supramolecule (also known as supermolecule). As used herein, a “monovalent variant of a compound” refers to a moiety that is identical to the compound except that one hydrogen has been removed and replaced with a bond to the rest of the chemical structure. As used herein, a “polyvalent variant of a compound” refers to a moiety that is identical to the compound except that more than one hydrogen has been removed and replaced with a bond or bonds to the rest of the chemical structure. In the instance of a supramolecule, the inventive compound can also be incorporated into the supramolecule complex without covalent bonds.

D. Combination of the Compounds of the Present Disclosure with Other Materials

(132) The materials described herein as useful for a particular layer in an organic light emitting device may be used in combination with a wide variety of other materials present in the device. For example, emissive dopants disclosed herein may be used in conjunction with a wide variety of hosts, transport layers, blocking layers, injection layers, electrodes and other layers that may be present. The materials described or referred to below are non-limiting examples of materials that may be useful in combination with the compounds disclosed herein, and one of skill in the art can readily consult the literature to identify other materials that may be useful in combination.

(133) a) Conductivity Dopants:

(134) A charge transport layer can be doped with conductivity dopants to substantially alter its density of charge carriers, which will in turn alter its conductivity. The conductivity is increased by generating charge carriers in the matrix material, and depending on the type of dopant, a change in the Fermi level of the semiconductor may also be achieved. Hole-transporting layer can be doped by p-type conductivity dopants and n-type conductivity dopants are used in the electron-transporting layer.

(135) Non-limiting examples of the conductivity dopants that may be used in an OLED in combination with materials disclosed herein are exemplified below together with references that disclose those materials: EP01617493, EP01968131, EP2020694, EP2684932, US20050139810, US20070160905, US20090167167, US2010288362, WO06081780, WO2009003455, WO2009008277, WO2009011327, WO2014009310, US2007252140, US2015060804, US20150123047, and US2012146012.

(136) ##STR00084## ##STR00085##

b) HIL/HTL:

(137) A hole injecting/transporting material to be used in the present disclosure is not particularly limited, and any compound may be used as long as the compound is typically used as a hole injecting/transporting material. Examples of the material include, but are not limited to: a phthalocyanine or porphyrin derivative; an aromatic amine derivative; an indolocarbazole derivative; a polymer containing fluorohydrocarbon; a polymer with conductivity dopants; a conducting polymer, such as PEDOT/PSS; a self-assembly monomer derived from compounds such as phosphoric acid and silane derivatives; a metal oxide derivative, such as MoO_{sub}.x; a p-type semiconducting organic compound, such as 1,4,5,8,9,12-

Hexaazatriphenylenehexacarbonitrile; a metal complex, and a cross-linkable compounds.

(138) Examples of aromatic amine derivatives used in HIL or HTL include, but not limit to the following general structures:

(139) ##STR00086##

(140) Each of Ar_{sup}.1 to Ar_{sup}.9 is selected from the group consisting of aromatic hydrocarbon cyclic compounds such as benzene, biphenyl, triphenyl, triphenylene, naphthalene, anthracene, phenalene, phenanthrene, fluorene, pyrene, chrysene, perylene, and azulene; the group consisting of aromatic heterocyclic compounds such as dibenzothiophene, dibenzofuran, dibenzoselenophene,

furan, thiophene, benzofuran, benzothiophene, cathazole, indolocarbazole, pyridylindole, pyrrolodipyridine, pyrazole, imidazole, thazole, oxazole, thiazole, oxadiazole, oxatriazole, dioxazole, thiadiazole, pyridine, pyridazine, pyrimidine, pyrazine, triazine, oxazine, oxathiazine, oxadiazine, indole, benzimidazole, indazole, indoxazine, benzoxazole, benzisoxazole, benzothiazole, quinoline, isoquinoline, cinnoline, quinazoline, quinoxaline, naphthyridine, phthalazine, pteridine, xanthene, acridine, phenazine, phenothiazine, phenoxazine, benzofuropyridine, furodipyridine, benzothienopyridine, thienodipyridine, benzoselenophenopyridine, and selenophenodipyridine; and the group consisting of 2 to 10 cyclic structural units which are groups of the same type or different types selected from the aromatic hydrocarbon cyclic group and the aromatic heterocyclic group and are bonded to each other directly or via at least one of oxygen atom, nitrogen atom, sulfur atom, silicon atom, phosphorus atom, boron atom, chain structural unit and the aliphatic cyclic group. Each Ar may be unsubstituted or may be substituted by a substituent selected from the group consisting of deuterium, halogen, alkyl, cycloalkyl, heteroalkyl, heterocycloalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carboxylic acids, ether, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof.

(141) In one aspect, Ar.sup.1 to Ar.sup.9 is independently selected from the group consisting of:

(142) ##STR00087##

wherein k is an integer from 1 to 20; X.sup.101 to X.sup.108 is C (including CH) or N; Z.sup.101 is NAr.sup.1, O, or S; Ar.sup.1 has the same group defined above.

(143) Examples of metal complexes used in HIL or HTL include, but are not limited to the following general formula:

(144) ##STR00088##

wherein Met is a metal, which can have an atomic weight greater than 40; (Y.sup.101-Y.sup.102) is a bidentate ligand, Y.sup.101 and Y.sup.102 are independently selected from C, N, O, P, and S; Y.sup.101 is an ancillary ligand; k' is an integer value from 1 to the maximum number of ligands that may be attached to the metal; and k'+k'' is the maximum number of ligands that may be attached to the metal.

(145) In one aspect, (Y.sup.101-Y.sup.102) is a 2-phenylpyridine derivative. In another aspect, (Y.sup.101-Y.sup.102) is a carbene ligand. In another aspect, Met is selected from Ir, Pt, Os, and Zn. In a further aspect, the metal complex has a smallest oxidation potential in solution vs. Fc.sup.+ / Fc couple less than about 0.6 V.

(146) Non-limiting examples of the HIL and HTL materials that may be used in an OLED in combination with materials disclosed herein are exemplified below together with references that disclose those materials: CN102702075, DE102012005215, EP01624500, EP01698613, EP01806334, EP01930964, EP01972613, EP01997799, EP02011790, EP02055700, EP02055701, EP1725079, EP2085382, EP2660300, EP650955, JP07-073529, JP2005112765, JP2007091719, JP2008021687, JP2014-009196, KR20110088898, KR20130077473, TW201139402, U.S. Ser. No. 06/517,957, US20020158242, US20030162053, US20050123751, US20060182993, US20060240279, US20070145888, US20070181874, US20070278938, US20080014464, US20080091025, US20080106190, US20080124572, US20080145707, US20080220265, US20080233434, US20080303417, US2008107919, US20090115320, US20090167161, US2009066235, US2011007385, US20110163302, US2011240968, US2011278551, US2012205642, US2013241401, US20140117329, US2014183517, U.S. Pat. Nos. 5,061,569, 5,639,914, WO05075451, WO07125714, WO08023550, WO08023759, WO2009145016, WO2010061824, WO2011075644, WO2012177006, WO2013018530, WO2013039073, WO2013087142, WO2013118812, WO2013120577, WO2013157367, WO2013175747, WO2014002873, WO2014015935, WO2014015937, WO2014030872, WO2014030921, WO2014034791, WO2014104514, WO2014157018.

(147) ##STR00089## ##STR00090## ##STR00091## ##STR00092## ##STR00093##

##STR00094## ##STR00095## ##STR00096## ##STR00097## ##STR00098## ##STR00099##
##STR00100## ##STR00101## ##STR00102## ##STR00103##

c) EBL:

(148) An electron blocking layer (EBL) may be used to reduce the number of electrons and/or excitons that leave the emissive layer. The presence of such a blocking layer in a device may result in substantially higher efficiencies, and/or longer lifetime, as compared to a similar device lacking a blocking layer. Also, a blocking layer may be used to confine emission to a desired region of an OLED. In some embodiments, the EBL material has a higher LUMO (closer to the vacuum level) and/or higher triplet energy than the emitter closest to the EBL interface. In some embodiments, the EBL material has a higher LUMO (closer to the vacuum level) and/or higher triplet energy than one or more of the hosts closest to the EBL interface. In one aspect, the compound used in EBL contains the same molecule or the same functional groups used as one of the hosts described below.

(149) d) Hosts:

(150) The light emitting layer of the organic EL device of the present disclosure preferably contains at least a metal complex as light emitting material, and may contain a host material using the metal complex as a dopant material. Examples of the host material are not particularly limited, and any metal complexes or organic compounds may be used as long as the triplet energy of the host is larger than that of the dopant. Any host material may be used with any dopant so long as the triplet criteria is satisfied.

(151) Examples of metal complexes used as host are preferred to have the following general formula:

(152) ##STR00104##

wherein Met is a metal; (Y^{sup.103}-Y^{sup.104}) is a bidentate ligand, Y^{sup.103} and Y^{sup.104} are independently selected from C, N, O, P, and S; L^{sup.101} is another ligand; k' is an integer value from 1 to the maximum number of ligands that may be attached to the metal; and k'+k'' is the maximum number of ligands that may be attached to the metal.

(153) In one aspect, the metal complexes are:

(154) ##STR00105##

wherein (O—N) is a bidentate ligand, having metal coordinated to atoms O and N.

(155) In another aspect, Met is selected from Ir and Pt. In a further aspect, (Y^{sup.103}-Y^{sup.104}) is a carbene ligand

(156) In one aspect, the host compound contains at least one of the following groups selected from the group consisting of aromatic hydrocarbon cyclic compounds such as benzene, biphenyl, triphenyl, triphenylene, tetraphenylene, naphthalene, anthracene, phenalene, phenanthrene, fluorene, pyrene, chrysene, perylene, and azulene; the group consisting of aromatic heterocyclic compounds such as dibenzothiophene, dibenzofuran, dibenzoselenophene, furan, thiophene, benzofuran, benzothiophene, benzoselenophene, carbazole, indolocarbazole, pyridylindole, pyrrolodipyridine, pyrazole, imidazole, triazole, oxazole, thiazole, oxadiazole, oxatriazole, dioxazole, thiadiazole, pyridine, pyridazine, pyrimidine, pyrazine, triazine, oxazine, oxathiazine, oxadiazine, indole, benzimidazole, indazole, indoxazine, benzoxazole, benzisoxazole, benzothiazole, quinoline, isoquinoline, cinnoline, quinazoline, quinoxaline, naphthyridine, phthalazine, pteridine, xanthene, acridine, phenazine, phenothiazine, phenoxazine, benzofuropyridine, furodipyridine, benzothenopyridine, thienodipyridine, benzoselenophenopyridine, and selenophenodipyridine; and the group consisting of 2 to 10 cyclic structural units which are groups of the same type or different types selected from the aromatic hydrocarbon cyclic group and the aromatic heterocyclic group and are bonded to each other directly or via at least one of oxygen atom, nitrogen atom, sulfur atom, silicon atom, phosphorus atom, boron atom, chain structural unit and the aliphatic cyclic group. Each option within each group may be unsubstituted or may be substituted by a substituent selected from the group consisting of deuterium, halogen, alkyl, cycloalkyl, heteroalkyl, heterocycloalkyl, arylalkyl, alkoxy,

aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carboxylic acids, ether, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof.

(157) In one aspect, the host compound contains at least one of the following groups in the molecule:

(158) ##STR00106## ##STR00107##

wherein R.sup.101 is selected from the group consisting of hydrogen, deuterium, halogen, alkyl, cycloalkyl, heteroalkyl, heterocycloalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carboxylic acids, ether, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof, and when it is aryl or heteroaryl, it has the similar definition as Ar's mentioned above. k is an integer from 0 to 20 or 1 to 20. X.sup.101 to X.sup.108 are independently selected from C (including CH) or N. Z.sup.101 and Z.sup.102 are independently selected from NR.sup.101, O, or S.

(159) Non-limiting examples of the host materials that may be used in an OLED in combination with materials disclosed herein are exemplified below together with references that disclose those materials: EP2034538, EP2034538A, EP2757608, JP2007254297, KR20100079458, KR20120088644, KR20120129733, KR20130115564, TW201329200, US20030175553, US20050238919, US20060280965, US20090017330, US20090030202, US20090167162, US20090302743, US20090309488, US20100012931, US20100084966, US20100187984, US2010187984, US2012075273, US2012126221, US2013009543, US2013105787, US2013175519, US2014001446, US20140183503, US20140225088, US2014034914, U.S. Pat. No. 7,154,114, WO2001039234, WO2004093207, WO2005014551, WO2005089025, WO2006072002, WO2006114966, WO2007063754, WO2008056746, WO2009003898, WO2009021126, WO2009063833, WO2009066778, WO2009066779, WO2009086028, WO2010056066, WO2010107244, WO2011081423, WO2011081431, WO2011086863, WO2012128298, WO2012133644, WO2012133649, WO2013024872, WO2013035275, WO2013081315, WO2013191404, WO2014142472, US20170263869, US20160163995, U.S. Pat. No. 9,466,803,

(160) ##STR00108## ##STR00109## ##STR00110## ##STR00111## ##STR00112##
##STR00113## ##STR00114## ##STR00115## ##STR00116## ##STR00117## ##STR00118##
e) Additional Emitters:

(161) One or more additional emitter dopants may be used in conjunction with the compound of the present disclosure. Examples of the additional emitter dopants are not particularly limited, and any compounds may be used as long as the compounds are typically used as emitter materials. Examples of suitable emitter materials include, but are not limited to, compounds which can produce emissions via phosphorescence, fluorescence, thermally activated delayed fluorescence, i.e., TADF (also referred to as E-type delayed fluorescence), triplet-triplet annihilation, or combinations of these processes.

(162) Non-limiting examples of the emitter materials that may be used in an OLED in combination with materials disclosed herein are exemplified below together with references that disclose those materials: CN103694277, CN1696137, EB01238981, EP01239526, EP01961743, EP1239526, EP1244155, EP1642951, EP1647554, EP1841834, EP1841834B, EP2062907, EP2730583, JP2012074444, JP2013110263, JP4478555, KR1020090133652, KR20120032054, KR20130043460, TW201332980, U.S. Ser. No. 06/699,599, U.S. Ser. No. 06/916,554, US20010019782, US20020034656, US20030068526, US20030072964, US20030138657, US20050123788, US20050244673, US2005123791, US2005260449, US20060008670, US20060065890, US20060127696, US20060134459, US20060134462, US20060202194, US20060251923, US20070034863, US20070087321, US20070103060, US20070111026, US20070190359, US20070231600, US2007034863, US2007104979, US2007104980, US2007138437, US2007224450, US2007278936, US20080020237, US20080233410,

US20080261076, US20080297033, US200805851, US2008161567, US2008210930, US20090039776, US20090108737, US20090115322, US20090179555, US2009085476, US2009104472, US20100090591, US20100148663, US20100244004, US20100295032, US2010102716, US2010105902, US2010244004, US2010270916, US20110057559, US20110108822, US20110204333, US2011215710, US2011227049, US2011285275, US2012292601, US20130146848, US2013033172, US2013165653, US2013181190, US2013334521, US20140246656, US2014103305, U.S. Pat. Nos. 6,303,238, 6,413,656, 6,653,654, 6,670,645, 6,687,266, 6,835,469, 6,921,915, 7,279,704, 7,332,232, 7,378,162, 7,534,505, 7,675,228, 7,728,137, 7,740,957, 7,759,489, 7,951,947, 8,067,099, 8,592,586, 8,871,361, WO06081973, WO06121811, WO07018067, WO07108362, WO07115970, WO07115981, WO08035571, WO2002015645, WO2003040257, WO2005019373, WO2006056418, WO2008054584, WO2008078800, WO2008096609, WO2008101842, WO2009000673, WO2009050281, WO2009100991, WO2010028151, WO2010054731, WO2010086089, WO2010118029, WO2011044988, WO2011051404, WO2011107491, WO2012020327, WO2012163471, WO2013094620, WO2013107487, WO2013174471, WO2014007565, WO2014008982, WO2014023377, WO2014024131, WO2014031977, WO2014038456, WO2014112450.

(163) ##STR00119## ##STR00120## ##STR00121## ##STR00122## ##STR00123## ##STR00124## ##STR00125## ##STR00126## ##STR00127## ##STR00128## ##STR00129## ##STR00130## ##STR00131## ##STR00132## ##STR00133## ##STR00134## ##STR00135## ##STR00136##

f) HBL:

(164) A hole blocking layer (HBL) may be used to reduce the number of holes and/or excitons that leave the emissive layer. The presence of such a blocking layer in a device may result in substantially higher efficiencies and/or longer lifetime as compared to a similar device lacking a blocking layer. Also, a blocking layer may be used to confine emission to a desired region of an OLED. In some embodiments, the HBL material has a lower HOMO (further from the vacuum level) and/or higher triplet energy than the emitter closest to the HBL interface. In some embodiments, the HBL material has a lower HOMO (further from the vacuum level) and/or higher triplet energy than one or more of the hosts closest to the HBL interface.

(165) In one aspect, compound used in HBL contains the same molecule or the same functional groups used as host described above.

(166) In another aspect, compound used in HBL contains at least one of the following groups in the molecule:

(167) ##STR00137##

wherein k is an integer from 1 to 20; L.sup.101 is another ligand, k' is an integer from 1 to 3.

g) ETL:

(168) Electron transport layer (ETL) may include a material capable of transporting electrons. Electron transport layer may be intrinsic (undoped), or doped. Doping may be used to enhance conductivity. Examples of the ETL material are not particularly limited, and any metal complexes or organic compounds may be used as long as they are typically used to transport electrons.

(169) In one aspect, compound used in ETL contains at least one of the following groups in the molecule:

(170) ##STR00138##

wherein R.sup.101 is selected from the group consisting of hydrogen, deuterium, halogen, alkyl, cycloalkyl, heteroalkyl, heterocycloalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carboxylic acids, ether, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof, when it is aryl or heteroaryl, it has the similar definition as Ar's mentioned above. Ar.sup.1 to Ar.sup.3 has the similar definition as Ar's mentioned above. k is an integer from 1 to 20. X.sup.10' to X.sup.108 is selected

from C (including CH) or N.

(171) In another aspect, the metal complexes used in ETL contains, but not limit to the following general formula:

(172) ##STR00139##

wherein (O—N) or (N—N) is a bidentate ligand, having metal coordinated to atoms O, N or N, N; L.sup.101 is another ligand; k' is an integer value from 1 to the maximum number of ligands that may be attached to the metal.

(173) Non-limiting examples of the ETL materials that may be used in an OLED in combination with materials disclosed herein are exemplified below together with references that disclose those materials: CN103508940, EP01602648, EP01734038, EP01956007, JP2004-022334, JP2005149918, JP2005-268199, KR0117693, KR20130108183, US20040036077, US20070104977, US2007018155, US20090101870, US20090115316, US20090140637, US20090179554, US2009218940, US2010108990, US2011156017, US2011210320, US2012193612, US2012214993, US2014014925, US2014014927, US20140284580, U.S. Pat. Nos. 6,656,612, 8,415,031, WO2003060956, WO2007111263, WO2009148269, WO2010067894, WO2010072300, WO2011074770, WO2011105373, WO2013079217, WO2013145667, WO2013180376, WO2014104499, WO2014104535,

(174) ##STR00140## ##STR00141## ##STR00142## ##STR00143## ##STR00144## ##STR00145## ##STR00146## ##STR00147## ##STR00148##

h) Charge generation layer (CGL)

(175) In tandem or stacked OLEDs, the CGL plays an essential role in the performance, which is composed of an n-doped layer and a p-doped layer for injection of electrons and holes, respectively. Electrons and holes are supplied from the CGL and electrodes. The consumed electrons and holes in the CGL are refilled by the electrons and holes injected from the cathode and anode, respectively; then, the bipolar currents reach a steady state gradually. Typical CGL materials include n and p conductivity dopants used in the transport layers.

(176) In any above-mentioned compounds used in each layer of the OLED device, the hydrogen atoms can be partially or fully deuterated. Thus, any specifically listed substituent, such as, without limitation, methyl, phenyl, pyridyl, etc. may be undeuterated, partially deuterated, and fully deuterated versions thereof. Similarly, classes of substituents such as, without limitation, alkyl, aryl, cycloalkyl, heteroaryl, etc. also may be undeuterated, partially deuterated, and fully deuterated versions thereof.

(177) It is understood that the various embodiments described herein are by way of example only and are not intended to limit the scope of the invention. For example, many of the materials and structures described herein may be substituted with other materials and structures without deviating from the spirit of the invention. The present invention as claimed may therefore include variations from the particular examples and preferred embodiments described herein, as will be apparent to one of skill in the art. It is understood that various theories as to why the invention works are not intended to be limiting.

E. Experimental Data

(178) Synthesis of Materials

Synthesis of L.SUB.A122-1

(179) ##STR00149##

(180) 6-Chloropyrimidine-4,5-diamine (2.00 g, 13.83 mmol), benzo[b]thiophen-2-ylboronic acid (2.96 g, 16.60 mmol), tetrakis(triphenylphosphine)palladium(O) (0.799 g, 0.692 mmol), and potassium carbonate (3.82 g, 27.7 mmol) were charged into an oven dried two-necked round bottom flask fitted with a reflux condenser. The mixture was degassed by vacuum/nitrogen cycle three times, then anhydrous 1,4-dioxane (50 ml) and biacetyl (2.428 ml, 27.7 mmol) were added. The reaction mixture was stirred at 80° C. for 16 hours. After cooling down to room temperature, the reaction mixture was filtered off through a Celite cartridge and the volatiles were removed

under vacuum. The resulting crude mixture was purified by triethylamine-treated silica gel column chromatography using a mixture of dichloromethane (50-70%), heptane, and triethylamine (1%) as eluent, affording a yellow solid (1.24 g, 4.22 mmol, 31%).

Synthesis of [L.SUB.A122-1].SUB.2.Ir[L.SUB.C17-1.]

(181) ##STR00150##

(182) Iridium precursor (0.893 g, 2.078 mmol) and L.sub.A122-1 (0.81 g, 2.77 mmol) were dissolved in 1,2-dichlorobenzene (28 ml) in a schlenk tube. The solution was purged with nitrogen for 10 minutes, then 2,2,2-trifluoroacetic acid (0.477 ml, 6.23 mmol) was added. The reaction mixture was stirred at 130° C. for 16 hours and monitored by NMR analysis, which indicated the ligand was almost consumed. To the mixture was added K.sub.2CO.sub.3 (0.766 g, 5.54 mmol), 3,7-diethylnonane-4,6-dione (1.293 ml, 5.54 mmol), and 1,4-Dioxane (28 ml). The reaction mixture was purged with nitrogen for 10 minutes, then stirred at 60° C. for 3 days. After cooling down to room temperature, the reaction mixture was filtered off through a Celite cartridge and the volatiles were removed under vacuum. The resulting crude mixture was purified by silica gel column chromatography using a mixture of ethyl acetate (5%) and dichloromethane as eluent, affording a dark green solid (0.16 g, 0.16 mmol, 12%).

(183) TABLE-US-00003 TABLE 1 HOMO LUMO Gap T.sub.1 Example (eV) (eV) (eV) (nm)

Inventive example 1 -5.577 -2.981 2.596 785 Comparative example 1 -5.415 -2.538 2.877 708

Inventive example 2 -5.369 -3.160 2.209 905 Comparative example 2 -5.330 -2.643 2.687 723

Inventive example 3 -5.374 -3.428 1.946 1074 Comparative example 3 -5.221 -2.801 2.420 818

Inventive example 4 -5.285 -3.338 1.947 1112 Comparative example 4 -5.123 -2.810 2.313 904

Inventive example 5 -5.582 -3.825 1.757 1198 Comparative example 5 -5.315 -3.140 2.175 878





Table 1 above provides the results of the DFT calculations performed to determine the HOMO/LUMO level, HOMO-LUMO gap, and the energy of the lowest triplet (T.sub.1) excited state of various compounds. The data was gathered using the program Gaussian16. Geometries were optimized using B3LYP functional and CEP-31G basis set. Excited state energies were computed by TDDFT at the optimized ground state geometries. THE solvent was simulated using a self-consistent reaction field to further improve agreement with experiment. The T.sub.1 energies of the inventive examples 1~5 were calculated to be 785, 905, 1074, 1112, and 1198 nm. In comparison, the T.sub.1 energies of the comparative examples 1~5 were calculated to be 708, 723, 818, 904, and 878 nm, which are unexpectedly higher by 70-300 nm than the corresponding inventive examples. The emission maximum peak of the inventive example 1 [L.sub.A122-1].sub.2Ir[L.sub.C17-1] measured in 2-methylTHF solution at room temperature is 767 nm, which is in agreement with DFT calculation data. Without being bound by any theories, this is probably due to the polyaza group in the inventive examples to effectively lower the LUMO level and T.sub.1 of the metal complexes, resulting in phosphorescence in near-infrared (NIR) region. These inventive compounds are demonstrated here to be applicable as near-infrared emitter used in the OLED device.

(184) The calculations obtained with the above-identified DFT functional set and basis set are theoretical. Computational composite protocols, such as the Gaussian with B3LYP and CEP-31G protocol used herein, rely on the assumption that electronic effects are additive and, therefore, larger basis sets can be used to extrapolate to the complete basis set (CBS) limit. However, when the goal of a study is to understand variations in HOMO, LUMO, Si, Ti, bond dissociation energies, etc. over a series of structurally-related compounds, the additive effects are expected to be similar. Accordingly, while absolute errors from using the B3LYP may be significant compared to other computational methods, the relative differences between the HOMO, LUMO, S.sub.1, T.sub.1, and bond dissociation energy values calculated with B3LYP protocol are expected to reproduce experiment quite well. See, e.g., Hong et al., *Chem. Mater.* 2016, 28, 5791-98, 5792-93

and Supplemental Information (discussing the reliability of DFT calculations in the context of OLED materials). Moreover, with respect to iridium or platinum complexes that are useful in the OLED art, the data obtained from DFT calculations correlates very well to actual experimental data. See Tavasli et al., *J. Mater. Chem.* 2012, 22, 6419-29, 6422 (Table 3) (showing DFT calculations closely correlating with actual data for a variety of emissive complexes); Morello, G. R., *J. Mol. Model.* 2017, 23:174 (studying of a variety of DFT functional sets and basis sets and concluding the combination of B3LYP and CEP-31G is particularly accurate for emissive complexes).

Claims

1. A compound of Formula $\text{Ir}(\text{L.sub.A})_{\text{x}}(\text{L.sub.C})_{\text{y}}$, wherein: ligand L.sub.A has Formula I' ##STR00161## and ligand L.sub.C has Formula II' ##STR00162## wherein: x is 1, 2, or 3; y is 0, 1, or 2; x+y=3; X¹-X⁸ are each independently C, N, or CR, and at least four of X¹-X⁸ are N; the X¹-X⁴ that connects to ring A is C, and the X¹-X⁴ that coordinates to Ir is N; ring A is an unsaturated 5-membered or 6-membered carbocyclic or heterocyclic ring; R^A represents zero, mono, or up to the maximum number of allowed substitutions to its associated ring; each R, R¹-R³, and R^A is independently a hydrogen or a substituent selected from the group consisting of deuterium, halogen, alkyl, cycloalkyl, heteroalkyl, heterocycloalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, boryl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carboxylic acid, ether, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof; R¹ comprises at least two carbon atoms; two substituents can be joined or fused together to form a ring, wherein the ligand L.sub.A coordinates to Ir to form a 5-membered chelate ring as indicated by the two dashed lines of the Formula I'; wherein at least one pair of substituents are joined or fused together to form a ring; and with a proviso that if ring A is a 6-membered ring and two R^A are joined or fused to form a naphthalene that includes ring A, then said naphthalene is substituted.
2. The compound of claim 1, wherein each R, R¹-R³, and R^A is independently a hydrogen or a substituent selected from the group consisting of deuterium, fluorine, alkyl, cycloalkyl, heteroalkyl, alkoxy, aryloxy, amino, silyl, boryl, alkenyl, cycloalkenyl, heteroalkenyl, aryl, heteroaryl, nitrile, isonitrile, sulfanyl, and combinations thereof.
3. The compound of claim 1, wherein four of X¹-X⁸ are N.
4. The compound of claim 1, wherein X², X⁴, X⁵, and X⁸ are N, and X¹ and X³ are C.
5. The compound of claim 1, wherein X⁶ and X⁷ are CR wherein the two R substituents are joined to form a fused ring.
6. The compound of claim 1, wherein two of the X⁵-X⁸ are CR and adjacent to each other with the two R substituents joined to form a fused 6-membered aromatic ring.
7. The compound of claim 1, wherein ring A is a 6-membered aromatic ring.
8. The compound of claim 1, wherein ring A is a 5-membered heteroaryl ring.
9. The compound of claim 1, wherein two adjacent R^A substituents are joined to form a 6-membered aromatic ring fused to ring A.
10. The compound of claim 1, wherein R¹ and R² are independently an alkyl, cycloalkyl, partially or fully deuterated variants thereof, partially or fully fluorinated variants thereof, and combinations thereof.
11. The compound of claim 1, wherein the ligand L.sub.A is each independently selected from the group consisting of: ##STR00163## ##STR00164## ##STR00165## ##STR00166## wherein: Z¹-Z⁶ are each independently N or CR^F; Y is O or S; each of R^E and R^F is independently a hydrogen or a substituent selected from the group consisting of deuterium, fluorine, alkyl, cycloalkyl, heteroalkyl, alkoxy, aryloxy, amino, silyl, boryl, alkenyl, cycloalkenyl,

heteroalkenyl, aryl, heteroaryl, nitrile, isonitrile, sulfonyl, and combinations thereof.

12. The compound of claim 1, wherein the ligand L.sub.A is L.sub.Ai-m, wherein i is an integer from 61 to 200 and m is an integer from 1 to 20, wherein the ligand L.sub.A is selected from the group consisting of L.sub.A61-1 through L.sub.A200-1, L.sub.A61-2 through L.sub.A200-2, L.sub.A61-3 through L.sub.A200-3, L.sub.A1-4 through L.sub.A200-4, L.sub.A61-5 through L.sub.A200-5, L.sub.A61-6 through L.sub.A200-6, L.sub.A1-7 through L.sub.A200-7, L.sub.A61-8 through L.sub.A200-8, L.sub.A61-9 through L.sub.A200-9, L.sub.A61-10 through L.sub.A200-10, L.sub.A1-11 through L.sub.A200-11, L.sub.A61-12 through L.sub.A200-12, L.sub.A1-13 through L.sub.A200-13, L.sub.A61-14 through L.sub.A200-14, L.sub.A61-15 through L.sub.A200-15, L.sub.A61-16 through L.sub.A200-16, L.sub.A1-17 through L.sub.A200-17, L.sub.A61-18 through L.sub.A200-18, L.sub.A61-19 through L.sub.A200-19, and L.sub.A1-20 through L.sub.A200-20, wherein the structure of each of L.sub.Ai-1 through L.sub.Ai-20 is defined below:

##STR00167## ##STR00168## ##STR00169## ##STR00170## wherein for each L.sub.Ai in L.sub.Ai-1 to L.sub.Ai-20, R.sup.E and G in formulas 1 to formula 20 are defined as follows:
TABLE-US-00004 L.sub.Ai R.sup.E G L.sub.A1 R.sup.E1 G.sup.1 L.sub.A2 R.sup.E2 G.sup.1 L.sub.A3 R.sup.E3 G.sup.1 L.sub.A4 R.sup.E4 G.sup.1 L.sub.A5 R.sup.E5 G.sup.1 L.sub.A6 R.sup.E6 G.sup.1 L.sub.A7 R.sup.E7 G.sup.1 L.sub.A8 R.sup.E8 G.sup.1 L.sub.A9 R.sup.E9 G.sup.1 L.sub.A10 R.sup.E10 G.sup.1 L.sub.A11 R.sup.E11 G.sup.1 L.sub.A12 R.sup.E12 G.sup.1 L.sub.A13 R.sup.E13 G.sup.1 L.sub.A14 R.sup.E14 G.sup.1 L.sub.A15 R.sup.E15 G.sup.1 L.sub.A16 R.sup.E16 G.sup.1 L.sub.A17 R.sup.E17 G.sup.1 L.sub.A18 R.sup.E18 G.sup.1 L.sub.A19 R.sup.E19 G.sup.1 L.sub.A20 R.sup.E20 G.sup.1 L.sub.A21 R.sup.E1 G.sup.2 L.sub.A22 R.sup.E2 G.sup.2 L.sub.A23 R.sup.E3 G.sup.2 L.sub.A24 R.sup.E4 G.sup.2 L.sub.A25 R.sup.E5 G.sup.2 L.sub.A26 R.sup.E6 G.sup.2 L.sub.A27 R.sup.E7 G.sup.2 L.sub.A28 R.sup.E8 G.sup.2 L.sub.A29 R.sup.E9 G.sup.2 L.sub.A30 R.sup.E10 G.sup.2 L.sub.A31 R.sup.E11 G.sup.2 L.sub.A32 R.sup.E12 G.sup.2 L.sub.A33 R.sup.E13 G.sup.2 L.sub.A34 R.sup.E14 G.sup.2 L.sub.A35 R.sup.E15 G.sup.2 L.sub.A36 R.sup.E16 G.sup.2 L.sub.A37 R.sup.E17 G.sup.2 L.sub.A38 R.sup.E18 G.sup.2 L.sub.A39 R.sup.E19 G.sup.2 L.sub.A40 R.sup.E20 G.sup.2 L.sub.A41 R.sup.E1 G.sup.3 L.sub.A42 R.sup.E2 G.sup.3 L.sub.A43 R.sup.E3 G.sup.3 L.sub.A44 R.sup.E4 G.sup.3 L.sub.A45 R.sup.E5 G.sup.3 L.sub.A46 R.sup.E6 G.sup.3 L.sub.A47 R.sup.E7 G.sup.3 L.sub.A48 R.sup.E8 G.sup.3 L.sub.A49 R.sup.E9 G.sup.3 L.sub.A50 R.sup.E10 G.sup.3 L.sub.A51 R.sup.E11 G.sup.3 L.sub.A52 R.sup.E12 G.sup.3 L.sub.A53 R.sup.E13 G.sup.3 L.sub.A54 R.sup.E14 G.sup.3 L.sub.A55 R.sup.E15 G.sup.3 L.sub.A56 R.sup.E16 G.sup.3 L.sub.A57 R.sup.E17 G.sup.3 L.sub.A58 R.sup.E15 G.sup.3 L.sub.A59 R.sup.E19 G.sup.3 L.sub.A60 R.sup.E20 G.sup.3 L.sub.A61 R.sup.E1 G.sup.4 L.sub.A62 R.sup.E2 G.sup.4 L.sub.A63 R.sup.E3 G.sup.4 L.sub.A64 R.sup.E4 G.sup.4 L.sub.A65 R.sup.E5 G.sup.4 L.sub.A66 R.sup.E6 G.sup.4 L.sub.A67 R.sup.E7 G.sup.4 L.sub.A68 R.sup.E8 G.sup.4 L.sub.A69 R.sup.E9 G.sup.4 L.sub.A70 R.sup.E10 G.sup.4 L.sub.A71 R.sup.E11 G.sup.4 L.sub.A72 R.sup.E12 G.sup.4 L.sub.A73 R.sup.E13 G.sup.4 L.sub.A74 R.sup.E14 G.sup.4 L.sub.A75 R.sup.E15 G.sup.4 L.sub.A76 R.sup.E16 G.sup.4 L.sub.A77 R.sup.E17 G.sup.4 L.sub.A78 R.sup.E18 G.sup.4 L.sub.A79 R.sup.E19 G.sup.4 L.sub.A80 R.sup.E20 G.sup.4 L.sub.A81 R.sup.E1 G.sup.5 L.sub.A82 R.sup.E2 G.sup.5 L.sub.A83 R.sup.E3 G.sup.5 L.sub.A84 R.sup.E4 G.sup.5 L.sub.A85 R.sup.E5 G.sup.5 L.sub.A86 R.sup.E6 G.sup.5 L.sub.A87 R.sup.E7 G.sup.5 L.sub.A88 R.sup.E8 G.sup.5 L.sub.A89 R.sup.E9 G.sup.5 L.sub.A90 R.sup.E10 G.sup.5 L.sub.A91 R.sup.E11 G.sup.5 L.sub.A92 R.sup.E12 G.sup.5 L.sub.A93 R.sup.E13 G.sup.5 L.sub.A94 R.sup.E14 G.sup.5 L.sub.A95 R.sup.E15 G.sup.5 L.sub.A96 R.sup.E16 G.sup.5 L.sub.A97 R.sup.E17 G.sup.5 L.sub.A98 R.sup.E18 G.sup.5 L.sub.A99 R.sup.E19 G.sup.5 L.sub.A100 R.sup.E20 G.sup.5 L.sub.A101 R.sup.E1 G.sup.6 L.sub.A102 R.sup.E2 G.sup.6 L.sub.A103 R.sup.E3 G.sup.6 L.sub.A104 R.sup.E4 G.sup.6 L.sub.A105 R.sup.E5 G.sup.6 L.sub.A106 R.sup.E6 G.sup.6 L.sub.A107 R.sup.E7 G.sup.6 L.sub.A108 R.sup.E8 G.sup.6 L.sub.A109 R.sup.E9 G.sup.6 L.sub.A110 R.sup.E10 G.sup.6 L.sub.A111 R.sup.E11 G.sup.6 L.sub.A112 R.sup.E12 G.sup.6

L.sub.A113 R.sup.E13 G.sup.6 L.sub.A114 R.sup.E14 G.sup.6 L.sub.A115 R.sup.E15 G.sup.6
L.sub.A116 R.sup.E16 G.sup.6 L.sub.A117 R.sup.E17 G.sup.6 L.sub.A118 R.sup.E18 G.sup.6
L.sub.A119 R.sup.E19 G.sup.6 L.sub.A120 R.sup.E20 G.sup.6 L.sub.A121 R.sup.E1 G.sup.7
L.sub.A122 R.sup.E2 G.sup.7 L.sub.A123 R.sup.E3 G.sup.7 L.sub.A124 R.sup.E4 G.sup.7
L.sub.A125 R.sup.E5 G.sup.7 L.sub.A126 R.sup.E6 G.sup.7 L.sub.A127 R.sup.E7 G.sup.7
L.sub.A128 R.sup.E8 G.sup.7 L.sub.A129 R.sup.E9 G.sup.7 L.sub.A130 R.sup.E10 G.sup.7
L.sub.A131 R.sup.E11 G.sup.7 L.sub.A132 R.sup.E12 G.sup.7 L.sub.A133 R.sup.E13 G.sup.7
L.sub.A134 R.sup.E14 G.sup.7 L.sub.A135 R.sup.E15 G.sup.7 L.sub.A136 R.sup.E16 G.sup.7
L.sub.A137 R.sup.E17 G.sup.7 L.sub.A138 R.sup.E18 G.sup.7 L.sub.A139 R.sup.E19 G.sup.7
L.sub.A140 R.sup.E20 G.sup.7 L.sub.A141 R.sup.E1 G.sup.8 L.sub.A142 R.sup.E2 G.sup.8
L.sub.A143 R.sup.E3 G.sup.8 L.sub.A144 R.sup.E4 G.sup.8 L.sub.A145 R.sup.E5 G.sup.8
L.sub.A146 R.sup.E6 G.sup.8 L.sub.A147 R.sup.E7 G.sup.8 L.sub.A148 R.sup.E8 G.sup.8
L.sub.A149 R.sup.E9 G.sup.8 L.sub.A150 R.sup.E10 G.sup.8 L.sub.A151 R.sup.E11 G.sup.8
L.sub.A152 R.sup.E32 G.sup.8 L.sub.A153 R.sup.E13 G.sup.8 L.sub.A154 R.sup.E14 G.sup.8
L.sub.A155 R.sup.E15 G.sup.8 L.sub.A156 R.sup.E16 G.sup.8 L.sub.A157 R.sup.E17 G.sup.8
L.sub.A158 R.sup.E15 G.sup.8 L.sub.A159 R.sup.E19 G.sup.8 L.sub.A160 R.sup.E20 G.sup.8
L.sub.A161 R.sup.E1 G.sup.9 L.sub.A162 R.sup.E2 G.sup.9 L.sub.A163 R.sup.E3 G.sup.9
L.sub.A164 R.sup.E4 G.sup.9 L.sub.A165 R.sup.E5 G.sup.9 L.sub.A166 R.sup.E6 G.sup.9
L.sub.A167 R.sup.E7 G.sup.9 L.sub.A168 R.sup.E8 G.sup.9 L.sub.A169 R.sup.E9 G.sup.9
L.sub.A170 R.sup.E10 G.sup.9 L.sub.A171 R.sup.E11 G.sup.9 L.sub.A172 R.sup.E12 G.sup.9
L.sub.A173 R.sup.E13 G.sup.9 L.sub.A174 R.sup.E14 G.sup.9 L.sub.A175 R.sup.E15 G.sup.9
L.sub.A176 R.sup.E16 G.sup.9 L.sub.A177 R.sup.E17 G.sup.9 L.sub.A178 R.sup.E18 G.sup.9
L.sub.A179 R.sup.E19 G.sup.9 L.sub.A180 R.sup.E20 G.sup.9 L.sub.A181 R.sup.E1 G.sup.10
L.sub.A182 R.sup.E2 G.sup.10 L.sub.A183 R.sup.E3 G.sup.10 L.sub.A184 R.sup.E4 G.sup.10
L.sub.A185 R.sup.E5 G.sup.10 L.sub.A186 R.sup.E6 G.sup.10 L.sub.A187 R.sup.E7 G.sup.10
L.sub.A188 R.sup.E8 G.sup.10 L.sub.A189 R.sup.E9 G.sup.10 L.sub.A190 R.sup.E10 G.sup.10
L.sub.A191 R.sup.E11 G.sup.10 L.sub.A192 R.sup.E12 G.sup.10 L.sub.A193 R.sup.E13 G.sup.10
L.sub.A194 R.sup.E14 G.sup.10 L.sub.A195 R.sup.E15 G.sup.10 L.sub.A196 R.sup.E16 G.sup.10
L.sub.A197 R.sup.E17 G.sup.10 L.sub.A198 R.sup.E18 G.sup.10 L.sub.A199 R.sup.E19 G.sup.10
L.sub.A200 R.sup.E20 G.sup.10 wherein R.sup.1 to R.sup.20 have the following structures:
##STR00171## ##STR00172## and wherein G.sup.1 to G.sup.10 have the following structures:
##STR00173## ##STR00174##

13. The compound of claim 1, wherein the ligand L.sub.A is selected from the group consisting of:
##STR00175## ##STR00176## ##STR00177## ##STR00178## ##STR00179## ##STR00180##
##STR00181## ##STR00182## ##STR00183## ##STR00184## ##STR00185## ##STR00186##
##STR00187## ##STR00188##

14. The compound of claim 12, wherein the compound has the formula of Ir(L.sub.Ai-m).sub.3, or
the formula of Ir(L.sub.Ai-m).sub.2(L.sub.Cj-n), wherein i is an integer from 61 to 200, m is an
integer from 1 to 20, j is an integer from 1 to 768, and n is a roman numeral from I to II, and the
compound is selected from the group consisting of Ir(L.sub.A61-1).sub.3 through Ir(L.sub.A200-
1).sub.3, Ir(L.sub.A61-2).sub.3 through Ir(L.sub.A200-2).sub.3, Ir(L.sub.A61-3).sub.3 through
Ir(L.sub.A200-3).sub.3, Ir(L.sub.A1-4).sub.3 through Ir(L.sub.A200-4).sub.3, Ir(L.sub.A61-
5).sub.3 through Ir(L.sub.A200-5).sub.3, Ir(L.sub.A61-6).sub.3 through Ir(L.sub.A200-6).sub.3,
Ir(L.sub.A1-7).sub.3 through Ir(L.sub.A200-7).sub.3, Ir(L.sub.A61-8).sub.3 through
Ir(L.sub.A200-8).sub.3, Ir(L.sub.A61-9).sub.3 through Ir(L.sub.A200-9).sub.3, Ir(L.sub.A61-
10).sub.3 through Ir(L.sub.A200-10).sub.3, Ir(L.sub.A1-11).sub.3 through Ir(L.sub.A200-
11).sub.3, Ir(L.sub.A61-12).sub.3 through Ir(L.sub.A200-12).sub.3, Ir(L.sub.A1-13).sub.3 through
Ir(L.sub.A200-13).sub.3, Ir(L.sub.A61-14).sub.3 through Ir(L.sub.A200-14).sub.3, Ir L.sub.A61-
15).sub.3 through Ir(L.sub.A200-15).sub.3, Ir(L.sub.A61-16).sub.3 through Ir(L.sub.A200-
16).sub.3, Ir(L.sub.A1-17).sub.3 through Ir(L.sub.A200-17).sub.3, Ir(L.sub.A61-18).sub.3 through

Ir(L.sub.A200-18).sub.3, Ir(L.sub.A61-19).sub.3 through Ir(L.sub.A200-19).sub.3, Ir(L.sub.A1-20).sub.3 through Ir(L.sub.A200-20).sub.3, Ir(L.sub.A61-1).sub.2 (L.sub.C1-1) through Ir(L.sub.A200-1).sub.2 (L.sub.C768-1), Ir(L.sub.A61-2).sub.2 (L.sub.C1-I) through Ir(L.sub.A200-2).sub.2(L.sub.C768-1), Ir(L.sub.A61-3).sub.2 (L.sub.C1-1) through Ir(L.sub.A200-3).sub.2 (L.sub.C768-1), Ir(L.sub.A1-4).sub.2 (L.sub.C1-1) through Ir(L.sub.A200-4).sub.2 (L.sub.C768-1), Ir(L.sub.A61-5).sub.2 (L.sub.C1-1) through Ir(L.sub.A200-5).sub.2 (L.sub.C768-1), Ir(L.sub.A61-6).sub.2 (L.sub.C1-1) through Ir(L.sub.A200-6).sub.2 (L.sub.C768-1), Ir(L.sub.A1-7).sub.2 (L.sub.C1-1) through Ir(L.sub.A200-7).sub.2 (L.sub.C768-1), Ir(L.sub.A61-8).sub.2 (L.sub.C1-1) through Ir(L.sub.A200-8).sub.2 (L.sub.C768-1), Ir(L.sub.A61-9).sub.2 (L.sub.C1-1) through Ir(L.sub.A200-9).sub.2 (L.sub.C768-1), Ir(L.sub.A61-10).sub.2 (L.sub.C1-1) through Ir(L.sub.A200-10).sub.2 (L.sub.C768-1), Ir(L.sub.A1-11).sub.2 (L.sub.C1-1) through Ir(L.sub.A200-11).sub.2 (L.sub.C768-1), Ir(L.sub.A61-12).sub.2 (L.sub.C1-1) through Ir(L.sub.A200-12).sub.2 (L.sub.C768-1), Ir(L.sub.A1-13).sub.2 (L.sub.C1-1) through Ir(L.sub.A200-13).sub.2 (L.sub.C768-1), Ir(L.sub.A61-14).sub.2 (L.sub.C1-1) through Ir(L.sub.A200-14).sub.2 (L.sub.C768-1), Ir(L.sub.A61-15).sub.2 (L.sub.C1-1) through Ir(L.sub.A200-15).sub.2 (L.sub.C768-1), Ir(L.sub.A61-16).sub.2 (L.sub.C1-1) through Ir(L.sub.A200-16).sub.2 (L.sub.C768-1), Ir(L.sub.A1-17).sub.2(L.sub.C1-1) through Ir(L.sub.A200-17).sub.2(L.sub.C768-1), Ir(L.sub.A61-18).sub.2(L.sub.C1-1) through Ir(L.sub.A200-18).sub.2(L.sub.C768-1), Ir(L.sub.A61-19).sub.2(L.sub.C1-1) through Ir(L.sub.A200-19).sub.2(L.sub.C768-1), Ir(L.sub.A1-20).sub.2(L.sub.C1-1) through Ir(L.sub.A200-20).sub.2(L.sub.C768-1), Ir(L.sub.A61-1).sub.2(L.sub.C1-II) through Ir(L.sub.A200-1).sub.2(L.sub.C768-II), Ir(L.sub.A61-2).sub.2(L.sub.C1-II) through Ir(L.sub.A200-2).sub.2(L.sub.C768-II), Ir(L.sub.A61-3).sub.2(L.sub.C1-II) through Ir(L.sub.A200-3).sub.2(L.sub.C768-II), Ir(L.sub.A1-4).sub.2(L.sub.C1-II) through Ir(L.sub.A200-4).sub.2(L.sub.C768-II), Ir(L.sub.A61-5).sub.2(L.sub.C1-II) through Ir(L.sub.A200-5).sub.2(L.sub.C768-II), Ir(L.sub.A61-6).sub.2 (L.sub.C1-II) through Ir(L.sub.A200-6).sub.2(L.sub.C768-II), Ir(L.sub.A1-7).sub.2(L.sub.C1-II) through Ir(L.sub.A200-7).sub.2(L.sub.C768-II), Ir(L.sub.A61-8).sub.2(L.sub.C1-II) through Ir(L.sub.A200-8).sub.2(L.sub.C768-II), Ir(L.sub.A61-9).sub.2(L.sub.C1-II) through Ir(L.sub.A200-9).sub.2(L.sub.C768-II), Ir(L.sub.A61-10).sub.2(L.sub.C1-II) through Ir(L.sub.A200-10).sub.2(L.sub.C768-II), Ir(L.sub.A1-11).sub.2(L.sub.C1-II) through Ir(L.sub.A200-11).sub.2(L.sub.C768-II), Ir(L.sub.A61-12).sub.2(L.sub.C1-II) through Ir(L.sub.A200-12).sub.2(L.sub.C768-II), Ir(L.sub.A1-13).sub.2(L.sub.C1-II) through Ir(L.sub.A200-13).sub.2(L.sub.C768-II), Ir(L.sub.A61-14).sub.2(L.sub.C1-II) through Ir(L.sub.A200-14).sub.2(L.sub.C768-II), Ir(L.sub.A61-15).sub.2(L.sub.C1-II) through Ir(L.sub.A200-15).sub.2(L.sub.C768-II), Ir(L.sub.A61-16).sub.2(L.sub.C1-II) through Ir(L.sub.A200-16).sub.2(L.sub.C768-11), Ir(L.sub.A1-17).sub.2(L.sub.C1-II) through Ir(L.sub.A200-17).sub.2(L.sub.C768-II), Ir(L.sub.A61-18).sub.2(L.sub.C1-II) through Ir(L.sub.A200-18).sub.2(L.sub.C768-II), Ir(L.sub.A61-19).sub.2(L.sub.C1-II) through Ir(L.sub.A200-19).sub.2(L.sub.C768-II), and Ir(L.sub.A1-20).sub.2(L.sub.C1-II) through Ir(L.sub.A200-20).sub.2(L.sub.C768-II), wherein L.sub.cj-I is based on a structure of ##STR00189## and L.sub.Cj-II is based on a structure of ##STR00190## wherein for each L.sub.Cj in L.sub.Cj-I and L.sub.Cj-II, R.sup.1 and R.sub.2 are as defined below: TABLE-US-00005 L.sub.Cj R.sup.1 R.sup.2 L.sub.Cjd R.sup.1 R.sup.2 L.sub.Cj R.sup.1 R.sup.2 L.sub.C1 R.sup.D1 R.sup.D1 L.sub.C193 R.sup.D1 R.sup.D3 L.sub.C385 R.sup.D17 R.sup.D40 L.sub.C577 R.sup.D143 R.sup.D120 L.sub.C2 R.sup.D2 R.sup.D2 L.sub.C194 R.sup.D1 R.sup.D4 L.sub.C386 R.sup.D17 R.sup.D41 L.sub.C578 R.sup.D143 R.sup.D133 L.sub.C3 R.sup.D3 R.sup.D3 L.sub.C195 R.sup.D1 R.sup.D5 L.sub.C387 R.sup.D17 R.sup.D42 L.sub.C579 R.sup.D143 R.sup.D134 L.sub.C4 R.sup.D4 R.sup.D4 L.sub.C196 R.sup.D1 R.sup.D9 L.sub.C388 R.sup.D17 R.sup.D43 L.sub.C580 R.sup.D143 R.sup.D135 L.sub.C5 R.sup.D5 R.sup.D5 L.sub.C197

R.sup.D1 R.sup.D10 L.sub.C389 R.sup.D17 R.sup.D58 L.sub.C581 R.sup.D143 R.sup.D136
L.sub.C6 R.sup.D6 R.sup.D6 L.sub.C198 R.sup.D1 R.sup.D17 L.sub.C390 R.sup.D17 R.sup.D49
L.sub.C582 R.sup.D143 R.sup.D144 L.sub.C7 R.sup.D7 R.sup.D7 L.sub.C199 R.sup.D1
R.sup.D18 L.sub.C391 R.sup.D17 R.sup.D50 L.sub.C583 R.sup.D143 R.sup.D145 L.sub.C8
R.sup.D8 R.sup.D8 L.sub.C200 R.sup.D1 R.sup.D20 L.sub.C392 R.sup.D17 R.sup.D54
L.sub.C584 R.sup.D143 R.sup.D146 L.sub.C9 R.sup.D9 R.sup.D9 L.sub.C201 R.sup.D1
R.sup.D22 L.sub.C393 R.sup.D17 R.sup.D55 L.sub.C585 R.sup.D143 R.sup.D147 L.sub.C10
R.sup.D10 R.sup.D10 L.sub.C202 R.sup.D1 R.sup.D37 L.sub.C394 R.sup.D17 R.sup.D58
L.sub.C586 R.sup.D143 R.sup.D149 L.sub.C11 R.sup.D11 R.sup.D11 L.sub.C203 R.sup.D1
R.sup.D40 L.sub.C395 R.sup.D17 R.sup.D59 L.sub.C587 R.sup.D143 R.sup.D151 L.sub.C12
R.sup.D12 R.sup.D12 L.sub.C204 R.sup.D1 R.sup.D41 L.sub.C396 R.sup.D17 R.sup.D78
L.sub.C588 R.sup.D143 R.sup.D154 L.sub.C13 R.sup.D13 R.sup.D13 L.sub.C205 R.sup.D1
R.sup.D42 L.sub.C397 R.sup.D17 R.sup.D79 L.sub.C589 R.sup.D143 R.sup.D155 L.sub.C14
R.sup.D14 R.sup.D14 L.sub.C206 R.sup.D1 R.sup.D43 L.sub.C398 R.sup.D17 R.sup.D81
L.sub.C590 R.sup.D143 R.sup.D161 L.sub.C15 R.sup.D15 R.sup.D15 L.sub.C207 R.sup.D1
R.sup.D48 L.sub.C399 R.sup.D17 R.sup.D87 L.sub.C591 R.sup.D143 R.sup.D175 L.sub.C16
R.sup.D16 R.sup.D16 L.sub.C208 R.sup.D1 R.sup.D49 L.sub.C400 R.sup.D17 R.sup.D88
L.sub.C592 R.sup.D144 R.sup.D3 L.sub.C17 R.sup.D17 R.sup.D17 L.sub.C209 R.sup.D1
R.sup.D50 L.sub.C401 R.sup.D17 R.sup.D89 L.sub.C593 R.sup.D144 R.sup.D5 L.sub.C18
R.sup.D18 R.sup.D18 L.sub.C210 R.sup.D1 R.sup.D54 L.sub.C402 R.sup.D17 R.sup.D93
L.sub.C594 R.sup.D144 R.sup.D17 L.sub.C19 R.sup.D19 R.sup.D19 L.sub.C211 R.sup.D1
R.sup.D55 L.sub.C403 R.sup.D17 R.sup.D116 L.sub.C595 R.sup.D144 R.sup.D18 L.sub.C20
R.sup.D20 R.sup.D20 L.sub.C212 R.sup.D1 R.sup.D58 L.sub.C404 R.sup.D17 R.sup.D117
L.sub.C596 R.sup.D144 R.sup.D20 L.sub.C21 R.sup.D21 R.sup.D21 L.sub.C213 R.sup.D1
R.sup.D59 L.sub.C405 R.sup.D17 R.sup.D118 L.sub.C597 R.sup.D144 R.sup.D22 L.sub.C22
R.sup.D22 R.sup.D22 L.sub.C214 R.sup.D1 R.sup.D78 L.sub.C406 R.sup.D17 R.sup.D119
L.sub.C598 R.sup.D144 R.sup.D37 L.sub.C23 R.sup.D23 R.sup.D23 L.sub.C215 R.sup.D1
R.sup.D79 L.sub.C407 R.sup.D17 R.sup.D120 L.sub.C599 R.sup.D144 R.sup.D40 L.sub.C24
R.sup.D24 R.sup.D24 L.sub.C216 R.sup.D1 R.sup.D81 L.sub.C408 R.sup.D17 R.sup.D133
L.sub.C600 R.sup.D144 R.sup.D41 L.sub.C25 R.sup.D25 R.sup.D25 L.sub.C217 R.sup.D1
R.sup.D87 L.sub.C409 R.sup.D17 R.sup.D134 L.sub.C601 R.sup.D144 R.sup.D42 L.sub.C26
R.sup.D26 R.sup.D26 L.sub.C218 R.sup.D1 R.sup.D88 L.sub.C410 R.sup.D17 R.sup.D135
L.sub.C602 R.sup.D144 R.sup.D43 L.sub.C27 R.sup.D27 R.sup.D27 L.sub.C219 R.sup.D1
R.sup.D89 L.sub.C411 R.sup.D17 R.sup.D136 L.sub.C603 R.sup.D144 R.sup.D48 L.sub.C28
R.sup.D28 R.sup.D28 L.sub.C220 R.sup.D1 R.sup.D93 L.sub.C412 R.sup.D17 R.sup.D143
L.sub.C604 R.sup.D144 R.sup.D49 L.sub.C29 R.sup.D29 R.sup.D29 L.sub.C221 R.sup.D1
R.sup.D116 L.sub.C413 R.sup.D17 R.sup.D144 L.sub.C605 R.sup.D144 R.sup.D54 L.sub.C30
R.sup.D30 R.sup.D30 L.sub.C222 R.sup.D1 R.sup.D117 L.sub.C414 R.sup.D17 R.sup.D145
L.sub.C606 R.sup.D144 R.sup.D58 L.sub.C31 R.sup.D31 R.sup.D31 L.sub.C223 R.sup.D1
R.sup.D118 L.sub.C415 R.sup.D17 R.sup.D146 L.sub.C607 R.sup.D144 R.sup.D59 L.sub.C32
R.sup.D32 R.sup.D32 L.sub.C224 R.sup.D1 R.sup.D119 L.sub.C416 R.sup.D17 R.sup.D147
L.sub.C608 R.sup.D144 R.sup.D78 L.sub.C33 R.sup.D33 R.sup.D33 L.sub.C225 R.sup.D1
R.sup.D120 L.sub.C417 R.sup.D17 R.sup.D149 L.sub.C609 R.sup.D144 R.sup.D79 L.sub.C34
R.sup.D34 R.sup.D34 L.sub.C226 R.sup.D1 R.sup.D133 L.sub.C418 R.sup.D17 R.sup.D151
L.sub.C610 R.sup.D144 R.sup.D81 L.sub.C35 R.sup.D35 R.sup.D35 L.sub.C227 R.sup.D1
R.sup.D134 L.sub.C419 R.sup.D17 R.sup.D154 L.sub.C611 R.sup.D144 R.sup.D87 L.sub.C36
R.sup.D36 R.sup.D36 L.sub.C228 R.sup.D1 R.sup.D135 L.sub.C420 R.sup.D17 R.sup.D155
L.sub.C612 R.sup.D144 R.sup.D88 L.sub.C37 R.sup.D37 R.sup.D37 L.sub.C229 R.sup.D1
R.sup.D136 L.sub.C421 R.sup.D17 R.sup.D161 L.sub.C613 R.sup.D144 R.sup.D89 L.sub.C38
R.sup.D38 R.sup.D38 L.sub.C230 R.sup.D1 R.sup.D143 L.sub.C422 R.sup.D17 R.sup.D175

L.sub.C614 R.sup.D144 R.sup.D93 L.sub.C39 R.sup.D39 R.sup.D39 L.sub.C231 R.sup.D1
R.sup.D144 L.sub.C423 R.sup.D50 R.sup.D3 L.sub.C615 R.sup.D144 R.sup.D116 L.sub.C40
R.sup.D40 R.sup.D40 L.sub.C232 R.sup.D1 R.sup.D145 L.sub.C424 R.sup.D50 R.sup.D5
L.sub.C616 R.sup.D144 R.sup.D117 L.sub.C41 R.sup.D41 R.sup.D41 L.sub.C233 R.sup.D1
R.sup.D146 L.sub.C425 R.sup.D50 R.sup.D18 L.sub.C617 R.sup.D144 R.sup.D118 L.sub.C42
R.sup.D42 R.sup.D42 L.sub.C234 R.sup.D1 R.sup.D147 L.sub.C426 R.sup.D50 R.sup.D20
L.sub.C618 R.sup.D144 R.sup.D119 L.sub.C43 R.sup.D43 R.sup.D43 L.sub.C235 R.sup.D1
R.sup.D149 L.sub.C427 R.sup.D50 R.sup.D22 L.sub.C619 R.sup.D144 R.sup.D120 L.sub.C44
R.sup.D44 R.sup.D44 L.sub.C236 R.sup.D1 R.sup.D151 L.sub.C428 R.sup.D50 R.sup.D37
L.sub.C620 R.sup.D144 R.sup.D133 L.sub.C45 R.sup.D45 R.sup.D45 L.sub.C237 R.sup.D1
R.sup.D154 L.sub.C429 R.sup.D50 R.sup.D40 L.sub.C621 R.sup.D144 R.sup.D134 L.sub.C46
R.sup.D46 R.sup.D46 L.sub.C238 R.sup.D1 R.sup.D155 L.sub.C430 R.sup.D50 R.sup.D41
L.sub.C622 R.sup.D144 R.sup.D135 L.sub.C47 R.sup.D47 R.sup.D47 L.sub.C239 R.sup.D1
R.sup.D161 L.sub.C431 R.sup.D50 R.sup.D42 L.sub.C623 R.sup.D144 R.sup.D136 L.sub.C48
R.sup.D48 R.sup.D48 L.sub.C240 R.sup.D1 R.sup.D175 L.sub.C432 R.sup.D50 R.sup.D43
L.sub.C624 R.sup.D144 R.sup.D145 L.sub.C49 R.sup.D49 R.sup.D49 L.sub.C241 R.sup.D4
R.sup.D3 L.sub.C433 R.sup.D50 R.sup.D48 L.sub.C625 R.sup.D144 R.sup.D146 L.sub.C50
R.sup.D50 R.sup.D50 L.sub.C242 R.sup.D4 R.sup.D5 L.sub.C434 R.sup.D50 R.sup.D49
L.sub.C626 R.sup.D144 R.sup.D147 L.sub.C51 R.sup.D51 R.sup.D51 L.sub.C243 R.sup.D4
R.sup.D9 L.sub.C435 R.sup.D50 R.sup.D54 L.sub.C627 R.sup.D144 R.sup.D149 L.sub.C52
R.sup.D52 R.sup.D52 L.sub.C244 R.sup.D4 R.sup.D10 L.sub.C436 R.sup.D50 R.sup.D55
L.sub.C628 R.sup.D144 R.sup.D151 L.sub.C53 R.sup.D53 R.sup.D53 L.sub.C245 R.sup.D4
R.sup.D17 L.sub.C437 R.sup.D50 R.sup.D58 L.sub.C629 R.sup.D144 R.sup.D154 L.sub.C54
R.sup.D54 R.sup.D54 L.sub.C246 R.sup.D4 R.sup.D18 L.sub.C438 R.sup.D50 R.sup.D59
L.sub.C630 R.sup.D144 R.sup.D155 L.sub.C55 R.sup.D55 R.sup.D55 L.sub.C247 R.sup.D4
R.sup.D20 L.sub.C439 R.sup.D50 R.sup.D78 L.sub.C631 R.sup.D144 R.sup.D161 L.sub.C56
R.sup.D56 R.sup.D56 L.sub.C248 R.sup.D4 R.sup.D22 L.sub.C440 R.sup.D50 R.sup.D79
L.sub.C632 R.sup.D144 R.sup.D175 L.sub.C57 R.sup.D57 R.sup.D57 L.sub.C249 R.sup.D4
R.sup.D37 L.sub.C441 R.sup.D50 R.sup.D81 L.sub.C633 R.sup.D145 R.sup.D3 L.sub.C58
R.sup.D58 R.sup.D58 L.sub.C250 R.sup.D4 R.sup.D40 L.sub.C442 R.sup.D50 R.sup.D87
L.sub.C634 R.sup.D145 R.sup.D5 L.sub.C59 R.sup.D59 R.sup.D59 L.sub.C251 R.sup.D4
R.sup.D41 L.sub.C443 R.sup.D50 R.sup.D88 L.sub.C635 R.sup.D145 R.sup.D17 L.sub.C60
R.sup.D60 R.sup.D60 L.sub.C252 R.sup.D4 R.sup.D42 L.sub.C444 R.sup.D50 R.sup.D89
L.sub.C636 R.sup.D145 R.sup.D18 L.sub.C61 R.sup.D61 R.sup.D61 L.sub.C253 R.sup.D4
R.sup.D43 L.sub.C445 R.sup.D50 R.sup.D93 L.sub.C637 R.sup.D145 R.sup.D20 L.sub.C62
R.sup.D62 R.sup.D62 L.sub.C254 R.sup.D4 R.sup.D48 L.sub.C446 R.sup.D50 R.sup.D116
L.sub.C638 R.sup.D145 R.sup.D22 L.sub.C63 R.sup.D63 R.sup.D63 L.sub.C255 R.sup.D4
R.sup.D49 L.sub.C447 R.sup.D50 R.sup.D117 L.sub.C639 R.sup.D145 R.sup.D37 L.sub.C64
R.sup.D64 R.sup.D64 L.sub.C256 R.sup.D4 R.sup.D50 L.sub.C448 R.sup.D50 R.sup.D118
L.sub.C640 R.sup.D145 R.sup.D40 L.sub.C65 R.sup.D65 R.sup.D65 L.sub.C257 R.sup.D4
R.sup.D54 L.sub.C449 R.sup.D50 R.sup.D119 L.sub.C641 R.sup.D145 R.sup.D41 L.sub.C66
R.sup.D66 R.sup.D66 L.sub.C258 R.sup.D4 R.sup.D55 L.sub.C450 R.sup.D50 R.sup.D120
L.sub.C642 R.sup.D145 R.sup.D42 L.sub.C67 R.sup.D67 R.sup.D67 L.sub.C259 R.sup.D4
R.sup.D58 L.sub.C451 R.sup.D50 R.sup.D133 L.sub.C643 R.sup.D145 R.sup.D43 L.sub.C68
R.sup.D68 R.sup.D68 L.sub.C260 R.sup.D4 R.sup.D59 L.sub.C452 R.sup.D50 R.sup.D134
L.sub.C644 R.sup.D145 R.sup.D48 L.sub.C69 R.sup.D69 R.sup.D69 L.sub.C261 R.sup.D4
R.sup.D78 L.sub.C453 R.sup.D50 R.sup.D135 L.sub.C645 R.sup.D145 R.sup.D49 L.sub.C70
R.sup.D70 R.sup.D70 L.sub.C262 R.sup.D4 R.sup.D79 L.sub.C454 R.sup.D50 R.sup.D136
L.sub.C646 R.sup.D145 R.sup.D54 L.sub.C71 R.sup.D71 R.sup.D71 L.sub.C263 R.sup.D4
R.sup.D81 L.sub.C455 R.sup.D50 R.sup.D143 L.sub.C647 R.sup.D145 R.sup.D58 L.sub.C72

R.sup.D72 R.sup.D72 L.sub.C264 R.sup.D4 R.sup.D57 L.sub.C456 R.sup.D50 R.sup.D144
L.sub.C648 R.sup.D145 R.sup.D59 L.sub.C73 R.sup.D73 R.sup.D73 L.sub.C265 R.sup.D4
R.sup.D88 L.sub.C457 R.sup.D50 R.sup.D145 L.sub.C649 R.sup.D145 R.sup.D78 L.sub.C74
R.sup.D74 R.sup.D74 L.sub.C266 R.sup.D4 R.sup.D89 L.sub.C458 R.sup.D50 R.sup.D146
L.sub.C650 R.sup.D145 R.sup.D79 L.sub.C75 R.sup.D75 R.sup.D75 L.sub.C267 R.sup.D4
R.sup.D93 L.sub.C459 R.sup.D50 R.sup.D147 L.sub.C651 R.sup.D145 R.sup.D81 L.sub.C76
R.sup.D76 R.sup.D76 L.sub.C268 R.sup.D4 R.sup.D116 L.sub.C460 R.sup.D50 R.sup.D149
L.sub.C652 R.sup.D145 R.sup.D87 L.sub.C77 R.sup.D77 R.sup.D77 L.sub.C269 R.sup.D4
R.sup.D117 L.sub.C461 R.sup.D50 R.sup.D151 L.sub.C653 R.sup.D145 R.sup.D88 L.sub.C78
R.sup.D78 R.sup.D78 L.sub.C270 R.sup.D4 R.sup.D118 L.sub.C462 R.sup.D50 R.sup.D154
L.sub.C654 R.sup.D145 R.sup.D89 L.sub.C79 R.sup.D79 R.sup.D79 L.sub.C271 R.sup.D4
R.sup.D119 L.sub.C463 R.sup.D50 R.sup.D155 L.sub.C655 R.sup.D145 R.sup.D93 L.sub.C80
R.sup.D80 R.sup.D80 L.sub.C272 R.sup.D4 R.sup.D120 L.sub.C464 R.sup.D50 R.sup.D161
L.sub.C656 R.sup.D145 R.sup.D116 L.sub.C81 R.sup.D81 R.sup.D81 L.sub.C273 R.sup.D4
R.sup.D133 L.sub.C465 R.sup.D50 R.sup.D175 L.sub.C657 R.sup.D145 R.sup.D117 L.sub.C82
R.sup.D82 R.sup.D82 L.sub.C274 R.sup.D4 R.sup.D134 L.sub.C466 R.sup.D55 R.sup.D3
L.sub.C658 R.sup.D145 R.sup.D118 L.sub.C83 R.sup.D83 R.sup.D83 L.sub.C275 R.sup.D4
R.sup.D135 L.sub.C467 R.sup.D55 R.sup.D5 L.sub.C659 R.sup.D145 R.sup.D119 L.sub.C84
R.sup.D84 R.sup.D84 L.sub.C276 R.sup.D4 R.sup.D136 L.sub.C468 R.sup.D55 R.sup.D18
L.sub.C660 R.sup.D145 R.sup.D120 L.sub.C85 R.sup.D85 R.sup.D85 L.sub.C277 R.sup.D4
R.sup.D143 L.sub.C469 R.sup.D55 R.sup.D20 L.sub.C661 R.sup.D145 R.sup.D133 L.sub.C86
R.sup.D86 R.sup.D86 L.sub.C278 R.sup.D4 R.sup.D144 L.sub.C470 R.sup.D55 R.sup.D22
L.sub.C662 R.sup.D145 R.sup.D134 L.sub.C87 R.sup.D87 R.sup.D87 L.sub.C279 R.sup.D4
R.sup.D145 L.sub.C471 R.sup.D55 R.sup.D37 L.sub.C663 R.sup.D145 R.sup.D135 L.sub.C88
R.sup.D88 R.sup.D88 L.sub.C280 R.sup.D4 R.sup.D146 L.sub.C472 R.sup.D55 R.sup.D40
L.sub.C664 R.sup.D145 R.sup.D136 L.sub.C89 R.sup.D89 R.sup.D89 L.sub.C281 R.sup.D4
R.sup.D147 L.sub.C473 R.sup.D55 R.sup.D41 L.sub.C665 R.sup.D145 R.sup.D146 L.sub.C90
R.sup.D90 R.sup.D90 L.sub.C282 R.sup.D4 R.sup.D149 L.sub.C474 R.sup.D55 R.sup.D42
L.sub.C666 R.sup.D145 R.sup.D147 L.sub.C91 R.sup.D91 R.sup.D91 L.sub.C283 R.sup.D4
R.sup.D151 L.sub.C475 R.sup.D55 R.sup.D43 L.sub.C667 R.sup.D145 R.sup.D149 L.sub.C92
R.sup.D92 R.sup.D92 L.sub.C284 R.sup.D4 R.sup.D154 L.sub.C476 R.sup.D55 R.sup.D48
L.sub.C668 R.sup.D145 R.sup.D151 L.sub.C93 R.sup.D93 R.sup.D93 L.sub.C285 R.sup.D4
R.sup.D155 L.sub.C477 R.sup.D55 R.sup.D49 L.sub.C669 R.sup.D145 R.sup.D154 L.sub.C94
R.sup.D94 R.sup.D94 L.sub.C286 R.sup.D4 R.sup.D161 L.sub.C478 R.sup.D55 R.sup.D54
L.sub.C670 R.sup.D145 R.sup.D155 L.sub.C95 R.sup.D95 R.sup.D95 L.sub.C287 R.sup.D4
R.sup.D175 L.sub.C479 R.sup.D55 R.sup.D58 L.sub.C671 R.sup.D145 R.sup.D161 L.sub.C96
R.sup.D96 R.sup.D96 L.sub.C288 R.sup.D9 R.sup.D3 L.sub.C480 R.sup.D55 R.sup.D59
L.sub.C672 R.sup.D145 R.sup.D175 L.sub.C97 R.sup.D97 R.sup.D97 L.sub.C289 R.sup.D9
R.sup.D5 L.sub.C481 R.sup.D55 R.sup.D78 L.sub.C673 R.sup.D146 R.sup.D3 L.sub.C98
R.sup.D98 R.sup.D98 L.sub.C290 R.sup.D9 R.sup.D10 L.sub.C482 R.sup.D55 R.sup.D79
L.sub.C674 R.sup.D146 R.sup.D5 L.sub.C99 R.sup.D99 R.sup.D99 L.sub.C291 R.sup.D9
R.sup.D17 L.sub.C483 R.sup.D55 R.sup.D81 L.sub.C675 R.sup.D146 R.sup.D17 L.sub.C100
R.sup.D100 R.sup.D100 L.sub.C292 R.sup.D9 R.sup.D18 L.sub.C484 R.sup.D55 R.sup.D87
L.sub.C676 R.sup.D146 R.sup.D18 L.sub.C101 R.sup.D101 R.sup.D101 L.sub.C293 R.sup.D9
R.sup.D20 L.sub.C485 R.sup.D55 R.sup.D88 L.sub.C677 R.sup.D146 R.sup.D20 L.sub.C102
R.sup.D102 R.sup.D102 L.sub.C294 R.sup.D9 R.sup.D22 L.sub.C486 R.sup.D55 R.sup.D89
L.sub.C678 R.sup.D146 R.sup.D22 L.sub.C103 R.sup.D103 R.sup.D103 L.sub.C295 R.sup.D9
R.sup.D37 L.sub.C487 R.sup.D55 R.sup.D93 L.sub.C679 R.sup.D146 R.sup.D37 L.sub.C104
R.sup.D104 R.sup.D104 L.sub.C296 R.sup.D9 R.sup.D40 L.sub.C488 R.sup.D55 R.sup.D116
L.sub.C680 R.sup.D146 R.sup.D40 L.sub.C105 R.sup.D105 R.sup.D105 L.sub.C297 R.sup.D9

R.sup.D41 L.sub.C489 R.sup.D55 R.sup.D117 L.sub.C581 R.sup.D146 R.sup.D141 L.sub.C106
R.sup.D106 R.sup.D106 L.sub.C298 R.sup.D9 R.sup.D42 L.sub.C490 R.sup.D55 R.sup.D118
L.sub.C682 R.sup.D146 R.sup.D42 L.sub.C107 R.sup.D107 R.sup.D107 L.sub.C299 R.sup.D9
R.sup.D43 L.sub.C491 R.sup.D55 R.sup.D119 L.sub.C683 R.sup.D146 R.sup.D43 L.sub.C108
R.sup.D108 R.sup.D108 L.sub.C300 R.sup.D9 R.sup.D48 L.sub.C492 R.sup.D55 R.sup.D120
L.sub.C684 R.sup.D146 R.sup.D48 L.sub.C109 R.sup.D109 R.sup.D109 L.sub.C301 R.sup.D9
R.sup.D49 L.sub.C493 R.sup.D55 R.sup.D133 L.sub.C685 R.sup.D146 R.sup.D49 L.sub.C110
R.sup.D110 R.sup.D110 L.sub.C302 R.sup.D9 R.sup.D50 L.sub.C494 R.sup.D55 R.sup.D134
L.sub.C686 R.sup.D146 R.sup.D54 L.sub.C111 R.sup.D111 R.sup.D111 L.sub.C303 R.sup.D9
R.sup.D54 L.sub.C495 R.sup.D55 R.sup.D135 L.sub.C687 R.sup.D146 R.sup.D58 L.sub.C112
R.sup.D112 R.sup.D112 L.sub.C304 R.sup.D9 R.sup.D55 L.sub.C496 R.sup.D55 R.sup.D136
L.sub.C688 R.sup.D146 R.sup.D59 L.sub.C113 R.sup.D113 R.sup.D113 L.sub.C305 R.sup.D9
R.sup.D58 L.sub.C497 R.sup.D55 R.sup.D143 L.sub.C689 R.sup.D146 R.sup.D78 L.sub.C114
R.sup.D114 R.sup.D114 L.sub.C306 R.sup.D9 R.sup.D59 L.sub.C498 R.sup.D55 R.sup.D144
L.sub.C690 R.sup.D146 R.sup.D79 L.sub.C115 R.sup.D115 R.sup.D115 L.sub.C307 R.sup.D9
R.sup.D78 L.sub.C499 R.sup.D55 R.sup.D145 L.sub.C691 R.sup.D146 R.sup.D81 L.sub.C116
R.sup.D116 R.sup.D116 L.sub.C308 R.sup.D9 R.sup.D79 L.sub.C500 R.sup.D55 R.sup.D146
L.sub.C692 R.sup.D146 R.sup.D87 L.sub.C117 R.sup.D117 R.sup.D177 L.sub.C309 R.sup.D9
R.sup.D81 L.sub.C501 R.sup.D55 R.sup.D147 L.sub.C693 R.sup.D146 R.sup.D88 L.sub.C118
R.sup.D118 R.sup.D118 L.sub.C310 R.sup.D9 R.sup.D87 L.sub.C502 R.sup.D55 R.sup.D149
L.sub.C694 R.sup.D146 R.sup.D89 L.sub.C119 R.sup.D119 R.sup.D119 L.sub.C311 R.sup.D9
R.sup.D88 L.sub.C503 R.sup.D55 R.sup.D151 L.sub.C695 R.sup.D146 R.sup.D93 L.sub.C120
R.sup.D120 R.sup.D120 L.sub.C312 R.sup.D9 R.sup.D89 L.sub.C504 R.sup.D55 R.sup.D154
L.sub.C696 R.sup.D146 R.sup.D117 L.sub.C121 R.sup.D121 R.sup.D121 L.sub.C313 R.sup.D9
R.sup.D93 L.sub.C505 R.sup.D55 R.sup.D155 L.sub.C697 R.sup.D146 R.sup.D118 L.sub.C122
R.sup.D122 R.sup.D122 L.sub.C314 R.sup.D9 R.sup.D116 L.sub.C506 R.sup.D55 R.sup.D161
L.sub.C698 R.sup.D146 R.sup.D119 L.sub.C123 R.sup.D123 R.sup.D123 L.sub.C315 R.sup.D9
R.sup.D117 L.sub.C507 R.sup.D55 R.sup.D175 L.sub.C699 R.sup.D146 R.sup.D120 L.sub.C124
R.sup.D124 R.sup.D124 L.sub.C316 R.sup.D9 R.sup.D118 L.sub.C508 R.sup.D116 R.sup.D3
L.sub.C700 R.sup.D146 R.sup.D133 L.sub.C125 R.sup.D125 R.sup.D125 L.sub.C317 R.sup.D9
R.sup.D119 L.sub.C509 R.sup.D116 R.sup.D5 L.sub.C701 R.sup.D146 R.sup.D134 L.sub.C126
R.sup.D126 R.sup.D126 L.sub.C318 R.sup.D9 R.sup.D120 L.sub.C510 R.sup.D116 R.sup.D17
L.sub.C702 R.sup.D146 R.sup.D135 L.sub.C127 R.sup.D127 R.sup.D127 L.sub.C319 R.sup.D9
R.sup.D133 L.sub.C511 R.sup.D116 R.sup.D18 L.sub.C703 R.sup.D146 R.sup.D136 L.sub.C128
R.sup.D128 R.sup.D128 L.sub.C320 R.sup.D9 R.sup.D134 L.sub.C512 R.sup.D116 R.sup.D20
L.sub.C704 R.sup.D146 R.sup.D146 L.sub.C129 R.sup.D129 R.sup.D129 L.sub.C321 R.sup.D9
R.sup.D135 L.sub.C513 R.sup.D116 R.sup.D22 L.sub.C705 R.sup.D146 R.sup.D147 L.sub.C130
R.sup.D130 R.sup.D130 L.sub.C322 R.sup.D9 R.sup.D136 L.sub.C514 R.sup.D116 R.sup.D37
L.sub.C706 R.sup.D146 R.sup.D149 L.sub.C131 R.sup.D131 R.sup.D131 L.sub.C323 R.sup.D9
R.sup.D143 L.sub.C515 R.sup.D116 R.sup.D40 L.sub.C707 R.sup.D146 R.sup.D151 L.sub.C132
R.sup.D132 R.sup.D132 L.sub.C324 R.sup.D9 R.sup.D144 L.sub.C516 R.sup.D116 R.sup.D41
L.sub.C708 R.sup.D146 R.sup.D154 L.sub.C133 R.sup.D133 R.sup.D133 L.sub.C325 R.sup.D9
R.sup.D145 L.sub.C517 R.sup.D116 R.sup.D42 L.sub.C709 R.sup.D146 R.sup.D155 L.sub.C134
R.sup.D134 R.sup.D134 L.sub.C326 R.sup.D9 R.sup.D146 L.sub.C518 R.sup.D116 R.sup.D43
L.sub.C710 R.sup.D146 R.sup.D161 L.sub.C135 R.sup.D135 R.sup.D135 L.sub.C327 R.sup.D9
R.sup.D147 L.sub.C519 R.sup.D116 R.sup.D48 L.sub.C711 R.sup.D146 R.sup.D175 L.sub.C136
R.sup.D136 R.sup.D136 L.sub.C328 R.sup.D9 R.sup.D149 L.sub.C520 R.sup.D116 R.sup.D49
L.sub.C712 R.sup.D133 R.sup.D3 L.sub.C137 R.sup.D137 R.sup.D137 L.sub.C329 R.sup.D9
R.sup.D151 L.sub.C521 R.sup.D116 R.sup.D54 L.sub.C713 R.sup.D133 R.sup.D5 L.sub.C138
R.sup.D138 R.sup.D138 L.sub.C330 R.sup.D9 R.sup.D154 L.sub.C522 R.sup.D116 R.sup.D58

L.sub.C714 R.sup.D139 R.sup.D3 L.sub.C139 R.sup.D139 L.sub.C331 R.sup.D9
R.sup.D155 L.sub.C523 R.sup.D116 R .sup.D59 L.sub.C715 R.sup.D133 R.sup.D18 L.sub.C140
R.sup.D140 R.sup.D140 L.sub.C332 R.sup.D9 R.sup.D161 L.sub.C524 R.sup.D116 R.sup.D78
L.sub.C716 R.sup.D133 R.sup.D20 L.sub.C141 R.sup.D141 R.sup.D141 L.sub.C333 R.sup.D9
R.sup.D175 L.sub.C525 R.sup.D116 R.sup.D79 L.sub.C717 R.sup.D133 R.sup.D22 L.sub.C142
R.sup.D142 R.sup.D142 L.sub.C334 R.sup.D10 R.sup.D3 L.sub.C526 R.sup.D116 R.sup.D81
L.sub.C718 R.sup.D133 R.sup.D37 L.sub.C143 R.sup.D143 R.sup.D143 L.sub.C335 R.sup.D10
R.sup.D5 L.sub.C527 R.sup.D116 R.sup.D87 L.sub.C719 R.sup.D133 R.sup.D40 L.sub.C144
R.sup.D144 R.sup.D144 L.sub.C336 R.sup.D10 R.sup.D17 L.sub.C528 R.sup.D116 R.sup.D88
L.sub.C720 R.sup.D133 R.sup.D41 L.sub.C145 R.sup.D145 R.sup.D145 L.sub.C337 R.sup.D10
R.sup.D18 L.sub.C529 R.sup.D116 R.sup.D89 L.sub.C721 R.sup.D133 R.sup.D42 L.sub.C146
R.sup.D146 R.sup.D146 L.sub.C338 R.sup.D10 R.sup.D20 L.sub.C530 R.sup.D116 R.sup.D93
L.sub.C722 R.sup.D133 R.sup.D43 L.sub.C147 R.sup.D147 R.sup.D147 L.sub.C339 R.sup.D10
R.sup.D22 L.sub.C531 R.sup.D116 R.sup.D117 L.sub.C723 R.sup.D133 R.sup.D48 L.sub.C148
R.sup.D148 R.sup.D148 L.sub.C340 R.sup.D10 R.sup.D37 L.sub.C532 R.sup.D116 R.sup.D118
L.sub.C724 R.sup.D133 R.sup.D49 L.sub.C149 R.sup.D149 R.sup.D149 L.sub.C341 R.sup.D10
R.sup.D40 L.sub.C533 R.sup.D116 R.sup.D119 L.sub.C725 R.sup.D133 R.sup.D54 L.sub.C150
R.sup.D150 R.sup.D150 L.sub.C342 R.sup.D10 R.sup.D41 L.sub.C534 R.sup.D116 R.sup.D120
L.sub.C726 R.sup.D133 R.sup.D58 L.sub.C151 R.sup.D151 R.sup.D151 L.sub.C343 R.sup.D10
R.sup.D42 L.sub.C535 R.sup.D116 R.sup.D133 L.sub.C727 R.sup.D133 R.sup.D59 L.sub.C152
R.sup.D152 R.sup.D152 L.sub.C344 R.sup.D10 R.sup.D43 L.sub.C536 R.sup.D116 R.sup.D134
L.sub.C728 R.sup.D133 R.sup.D78 L.sub.C153 R.sup.D153 R.sup.D153 L.sub.C345 R.sup.D10
R.sup.D48 L.sub.C537 R.sup.D116 R.sup.D135 L.sub.C729 R.sup.D133 R.sup.D79 L.sub.C154
R.sup.D154 R.sup.D154 L.sub.C346 R.sup.D10 R.sup.D49 L.sub.C538 R.sup.D116 R.sup.D136
L.sub.C730 R.sup.D133 R.sup.D81 L.sub.C155 R.sup.D155 R.sup.D155 L.sub.C347 R.sup.D10
R.sup.D50 L.sub.C539 R.sup.D116 R.sup.D143 L.sub.C731 R.sup.D133 R.sup.D87 L.sub.C156
R.sup.D156 R.sup.D156 L.sub.C348 R.sup.D10 R.sup.D54 L.sub.C540 R.sup.D116 R.sup.D144
L.sub.C732 R.sup.D133 R.sup.D88 L.sub.C157 R.sup.D157 R.sup.D157 L.sub.C349 R.sup.D10
R.sup.D55 L.sub.C541 R.sup.D116 R.sup.D145 L.sub.C733 R.sup.D133 R.sup.D89 L.sub.C158
R.sup.D158 R.sup.D158 L.sub.C350 R.sup.D10 R.sup.D58 L.sub.C542 R.sup.D116 R.sup.D146
L.sub.C734 R.sup.D133 R.sup.D93 L.sub.C159 R.sup.D159 R.sup.D159 L.sub.C351 R.sup.D10
R.sup.D59 L.sub.C543 R.sup.D116 R.sup.D147 L.sub.C735 R.sup.D133 R.sup.D117 L.sub.C160
R.sup.D160 R.sup.D160 L.sub.C352 R.sup.D10 R.sup.D78 L.sub.C544 R.sup.D116 R.sup.D149
L.sub.C736 R.sup.D133 R.sup.D118 L.sub.C161 R.sup.D161 R.sup.D161 L.sub.C353 R.sup.D10
R.sup.D79 L.sub.C545 R.sup.D116 R.sup.D151 L.sub.C737 R.sup.D133 R.sup.D119 L.sub.C162
R.sup.D162 R.sup.D162 L.sub.C354 R.sup.D10 R.sup.D81 L.sub.C546 R.sup.D116 R.sup.D154
L.sub.C738 R.sup.D133 R.sup.D120 L.sub.C163 R.sup.D163 R.sup.D163 L.sub.C355 R.sup.D10
R.sup.D87 L.sub.C547 R.sup.D116 R.sup.D155 L.sub.C739 R.sup.D133 R.sup.D133 L.sub.C164
R.sup.D164 R.sup.D164 L.sub.C356 R.sup.D10 R.sup.D88 L.sub.C548 R.sup.D116 R.sup.D161
L.sub.C740 R.sup.D133 R.sup.D134 L.sub.C165 R.sup.D165 R.sup.D165 L.sub.C357 R.sup.D10
R.sup.D89 L.sub.C549 R.sup.D116 R.sup.D175 L.sub.C741 R.sup.D133 R.sup.D135 L.sub.C166
R.sup.D166 R.sup.D166 L.sub.C358 R.sup.D10 R.sup.D93 L.sub.C550 R.sup.D143 R.sup.D3
L.sub.C742 R.sup.D133 R.sup.D136 L.sub.C167 R.sup.D167 R.sup.D167 L.sub.C359 R.sup.D10
R.sup.D116 L.sub.C551 R.sup.D143 R.sup.D5 L.sub.C743 R.sup.D133 R.sup.D146 L.sub.C168
R.sup.D168 R.sup.D168 L.sub.C360 R.sup.D10 R.sup.D117 L.sub.C552 R.sup.D143 R.sup.D17
L.sub.C744 R.sup.D133 R.sup.D147 L.sub.C169 R.sup.D169 R.sup.D169 L.sub.C361 R.sup.D10
R.sup.D118 L.sub.C553 R.sup.D143 R.sup.D18 L.sub.C745 R.sup.D133 R.sup.D149 L.sub.C170
R.sup.D170 R.sup.D170 L.sub.C362 R.sup.D10 R.sup.D119 L.sub.C554 R.sup.D143 R.sup.D20
L.sub.C746 R.sup.D133 R.sup.D151 L.sub.C171 R.sup.D171 R.sup.D171 L.sub.C363 R.sup.D10
R.sup.D120 L.sub.C555 R.sup.D143 R.sup.D22 L.sub.C747 R.sup.D133 R.sup.D154 L.sub.C172

R.sup.D172 R.sup.D172 L.sub.C364 R.sup.D10 R.sup.D133 L.sub.C556 R.sup.D143 R.sup.D37
L.sub.C748 R.sup.D133 R.sup.D155 L.sub.C173 R.sup.D173 R.sup.D173 L.sub.C365 R.sup.D10
R.sup.D134 L.sub.C557 R.sup.D143 R.sup.D40 L.sub.C749 R.sup.D133 R.sup.D161 L.sub.C174
R.sup.D174 R.sup.D174 L.sub.C366 R.sup.D10 R.sup.D135 L.sub.C558 R.sup.D143 R.sup.D41
L.sub.C750 R.sup.D133 R.sup.D175 L.sub.C175 R.sup.D175 R.sup.D175 L.sub.C367 R.sup.D10
R.sup.D136 L.sub.C559 R.sup.D143 R.sup.D42 L.sub.C751 R.sup.D175 R.sup.D3 L.sub.C176
R.sup.D176 R.sup.D176 L.sub.C368 R.sup.D10 R.sup.D143 L.sub.C560 R.sup.D143 R.sup.D43
L.sub.C752 R.sup.D175 R.sup.D5 L.sub.C177 R.sup.D177 R.sup.D177 L.sub.C369 R.sup.D10
R.sup.D144 L.sub.C561 R.sup.D143 R.sup.D48 L.sub.C753 R.sup.D175 R.sup.D18 L.sub.C178
R.sup.D178 R.sup.D178 L.sub.C370 R.sup.D10 R.sup.D145 L.sub.C562 R.sup.D143 R.sup.D49
L.sub.C754 R.sup.D175 R.sup.D20 L.sub.C179 R.sup.D179 R.sup.D179 L.sub.C371 R.sup.D10
R.sup.D146 L.sub.C563 R.sup.D143 R.sup.D54 L.sub.C755 R.sup.D175 R.sup.D22 L.sub.C180
R.sup.D180 R.sup.D180 L.sub.C372 R.sup.D10 R.sup.D147 L.sub.C564 R.sup.D143 R.sup.D58
L.sub.C756 R.sup.D175 R.sup.D37 L.sub.C181 R.sup.D181 R.sup.D181 L.sub.C373 R.sup.D10
R.sup.D149 L.sub.C565 R.sup.D143 R.sup.D59 L.sub.C757 R.sup.D175 R.sup.D40 L.sub.C182
R.sup.D182 R.sup.D182 L.sub.C374 R.sup.D10 R.sup.D151 L.sub.C566 R.sup.D143 R.sup.D78
L.sub.C758 R.sup.D175 R.sup.D41 L.sub.C183 R.sup.D183 R.sup.D183 L.sub.C375 R.sup.D10
R.sup.D154 L.sub.C567 R.sup.D143 R.sup.D79 L.sub.C759 R.sup.D175 R.sup.D42 L.sub.C184
R.sup.D184 R.sup.D184 L.sub.C376 R.sup.D10 R.sup.D155 L.sub.C568 R.sup.D143 R.sup.D81
L.sub.C760 R.sup.D175 R.sup.D43 L.sub.C185 R.sup.D185 R.sup.D185 L.sub.C377 R.sup.D10
R.sup.D161 L.sub.C569 R.sup.D143 R.sup.D87 L.sub.C761 R.sup.D175 R.sup.D48 L.sub.C186
R.sup.D186 R.sup.D186 L.sub.C378 R.sup.D10 R.sup.D175 L.sub.C570 R.sup.D143 R.sup.D88
L.sub.C762 R.sup.D175 R.sup.D49 L.sub.C187 R.sup.D187 R.sup.D187 L.sub.C379 R.sup.D17
R.sup.D3 L.sub.C571 R.sup.D143 R.sup.D89 L.sub.C763 R.sup.D175 R.sup.D54 L.sub.C188
R.sup.D188 R.sup.D188 L.sub.C380 R.sup.D17 R.sup.D5 L.sub.C572 R.sup.D143 R.sup.D93
L.sub.C764 R.sup.D175 R.sup.D58 L.sub.C189 R.sup.D189 R.sup.D189 L.sub.C381 R.sup.D17
R.sup.D18 L.sub.C573 R.sup.D143 R.sup.D116 L.sub.C765 R.sup.D175 R.sup.D59 L.sub.C190
R.sup.D190 R.sup.D190 L.sub.C382 R.sup.D17 R.sup.D20 L.sub.C574 R.sup.D143 R.sup.D117
L.sub.C766 R.sup.D175 R.sup.D78 L.sub.C191 R.sup.D191 R.sup.D191 L.sub.C383 R.sup.D17
R.sup.D22 L.sub.C575 R.sup.D143 R.sup.D118 L.sub.C767 R.sup.D175 R.sup.D79 L.sub.C192
R.sup.D192 R.sup.D192 L.sub.C384 R.sup.D17 R.sup.D37 L.sub.C576 R.sup.D143 R.sup.D119
L.sub.C768 R.sup.D175 R.sup.D81 wherein R.sup.D1 to R.sup.D192 have the following

structures: ##STR00191## ##STR00192## ##STR00193## ##STR00194## ##STR00195##
##STR00196## ##STR00197## ##STR00198## ##STR00199## ##STR00200## ##STR00201##
##STR00202## ##STR00203## ##STR00204## ##STR00205## ##STR00206## ##STR00207##
##STR00208## ##STR00209## ##STR00210## ##STR00211##

15. The compound of claim 14, wherein the compound is selected from the group consisting of:

##STR00212## ##STR00213## ##STR00214## ##STR00215## ##STR00216## ##STR00217##
##STR00218## ##STR00219## ##STR00220## ##STR00221##

16. An organic light emitting device (OLED) comprising: an anode; a cathode; and an organic layer disposed between the anode and the cathode, wherein the organic layer comprises a compound of Formula Ir(L.sub.A).sub.x(L.sub.C).sub.y, wherein: ligand L.sub.A has Formula I ##STR00222## and ligand L.sub.C has Formula II' ##STR00223## wherein: x is 1, 2, or 3; y is 0, 1, or 2; x+y=3; X.sup.1-X.sup.8 are each independently C, N, or CR, and at least four of X.sup.1-X.sup.8 are N; the X.sup.1-X.sup.4 that connects to ring A is C, and the X.sup.1-X.sup.4 that coordinates to Ir is N; the maximum number of N atoms that are bonded to one another within a ring is two; ring A is an unsaturated 5-membered or 6-membered carbocyclic or heterocyclic ring; R.sup.A represents zero, mono, or up to the maximum number of allowed substitutions to its associated ring; each R, R.sup.1-R.sup.3, and R.sup.A is independently a hydrogen or a substituent selected from the group consisting of deuterium, halogen, alkyl, cycloalkyl, heteroalkyl, heterocycloalkyl, arylalkyl, alkoxy,

aryloxy, amino, silyl, boryl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carboxylic acid, ether, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof; R^{sup.1} comprises at least two carbon atoms; two substituents can be joined or fused together to form a ring, wherein the ligand L_{sub.A} coordinates to Ir to form a 5-membered chelate ring as indicated by the two dashed lines of the Formula I'; wherein at least one pair of substituents are joined or fused together to form a ring; and with a proviso that if ring A is a 6-membered ring and two R^{sup.A} are joined or fused to form a naphthalene that includes ring A, then said naphthalene is substituted.

17. The OLED of claim 16, wherein the organic layer further comprises a host, wherein host comprises at least one chemical group selected from the group consisting of triphenylene, carbazole, dibenzothiophene, dibenzofuran, dibenzoselenophene, azatriphenylene, azacarbazole, aza-dibenzothiophene, aza-dibenzofuran, and aza-dibenzoselenophene.

18. The OLED of claim 17, wherein the host is selected from the group consisting of:

##STR00224## ##STR00225## ##STR00226## ##STR00227## ##STR00228## ##STR00229##
and combinations thereof.

19. A consumer product comprising an organic light-emitting device (OLED) comprising: an anode; a cathode; and an organic layer disposed between the anode and the cathode, wherein the organic layer comprises a compound of Formula Ir(L_{sub.A})_{sub.x}(L_{sub.C})_{sub.y}, wherein: ligand L_{sub.A} has Formula I' ##STR00230## and ligand L_{sub.C} has Formula II ##STR00231## wherein: x is 1, 2, or 3; y is 0, 1, or 2; x+y=3; X^{sup.1}-X^{sup.8} are each independently C, N, or CR, and at least four of X^{sup.1}-X^{sup.8} are N; the X^{sup.1}-X^{sup.4} that connects to ring A is C, and the X^{sup.1}-X^{sup.4} that coordinates to Ir is N; the maximum number of N atoms that are bonded to one another within a ring is two; ring A is an unsaturated 5-membered or 6-membered carbocyclic or heterocyclic ring; R^{sup.A} represents zero, mono, or up to the maximum number of allowed substitutions to its associated ring; each R, R^{sup.1}-R^{sup.3}, and R^{sup.A} is independently a hydrogen or a substituent selected from the group consisting of deuterium, halogen, alkyl, cycloalkyl, heteroalkyl, heterocycloalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, boryl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carboxylic acid, ether, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof; R^{sup.1} comprises at least two carbon atoms; two substituents can be joined or fused together to form a ring, wherein the ligand L_{sub.A} coordinates to Ir to form a 5-membered chelate ring as indicated by the two dashed lines of the Formula I'; wherein at least one pair of substituents are joined or fused together to form a ring; and with a proviso that if ring A is a 6-membered ring and two R^{sup.A} are joined or fused to form a naphthalene that includes ring A, then said naphthalene is substituted.

20. A formulation comprising a compound according to claim 1.
